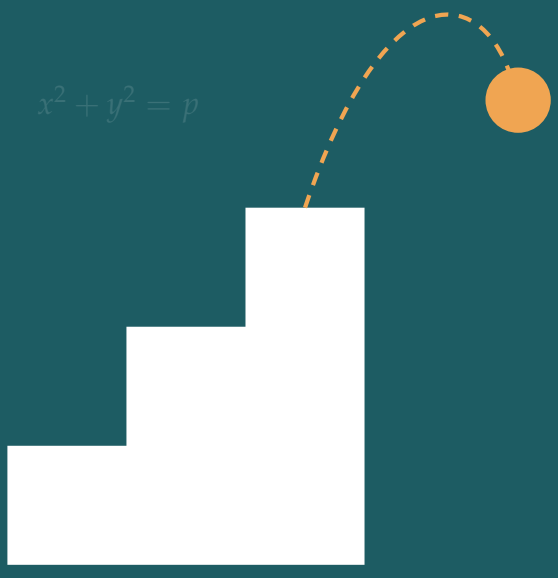
STEP PREPARATION · UNIVERSITY MATHEMATICS

Beyond STEP

STEP I, II AND III

1988 Worked Solutions



KEITH McNULTY

Beyond STEP

Past STEP questions as a doorway
to university mathematics

STEP I, II AND III

1988 Worked Solutions

Keith McNulty

Beyond STEP: Past STEP questions as a doorway to university mathematics

1988 Worked Solutions

Worked solutions to all 48 questions of the 1988 STEP papers (STEP I, II and III). Worked solutions to every STEP question since the specimen papers, the whole book *Beyond STEP* and every chapter on its own are free to read and download at `beyondstep.keithmcnulty.org`.

Have a proper go at a question before you read its solution. Every answer has been checked numerically, but with this many solutions some slips are inevitable: if something looks wrong, trust your own working and check it carefully.

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Corrections are welcome as issues at github.com/keithmcnulty/step_book_mcnulty.

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WORKED SOLUTION

STEP I 1988, Question 1

Pure • Curve Sketching

STEP I 1988, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/88-s1-q1

The key idea

The equation $n^m = m^n$ mixes the variables up between base and exponent, which is awkward. Taking logarithms gives $m \ln n = n \ln m$, and dividing by mn separates them completely: $\frac{\ln n}{n} = \frac{\ln m}{m}$. So we need two different integers at which h takes the same value, and the shape of the graph tells us exactly where such pairs can live.

Solution

The graph of h

Differentiating with the quotient rule,

$$\mathrm{h}'(x) = \frac{x \cdot \frac{1}{x} - \ln x}{x^2} = \frac{1 - \ln x}{x^2}.$$

Since $x^2 > 0$, the sign of $\mathrm{h}'(x)$ is the sign of $1 - \ln x$. So h is strictly increasing on $(0, \mathrm{e})$ and strictly decreasing on (e, ∞), with a maximum at $x = \mathrm{e}$ of value

$$\mathrm{h}(\mathrm{e}) = \frac{1}{\mathrm{e}}.$$

Differentiating again,

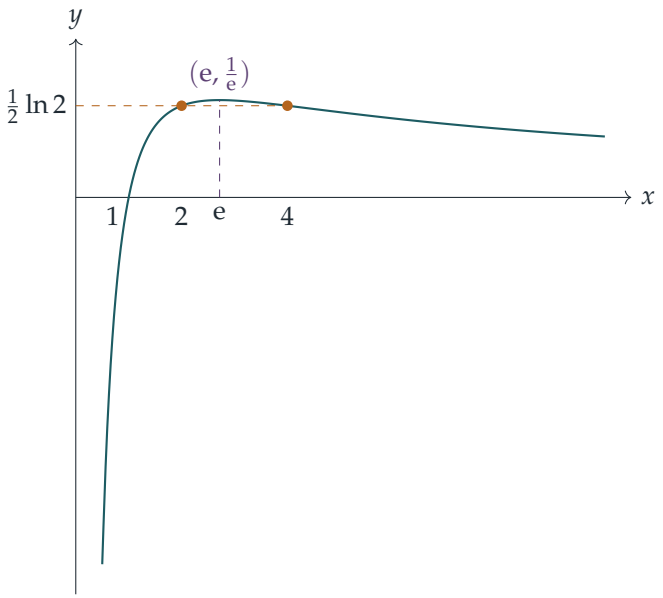
$$\mathrm{h}''(x) = \frac{-\frac{1}{x} \cdot x^2 - (1 - \ln x) \cdot 2x}{x^4} = \frac{2 \ln x - 3}{x^3},$$

so there is a point of inflection at $x = \mathrm{e}^{3/2} \approx 4.48$, where $\mathrm{h} = \frac{3}{2}\mathrm{e}^{-3/2} \approx 0.335$. The curve is concave to the left of this point and convex to the right.

Next, the behaviour at the ends and the sign:

- As $x \to 0^+$, $\ln x \to -\infty$ and $\frac{1}{x} \to +\infty$, so $\mathrm{h}(x) \to -\infty$: the y-axis is an asymptote.
- $\mathrm{h}(1) = 0$, and $\mathrm{h}(x)$ has the sign of $\ln x$: negative on $(0, 1)$, positive for $x > 1$.

- As $x \to \infty$, $\frac{\ln x}{x} \to 0$ (logarithms grow more slowly than any positive power of x), and $\mathrm{h}(x) > 0$ there, so the curve approaches the x-axis from above.



The equation $n^m = m^n$

Since m and n are positive, both sides are positive and we may take logarithms:

$$n^m = m^n \iff m \ln n = n \ln m \iff \frac{\ln n}{n} = \frac{\ln m}{m} \iff \mathrm{h}(n) = \mathrm{h}(m),$$

dividing by $mn > 0$ in the middle step. So we want distinct positive integers with $\mathrm{h}(m) = \mathrm{h}(n)$. By symmetry, suppose $m < n$.

Ask yourself Where on the graph can two different points have the same height?

Since h is strictly increasing on $(0, \mathrm{e}]$ and strictly decreasing on $[\mathrm{e}, \infty)$, it is one-to-one on each of these intervals. So two distinct inputs with the same output must lie on opposite sides of e: $m < \mathrm{e} < n$. Since m is a positive integer and $\mathrm{e} \approx 2.718$, $m = 1$ or $m = 2$.

- If $m = 1$ then $\mathrm{h}(n) = \mathrm{h}(1) = 0$, which forces $\ln n = 0$, so $n = 1 = m$, not allowed. (Directly: $n^1 = 1^n$ gives $n = 1$.)
- If $m = 2$ then we need an integer $n \geq 3$ with $\mathrm{h}(n) = \frac{1}{2}\ln 2$. Now $n = 4$ works, because

  $$\mathrm{h}(4) = \frac{\ln 4}{4} = \frac{2\ln 2}{4} = \frac{\ln 2}{2} = \mathrm{h}(2),$$

  that is, $4^2 = 16 = 2^4$. Since h is strictly decreasing on $[\mathrm{e}, \infty)$, it takes the value $\frac{1}{2}\ln 2$ at most once there, so $n = 4$ is the only possibility.

Hence the only pairs of distinct positive integers with $n^m = m^n$ are

$$(m,n) = (2,4) \quad \text{and} \quad (m,n) = (4,2).$$

Checks

A computer search over all $1 \le m,n \le 300$ with exact integer arithmetic finds only $(2,4)$ and $(4,2)$. Numerically, $\mathrm{h}(2) = \mathrm{h}(4) = 0.346574\ldots$, just below the maximum $1/\mathrm{e} = 0.367879\ldots$, as the graph suggests. You can also see why $\mathrm{h}(3)$ gives nothing: $\mathrm{h}(3) \approx 0.366$ exceeds $\mathrm{h}(2)$, and its partner on the left of e lies strictly between 2 and e, so it is not an integer.

Common slips

- Finding $(2,4)$ by trial and stopping there. The question asks for *all* pairs, so the monotonicity argument proving there are no others is the heart of the answer.
- Forgetting the asymptotic behaviour in the sketch: the curve plunges to $-\infty$ at the y-axis but only creeps towards 0 as $x \to \infty$, never crossing the axis again.

Beyond STEP

If you allow real numbers, every a in $(1,\mathrm{e})$ has exactly one partner $b > \mathrm{e}$ with $a^b = b^a$, and these pairs can be parametrised neatly as $a = (1+\frac{1}{t})^t$, $b = (1+\frac{1}{t})^{t+1}$ for $t > 0$; as $t \to \infty$ both tend to e, which is the meeting point at the top of the graph. The limit $\frac{\ln x}{x} \to 0$ that we used is a first example of comparing growth rates, made rigorous in a first analysis course; see Abbott, *Understanding Analysis*.

WORKED SOLUTION

STEP I 1988, Question 2

Pure • Stationary Points

STEP I 1988, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/88-s1-q2

The key idea

Think of g as a composition: $\mathrm{g} = h \circ \mathrm{f}$ with $h(t) = t + 1/t$. The chain rule then says g' vanishes either where $\mathrm{f}' = 0$ or where $h'(\mathrm{f}) = 0$, that is, where $\mathrm{f} = \pm 1$. Writing $\cos 2x$ in terms of $\sin x$ turns f into a quadratic in $\sin x$, which tells us everything about its size and shape.

Solution

The derivatives

Differentiating $\mathrm{g} = \mathrm{f} + \mathrm{f}^{-1}$ with the chain rule,

$$\mathrm{g}'(x) = \mathrm{f}'(x) - \frac{\mathrm{f}'(x)}{\mathrm{f}(x)^2} = \mathrm{f}'(x)\left(1 - \frac{1}{\mathrm{f}(x)^2}\right).$$

Differentiating again with the product rule, and using $\dfrac{\mathrm{d}}{\mathrm{d}x}\left(-\mathrm{f}^{-2}\right) = 2\mathrm{f}^{-3}\mathrm{f}'$,

$$\mathrm{g}''(x) = \mathrm{f}''(x)\left(1 - \frac{1}{\mathrm{f}(x)^2}\right) + \frac{2\,\mathrm{f}'(x)^2}{\mathrm{f}(x)^3}.$$

The size of f

Using $\cos 2x = 1 - 2\sin^2 x$ and writing $s = \sin x$,

$$\mathrm{f}(x) = 5 + 2s - 2s^2 = \tfrac{11}{2} - 2\left(s - \tfrac{1}{2}\right)^2.$$

As s runs over $[-1, 1]$, this is largest ($\frac{11}{2}$) at $s = \frac{1}{2}$ and smallest at the end of the interval furthest from $\frac{1}{2}$, namely $s = -1$, where it equals 1. So

$$1 \le \mathrm{f}(x) \le \tfrac{11}{2} \quad \text{for all } x,$$

with $\mathrm{f}(x) = 1$ only when $\sin x = -1$. In particular f is never zero, so g is defined everywhere.

Stationary points

$g'(x) = 0$ exactly when $f'(x) = 0$ or $f(x)^2 = 1$.

Zeros of f'. We have

$$f'(x) = -2\sin 2x + 2\cos x = 2\cos x\,(1 - 2\sin x),$$

which vanishes when $\cos x = 0$ or $\sin x = \frac{1}{2}$; in $[0, 2\pi]$ that is

$$x = \tfrac{\pi}{6},\ \tfrac{\pi}{2},\ \tfrac{5\pi}{6},\ \tfrac{3\pi}{2}.$$

Points where $f = \pm 1$. Since $f \geq 1$, this means $f = 1$, i.e. $\sin x = -1$, i.e. $x = \frac{3\pi}{2}$, which is already on the list.

So g has exactly four stationary points in $[0, 2\pi]$. (The endpoints are not among them: $g'(0) = g'(2\pi) = 2 \cdot \frac{24}{25} \neq 0$.)

Classifying them

We have $f''(x) = -4\cos 2x - 2\sin x$. At each point $f' = 0$, so the second term of g'' drops out and $g'' = f''\,(1 - f^{-2})$.

- $x = \frac{\pi}{6}$ and $x = \frac{5\pi}{6}$: here $\sin x = \frac{1}{2}$, $\cos 2x = \frac{1}{2}$, so $f = \frac{11}{2}$ and $f'' = -2 - 1 = -3$. Then $g'' = -3\big(1 - \frac{4}{121}\big) < 0$: both are **maxima**, with

$$g = \tfrac{11}{2} + \tfrac{2}{11} = \tfrac{125}{22}.$$

- $x = \frac{\pi}{2}$: here $\sin x = 1$, $\cos 2x = -1$, so $f = 5$ and $f'' = 4 - 2 = 2$. Then $g'' = 2\big(1 - \frac{1}{25}\big) = \frac{48}{25} > 0$: a **minimum**, with $g = 5 + \frac{1}{5} = \frac{26}{5}$.
- $x = \frac{3\pi}{2}$: here $f = 1$, so $1 - f^{-2} = 0$ and also $f' = 0$. Hence $g'' = 0$.

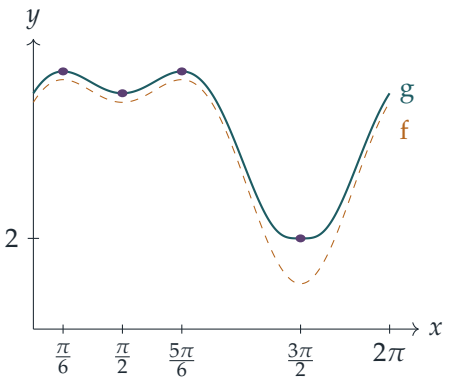
> **Ask yourself** The second-derivative test fails at $x = \frac{3\pi}{2}$. How else can we decide?

Go back to $h(t) = t + 1/t$. For $t > 0$,

$$t + \frac{1}{t} - 2 = \frac{(t-1)^2}{t} \geq 0,$$

with equality only at $t = 1$. Since $f > 0$, this gives $g(x) \geq 2$ for every x, with equality exactly when $f(x) = 1$. At $x = \frac{3\pi}{2}$ we have $g = 2$, so this is a **minimum** (indeed the global minimum of g).

Summary. Maxima at $x = \frac{\pi}{6}, \frac{5\pi}{6}$ (value $\frac{125}{22}$); minima at $x = \frac{\pi}{2}$ (value $\frac{26}{5}$) and $x = \frac{3\pi}{2}$ (value 2).



Checks

- Since h is increasing for $t > 1$, g should rise and fall with f wherever $\mathrm{f} > 1$; the graph confirms that f (dashed) and g have the same turning points.
- Numerically, $\mathrm{g}(\frac{\pi}{6}) \approx 5.6818 = \frac{125}{22}$ and $\mathrm{g}(\frac{\pi}{2}) = 5.2$; a fine scan of g over $[0, 2\pi]$ finds turning points only at $\frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}, \frac{3\pi}{2}$, with minimum value 2.000.
- The formula for g'' agrees with a finite-difference second derivative at random points.

Common slips

- Forgetting the factor $1 - \mathrm{f}^{-2}$: here it adds no new stationary point, but only because $\mathrm{f} = 1$ happens where $\mathrm{f}' = 0$ anyway. You must still check it.
- Concluding from $\mathrm{g}''(\frac{3\pi}{2}) = 0$ that the point is a point of inflection. A zero second derivative tells you nothing on its own.

Beyond STEP

The degenerate point $x = \frac{3\pi}{2}$ is typical of compositions: if f has a minimum where h also has a minimum, $h \circ \mathrm{f}$ is "flatter" there, and the second derivative vanishes. Undergraduate analysis replaces the second-derivative test with Taylor's theorem, which classifies such points by the first non-zero derivative, or (as we did) with a direct inequality. Abbott's *Understanding Analysis* is a good place to see the mean value theorem and Taylor's theorem developed carefully.

WORKED SOLUTION

STEP I 1988, Question 3

Pure • Geometric Extremal Bounds

STEP I 1988, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/88-s1-q3

The key idea

Ask where a point of the square can be if it is *far* from O: it must be squeezed into a small region near the opposite corner. If both P and Q are far from O, they are both in that small region, so they are close to each other. The number $\sqrt{6}-\sqrt{2}$ is exactly the size that makes this work, and it comes from an equilateral triangle sitting symmetrically in the square.

Solution

Work in centimetres with O at the origin and the square $0 \le x \le 1$, $0 \le y \le 1$. Write

$$s = \sqrt{6}-\sqrt{2}, \qquad s^2 = 6+2-2\sqrt{12} = 8-4\sqrt{3} \approx 1.072,$$

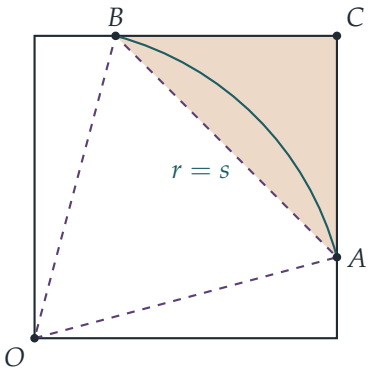
and let $t = 2-\sqrt{3} \approx 0.268$. Define the points

$$A = (1,t), \qquad B = (t,1), \qquad C = (1,1).$$

Then

$$\begin{aligned} OA^2 = OB^2 &= 1+(2-\sqrt{3})^2 = 8-4\sqrt{3} = s^2, \\ AB^2 &= 2(1-t)^2 = 2(\sqrt{3}-1)^2 = 8-4\sqrt{3} = s^2, \\ AC = BC &= 1-t = \sqrt{3}-1 \approx 0.732 < s. \end{aligned}$$

So OAB is an equilateral triangle of side s inside the square.



The teal arc is the circle of radius s about O; the part of the square beyond it lies inside the shaded triangle ABC.

Step 1: points far from O lie in triangle ABC

The line AB is $x + y = 1 + t$. Consider the part of the square on the O side of this line, $\{x + y \leq 1 + t\}$. It is the pentagon with vertices

$$O = (0,0), \quad (1,0), \quad A, \quad B, \quad (0,1).$$

The function $x^2 + y^2$ is convex, so its maximum over this convex pentagon is attained at a vertex. The values at the vertices are 0, 1, s^2, s^2, 1, and the largest is s^2. Hence

$$x + y \leq 1 + t \implies x^2 + y^2 \leq s^2$$

for every point of the square. Turning this round: if a point (x, y) of the square has distance from O *greater* than s, then $x + y > 1 + t$, and since $x \leq 1$ and $y \leq 1$ the point lies in the triangle ABC.

> **Ask yourself** Why is the largest distance between two points of a triangle one of its sides?

Fix Q. The distance $|PQ|$ is a convex function of P, so over the triangle it is largest when P is a vertex. Now fix that vertex and vary Q: by the same argument Q may also be taken to be a vertex. So the largest distance between two points of a triangle is the distance between two vertices, which is its longest side.

Step 2: the contradiction

Suppose that $OP > s$ and $OQ > s$; otherwise we are already done. By Step 1, both P and Q lie in triangle ABC. The longest side of that triangle is $AB = s$ (the other two sides are $\sqrt{3} - 1 < s$). So by the box above,

$$PQ \leq s.$$

In every case, then, at least one of OP, PQ, QO is at most $s = (\sqrt{6} - \sqrt{2})$ cm, which is what we had to show.

The bound is best possible

Taking $P = A$ and $Q = B$ gives $OP = PQ = QO = s$ exactly, so no smaller number would do. Since $\cos 15^\circ = \frac{\sqrt{6}+\sqrt{2}}{4}$, we have $s = \sec 15^\circ$: the extreme triangle is the equilateral triangle placed symmetrically about the diagonal, with each of OA and OB making 15° with a side of the square.

Checks

- $s \approx 1.0353$, a little more than the side of the square and well under the diagonal $\sqrt{2} \approx 1.414$, which is sensible.
- Numerically, 10^6 random pairs P, Q in the square never gave $\min(OP, PQ, QO)$ above s (the largest was about 1.0235). A hill-climbing search for the largest possible minimum settled on $P \approx (1, 0.26795)$, $Q \approx (0.26795, 1)$ with value 1.035276, matching A, B and s.

Common slips

- Arguing only that $\angle POQ < 60^\circ$ when $OP, OQ > s$. That is true, but it does not by itself make PQ the shortest side, so it does not finish the proof without more work.
- Checking only the symmetric configuration. It shows the bound cannot be improved; it does not prove that the bound holds for every P and Q.

Beyond STEP

This is a small extremal problem about packing points: how far apart can three points of a region be, all at once? Questions like this lead to packing and covering problems, and to the idea of the *diameter* of a set, which is the same for a set as for its convex hull. That fact is what made Step 2 painless. For more problems in this spirit, see Siklos, *Advanced Problems in Mathematics*.

WORKED SOLUTION

STEP I 1988, Question 4

Pure • Counting Line Intersections

STEP I 1988, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/88-s1-q4

The key idea

Don't try to picture all the segments at once. Ask instead: when do *two* segments cross? The answer depends only on the order of their endpoints along the axes, and it turns every crossing into a choice of two points on each axis. Once we can count the crossings on a single segment, the extreme segments and the final average follow by careful bookkeeping.

Solution

Label the points on the y-axis $B_1, \ldots, B_m$ in order of increasing distance from the origin O, and the points on the x-axis $A_1, \ldots, A_n$ in the same way. Write s_{ij} for the segment B_iA_j.

Part (i)

Each of the m points B_i is joined to each of the n points A_j, so there are

$$mn$$

segments.

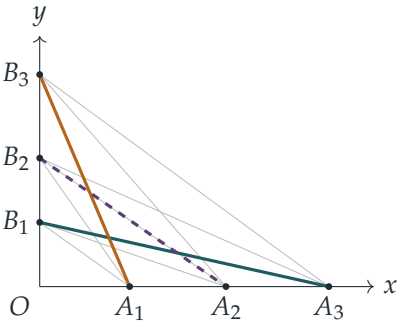
Part (ii)

Take two different segments s_{ij} and s_{kl}.

- If $i = k$ or $j = l$, they share an endpoint. Two distinct segments from a common point meet only there, and those meetings are ignored.
- If $i < k$ and $j < l$, both B_i and A_j lie on the same side of the line B_kA_l as O (they are on the legs OB_k and OA_l of the triangle OB_kA_l, not at B_k or A_l). A segment whose endpoints lie strictly on one side of a line stays on that side, so s_{ij} never meets s_{kl}: it is "nested" underneath.

- If $i < k$ and $j > l$, then B_i is on the same side of the line B_kA_l as O, while A_j is on the far side. So s_{ij} crosses that line. In the same way, A_l is on the O side of the line B_iA_j and B_k is on the far side. Each segment separates the endpoints of the other, so the two segments cross at exactly one point, which is not an endpoint.

The case $i > k$ is the same with the roles swapped.



In the figure ($m = n = 3$), the teal s_{13} and amber s_{31} have their endpoints in opposite orders, so they cross. The dashed s_{22} crosses each of them too.

So each pair of segments crossing at an interior point comes from choosing two points B_i, B_k ($i < k$) and two points A_j, A_l ($j < l$), and then taking s_{il} and s_{kj}. Conversely, each such choice gives exactly one crossing pair. Since no three segments pass through a common interior point, different pairs give different points. Hence the number of points of intersection is

$$\binom{m}{2}\binom{n}{2} = \frac{m(m-1)n(n-1)}{4}.$$

The average

Crossings on one segment. By the rule above, s_{ij} crosses s_{kl} exactly when $k < i$, $l > j$ or $k > i$, $l < j$. So the number of points of intersection on s_{ij} is

$$c_{ij} = (i-1)(n-j) + (m-i)(j-1).$$

The extreme segments when $m = n$. Put $N = n-1$, $a = i-1$, $b = j-1$, so $0 \le a, b \le N$ and

$$c_{ij} = a(N-b) + (N-a)b = N(a+b) - 2ab = \tfrac{1}{2}\big(N^2 - (N-2a)(N-2b)\big),$$

as you can check by expanding. Since $|N-2a| \le N$ and $|N-2b| \le N$, the product $(N-2a)(N-2b)$ lies between $-N^2$ and N^2. Therefore $0 \le c_{ij} \le N^2$, and:

- $c_{ij} = N^2$ only when $(N-2a)(N-2b) = -N^2$, that is, $\{a, b\} = \{0, N\}$. These are the two segments s_{1n} and s_{n1}.
- $c_{ij} = 0$ only when $(N-2a)(N-2b) = N^2$, that is, $a = b = 0$ or $a = b = N$. These are the two segments s_{11} and s_{nn}.

(Here $N \geq 2$, so the four segments are distinct.) So exactly two segments carry the greatest number, $(n-1)^2$, of points, and exactly two carry the least, 0. Removing them leaves $n^2 - 4$ segments.

Ask yourself What exactly is being averaged?

We count the points of intersection that remain once the four segments are gone, i.e. crossings between two remaining segments, each point counted once. Then we divide by the number of remaining segments. (If you instead add up the crossings on each remaining segment, every point is counted twice and some lost points are still counted. That gives $\frac{1}{2}(n-1)^2$, which is not the stated answer. See Common slips.)

Points lost. The segments s_{11} and s_{nn} carry no points of intersection, so removing them loses nothing. The segments s_{1n} and s_{n1} carry $(n-1)^2$ points each. They cross each other ($1 < n$ and $n > 1$), so exactly one point lies on both. By inclusion–exclusion, the number of points lost is $2(n-1)^2 - 1$.

Points remaining. From part (ii) with $m = n$, the number of points remaining is

$$\begin{aligned}\frac{n^2(n-1)^2}{4} - 2(n-1)^2 + 1 &= \frac{(n-1)^2(n^2-8)+4}{4} \\ &= \frac{n^4 - 2n^3 - 7n^2 + 16n - 4}{4} \\ &= \frac{(n-2)(n^3 - 7n + 2)}{4}.\end{aligned}$$

For the factorisation, note that $n = 2$ makes the quartic $16 - 16 - 28 + 32 - 4 = 0$, so $n - 2$ is a factor. Multiplying out $(n-2)(n^3 - 7n + 2) = n^4 - 2n^3 - 7n^2 + 16n - 4$ confirms it.

The average. Dividing by the $n^2 - 4 = (n-2)(n+2)$ remaining segments, and cancelling $n - 2 \neq 0$ (since $n \geq 3$), gives an average of

$$\frac{(n-2)(n^3 - 7n + 2)}{4(n-2)(n+2)} = \frac{n^3 - 7n + 2}{4(n+2)},$$

as required.

Checks

For $n = 3$ the formula gives $\frac{8}{20} = \frac{2}{5}$. Directly: removing $s_{11}, s_{33}, s_{13}, s_{31}$ leaves $s_{12}, s_{21}, s_{22}, s_{23}, s_{32}$. Among these, only s_{12} with s_{21}, and s_{23} with s_{32}, have endpoints in opposite orders, so there are 2 points on 5 segments, which gives $\frac{2}{5}$. A computer check placed random points on the axes, found every crossing with exact arithmetic, and confirmed mn, $\binom{m}{2}\binom{n}{2}$ and c_{ij}. It also confirmed the average formula for every n from 3 to 29.

Common slips

- Adding the crossings on each remaining segment. Each point lies on two segments, so this double counts, and it still includes points shared with the removed segments. The result is $\frac{1}{2}(n-1)^2$, not the given formula.
- Forgetting that s_{1n} and s_{n1} cross each other, and so subtracting $2(n-1)^2$ instead of $2(n-1)^2-1$.
- Counting crossings as $\binom{mn}{2}$ pairs of segments: most pairs share an endpoint or are nested and never cross.

Beyond STEP

The heart of this question is a bijection: crossings correspond exactly to choices of two points on each axis. This is the basic tool of combinatorics. The same picture appears in the "crossing number" of the complete bipartite graph $K_{m,n}$. Zarankiewicz's drawing (exactly this one, with points on both sides of each axis) uses $\lfloor \frac{m}{2} \rfloor \lfloor \frac{m-1}{2} \rfloor \lfloor \frac{n}{2} \rfloor \lfloor \frac{n-1}{2} \rfloor$ crossings, and whether this is always the least possible is still an open problem. For more counting by bijection, try Siklos, *Advanced Problems in Mathematics*, and for graph theory, Wilson, *Introduction to Graph Theory*.

WORKED SOLUTION

STEP I 1988, Question 5

Pure • Series and Generating Functions

STEP I 1988, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/88-s1-q5

The key idea

Expand each factor as a geometric series (the binomial theorem with index -1) and multiply: c_m is then a finite geometric sum, which has a neat closed form. Once c_m^2 is written as a combination of three pure powers, $(a^2)^m$, $(ab)^m$ and $(b^2)^m$, the final series splits into three geometric series. The rest is careful algebra with fractions, and keeping track of where each series converges.

Solution

Finding c_m

By the binomial theorem with index -1,

$$(1-ax)^{-1} = \sum_{k=0}^{\infty} a^k x^k \quad (|ax|<1), \qquad (1-bx)^{-1} = \sum_{j=0}^{\infty} b^j x^j \quad (|bx|<1).$$

Since $0 < a < b$, the condition $b|x| < 1$ gives $a|x| < b|x| < 1$, so both expansions are valid. Both series converge absolutely, so we may multiply them and collect powers of x. The coefficient of x^m comes from the pairs $k + j = m$:

$$c_m = \sum_{k=0}^{m} a^k b^{m-k} = b^m + ab^{m-1} + \cdots + a^{m-1}b + a^m.$$

This is a geometric series with first term b^m, common ratio a/b and $m+1$ terms. Since $a \neq b$, its sum is

$$c_m = \frac{b^{m+1} - a^{m+1}}{b-a}.$$

(Equivalently, $(b-a)c_m = b^{m+1} - a^{m+1}$, because the sum telescopes when multiplied out.)

The formula for c_m^2

Squaring, and using $(b-a)^2 = (a-b)^2$,

$$c_m^2 = \frac{\left(b^{m+1} - a^{m+1}\right)^2}{(b-a)^2} = \frac{a^{2m+2} - 2(ab)^{m+1} + b^{2m+2}}{(a-b)^2},$$

as required.

The series $\sum c_m^2 x^m$

The point of the previous formula is that each term is a constant times an mth power:

$$c_m^2 x^m = \frac{1}{(a-b)^2}\left(a^2\,(a^2x)^m - 2ab\,(abx)^m + b^2\,(b^2x)^m\right).$$

So $\sum c_m^2 x^m$ is a combination of three geometric series, with ratios a^2x, abx and b^2x.

? **Ask yourself** For which x do all three geometric series converge?

We need $|a^2x| < 1$, $|abx| < 1$ and $|b^2x| < 1$. Since $0 < a^2 < ab < b^2$, the last condition is the strongest and implies the other two. So all three converge exactly when $|x| < 1/b^2$. Then, summing each geometric series,

$$\sum_{m=0}^{\infty} c_m^2 x^m = \frac{1}{(a-b)^2}\left(\frac{a^2}{1-a^2x} - \frac{2ab}{1-abx} + \frac{b^2}{1-b^2x}\right).$$

Put everything over the common denominator $(1-abx)(1-a^2x)(1-b^2x)$. The numerator is

$$N = a^2(1-abx)(1-b^2x) - 2ab(1-a^2x)(1-b^2x) + b^2(1-a^2x)(1-abx).$$

We collect it by powers of x:

$$\begin{aligned}
\text{constant term:} \quad & a^2 - 2ab + b^2 = (a-b)^2, \\
\text{coefficient of } x: \quad & -a^2(ab+b^2) + 2ab(a^2+b^2) - b^2(a^2+ab) \\
& \qquad = a^3b - 2a^2b^2 + ab^3 = ab(a-b)^2, \\
\text{coefficient of } x^2: \quad & a^2 \cdot ab^3 - 2ab \cdot a^2b^2 + b^2 \cdot a^3b = 0.
\end{aligned}$$

So $N = (a-b)^2(1+abx)$. The factor $(a-b)^2$ cancels, which leaves

$$c_0^2 + c_1^2 x + c_2^2 x^2 + \cdots = \frac{1+abx}{(1-abx)(1-a^2x)(1-b^2x)} \qquad \text{for } -\frac{1}{b^2} < x < \frac{1}{b^2}.$$

The interval can't be any larger. At $x = \pm 1/b^2$ the terms satisfy

$$|c_m^2 x^m| = \frac{\left(b^{m+1} - a^{m+1}\right)^2}{(b-a)^2\, b^{2m}} = \frac{b^2\left(1 - (a/b)^{m+1}\right)^2}{(b-a)^2} \to \frac{b^2}{(b-a)^2} \neq 0,$$

so the series diverges there, and it diverges for $|x| > 1/b^2$ too, where the terms grow without bound.

Checks

Take $a = 1$, $b = 2$. Then $c_m = 2^{m+1} - 1$, so $c_0, c_1, c_2 = 1, 3, 7$, and directly $\frac{1}{(1-x)(1-2x)} = (1 + x + x^2 + \dots)(1 + 2x + 4x^2 + \dots) = 1 + 3x + 7x^2 + \cdots$. The squares are $1, 9, 49$. On the other side, $\frac{1}{(1-x)(1-2x)(1-4x)} = 1 + 7x + 35x^2 + \cdots$, and multiplying by $1 + 2x$ gives $1 + 9x + 49x^2 + \cdots$, as it should. Numerically, with $a = 0.7$, $b = 1.3$ and $x = 0.3$, summing $c_m^2 x^m$ to many terms gives $4.163878\ldots$, which matches the closed form.

Common slips

- Giving the interval as $b|x| < 1$, copied from the first part. The new series has ratios a^2x, abx, b^2x, so the condition is $b^2|x| < 1$.
- Leaving c_m as a sum. The closed form $(b^{m+1} - a^{m+1})/(b - a)$ is what makes the rest of the question work.
- Sign errors in the coefficient of x in N. Expanding each product carefully, term by term, is quicker than trying to be clever.

Beyond STEP

The series $\sum c_m x^m$ is the *generating function* of the sequence c_m, and the question shows that if a sequence has a rational generating function, so does the sequence of its squares. This is a special case of the *Hadamard product* of two rational power series, which is always rational. The interval found here is an instance of the *radius of convergence*: a power series of a rational function converges up to the nearest pole, here $x = 1/b^2$. Generating functions are developed beautifully in Wilf's *generatingfunctionology*, and power series and their convergence are treated carefully in Abbott, *Understanding Analysis*.

WORKED SOLUTION

STEP I 1988, Question 6

Pure • Complex Numbers and Geometry

STEP I 1988, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/88-s1-q6

The key idea

In the Argand diagram the vector $\overrightarrow{P_jP_k}$ is just the complex number $z_k - z_j$, so equal vectors become equal differences, and perpendicularity becomes "one difference divided by the other is purely imaginary". Once the given values are substituted, condition (i) pins down z_4 and w in terms of z, and condition (ii) fixes the imaginary part of z. The hexagon then turns out to be symmetric about its centre, which makes the convexity and the final angle easy to handle.

Solution

Translating the statements

Since $\overrightarrow{P_jP_k}$ is represented by $z_k - z_j$, statement (i) says

$$z_2 - z_1 = z_4 - z_5 \qquad \text{and} \qquad z_3 - z_2 = z_5 - z_6.$$

For (ii), two non-zero vectors u and v are perpendicular exactly when their arguments differ by $\pm\frac{\pi}{2}$, that is, when u/v is purely imaginary. The points are distinct, so $z_4 - z_2 \neq 0$ and $z_6 - z_3 \neq 0$, and statement (ii) says

$$\frac{z_4 - z_2}{z_6 - z_3} \text{ is purely imaginary,} \qquad \text{equivalently} \qquad \operatorname{Re}\left[(z_4 - z_2)\,\overline{(z_6 - z_3)}\right] = 0.$$

The consequence of (i)

Adding the two equations in (i), the z_2 and z_5 terms cancel:

$$z_3 - z_1 = z_4 - z_6, \qquad \text{so} \qquad z_4 - z_3 = z_6 - z_1,$$

which is the statement $\overrightarrow{P_3P_4} = \overrightarrow{P_1P_6}$, as required.

The hexagon

Now put $z_1 = 0$, $z_2 = 1$, $z_3 = z$, $z_5 = \mathrm{i}$ and $z_6 = w$. The first equation of (i) gives

$$z_4 = z_5 + z_2 - z_1 = 1 + \mathrm{i},$$

and the second gives $z - 1 = \mathrm{i} - w$, so

$$w = 1 + \mathrm{i} - z.$$

Notice that $z_1 + z_4 = z_2 + z_5 = z_3 + z_6 = 1 + \mathrm{i}$: opposite vertices are symmetric about the point $\frac{1}{2}(1 + \mathrm{i})$, so the hexagon has a centre of symmetry.

For (ii), $z_4 - z_2 = \mathrm{i}$ and $z_6 - z_3 = w - z = 1 + \mathrm{i} - 2z$. The quotient $\mathrm{i}/(1 + \mathrm{i} - 2z)$ is purely imaginary exactly when $1 + \mathrm{i} - 2z$ is real, that is, when $1 - 2\,\mathrm{Im}(z) = 0$. So

$$\mathrm{Im}(z) = \tfrac{1}{2}, \qquad \text{and then} \qquad \mathrm{Im}(w) = 1 - \tfrac{1}{2} = \tfrac{1}{2}.$$

Write $z = a + \frac{1}{2}\mathrm{i}$ with a real. Then $w = (1 - a) + \frac{1}{2}\mathrm{i}$, so $\mathrm{Re}(w) = 1 - \mathrm{Re}(z)$ always. The six vertices are

$$(0,0), \quad (1,0), \quad (a, \tfrac{1}{2}), \quad (1,1), \quad (0,1), \quad (1-a, \tfrac{1}{2}) :$$

the unit square with a "point" added halfway up each vertical side.

Ask yourself How can we test convexity without relying on a picture?

Walk round the hexagon in order. It is convex exactly when we always turn the same way at each vertex. The turn at a vertex is measured by the sign of the cross product $\mathrm{Im}(\bar{u}v) = u_x v_y - u_y v_x$ of the incoming edge u and the outgoing edge v. The edges are

$$P_1P_2 = (1,0), \quad P_2P_3 = (a-1, \tfrac{1}{2}), \quad P_3P_4 = (1-a, \tfrac{1}{2}),$$
$$P_4P_5 = (-1,0), \quad P_5P_6 = (1-a, -\tfrac{1}{2}), \quad P_6P_1 = (a-1, -\tfrac{1}{2}),$$

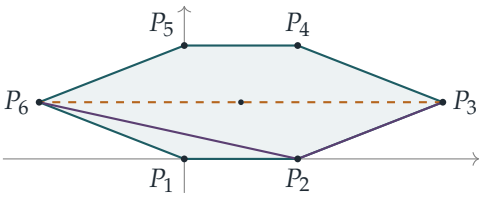
and the cross products at $P_2, P_3, P_4, P_5, P_6, P_1$ in turn are

$$\tfrac{1}{2}, \qquad a - 1, \qquad \tfrac{1}{2}, \qquad \tfrac{1}{2}, \qquad a - 1, \qquad \tfrac{1}{2}.$$

Four of these are positive whatever a is, so the hexagon is convex if and only if $a - 1 > 0$. (If $a = 1$, then P_3 lies on the segment P_2P_4 and we do not have a genuine hexagon.) Geometrically, P_3 must lie to the right of the side $x = 1$ of the square and, by the symmetry, P_6 to the left of $x = 0$. So the conditions are

$$\boxed{\mathrm{Re}(z) > 1 \quad \text{and} \quad \mathrm{Re}(w) = 1 - \mathrm{Re}(z) \ \ (\text{so } \mathrm{Re}(w) < 0)}$$

(together with $\mathrm{Im}(z) = \mathrm{Im}(w) = \frac{1}{2}$, which (ii) already forces).



The distance P_3P_6

At P_2 the two arms are

$$z_3 - z_2 = (a-1) + \tfrac{1}{2}\mathrm{i}, \qquad z_6 - z_2 = -a + \tfrac{1}{2}\mathrm{i}.$$

For the angle $\theta = \angle P_3P_2P_6$ (between 0 and π), the dot product of the arms gives $|P_2P_3|\,|P_2P_6|\cos\theta$, and the size of the cross product gives $|P_2P_3|\,|P_2P_6|\sin\theta$:

$$\text{dot} = -a(a-1) + \tfrac{1}{4} = \tfrac{1}{4}(1 + 4a - 4a^2), \qquad \text{cross} = (a-1)\cdot\tfrac{1}{2} - \tfrac{1}{2}\cdot(-a) = \tfrac{1}{2}(2a-1).$$

With $a > 1$ the cross product is positive, so it is exactly the $\sin\theta$ term, and

$$\tan\theta = \frac{\frac{1}{2}(2a-1)}{\frac{1}{4}(1+4a-4a^2)} = \frac{2(2a-1)}{1+4a-4a^2}.$$

Setting this equal to $-\frac{2}{3}$:

$$3(2a-1) = -(1+4a-4a^2) \quad\Longrightarrow\quad 4a^2 - 10a + 2 = 0 \quad\Longrightarrow\quad 2a^2 - 5a + 1 = 0,$$

so $a = \frac{1}{4}(5 \pm \sqrt{17})$. The minus sign gives $a \approx 0.22$, which breaks the convexity condition $a > 1$; the plus sign gives $a \approx 2.28 > 1$. (There the denominator $1 + 4a - 4a^2$ is negative, so θ is obtuse, as a negative tangent requires.)

Finally P_3 and P_6 have the same imaginary part, so

$$P_3P_6 = |z - w| = |2a - 1| = 2a - 1 = \frac{5+\sqrt{17}}{2} - 1 = \boxed{\frac{3+\sqrt{17}}{2}} \approx 3.56.$$

If we drop the convexity requirement, the same method (now with $|2a-1|$ in the numerator) gives one further solution, $a = \frac{1}{4}(-1 - \sqrt{17})$. This is just the mirror image $a \mapsto 1 - a$, which swaps z and w, and it gives the same distance $\frac{1}{2}(3 + \sqrt{17})$.

Checks

- Symmetry: $z_1 + z_4 = z_2 + z_5 = z_3 + z_6$, so the result $\overrightarrow{P_3P_4} = \overrightarrow{P_1P_6}$ is just the third pair of opposite sides of a centrally symmetric hexagon.
- With $a = \frac{1}{4}(5 + \sqrt{17}) \approx 2.2808$, a direct computation of $\arg\big((z_6 - z_2)/(z_3 - z_2)\big)$ gives an angle whose tangent is -0.66667, and $|z - w| \approx 3.5616 = \frac{1}{2}(3 + \sqrt{17})$.
- Testing convexity numerically for a from -5 to 5 confirms that the hexagon is convex exactly when $a > 1$.

Common slips

- Getting the direction of a vector wrong: $\overrightarrow{P_5P_4}$ is $z_4 - z_5$, not $z_5 - z_4$.

- Writing perpendicularity as "the product of gradients is -1" and then losing the vertical case. Here $\overrightarrow{P_2P_4}$ is vertical, and the quotient test handles this cleanly.
- Keeping the root $a = \frac{1}{4}(5 - \sqrt{17})$, which does not give a convex hexagon (and in fact gives $\tan\theta > 0$ there).

Beyond STEP

Treating points as complex numbers turns geometry into algebra: translations are additions, and rotations and scalings are multiplications. The test "u/v is purely imaginary" is one example of how angles become arguments. The cross-product sign test for convexity is the basis of the standard convex-hull algorithms in computational geometry. Needham's *Visual Complex Analysis* develops this geometric view of complex numbers beautifully.

WORKED SOLUTION

STEP I 1988, Question 7

Pure • Bounding Quadratic Derivatives

STEP I 1988, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/88-s1-q7

This question has a full guided treatment in *Beyond STEP*, Section 3.6, *A Bound on Slopes*: https://beyondstep.keithmcnulty.org/ch03/markov-bound.html

The key idea

A quadratic is pinned down by its values at three points, so f' can be rebuilt from $\mathrm{f}(1)$, $\mathrm{f}(-1)$ and $\mathrm{f}(0)$. Once f' is written that way, the hypothesis $|\mathrm{f}| \le 1$ controls each of the three coefficients, and the triangle inequality does the rest. For the final part, we look at which values of f make every inequality in the proof an equality.

Solution

The identity for $\mathrm{f}'(x)$

We have $\mathrm{f}(1) = a+b+c$, $\mathrm{f}(-1) = a-b+c$ and $\mathrm{f}(0) = c$. Collecting the right-hand side by powers of x:

$$\begin{aligned}&\mathrm{f}(1)\left(x+\tfrac{1}{2}\right) + \mathrm{f}(-1)\left(x-\tfrac{1}{2}\right) - 2\mathrm{f}(0)x \\ &\quad = \big[(a+b+c)+(a-b+c)-2c\big]x + \tfrac{1}{2}\big[(a+b+c)-(a-b+c)\big] \\ &\quad = 2ax + b.\end{aligned}$$

Since $\mathrm{f}'(x) = 2ax+b$, this is the required identity.

The bound $|\mathrm{f}'(x)| \le 4$

Suppose $|\mathrm{f}(x)| \le 1$ whenever $|x| \le 1$. In particular $|\mathrm{f}(1)|$, $|\mathrm{f}(-1)|$ and $|\mathrm{f}(0)|$ are all at most 1. Take any x with $|x| \le 1$. By the identity and the triangle inequality,

$$\begin{aligned}|\mathrm{f}'(x)| &\le |\mathrm{f}(1)|\,\big|x+\tfrac{1}{2}\big| + |\mathrm{f}(-1)|\,\big|x-\tfrac{1}{2}\big| + 2|\mathrm{f}(0)|\,|x| \\ &\le \big|x+\tfrac{1}{2}\big| + \big|x-\tfrac{1}{2}\big| + 2|x|.\end{aligned}$$

Ask yourself How big can $\left|x+\frac{1}{2}\right|+\left|x-\frac{1}{2}\right|$ be when $|x|\le 1$?

This is the sum of the distances from x to $\frac{1}{2}$ and to $-\frac{1}{2}$. If $|x|\le\frac{1}{2}$, then x lies between the two points and the sum is exactly 1. If $|x|>\frac{1}{2}$, both points lie on the same side of x, and the sum is $2|x|$. In either case it is at most 2 for $|x|\le 1$. Also $2|x|\le 2$, so

$$|f'(x)|\le 2+2=4 \qquad \text{for } |x|\le 1,$$

as required.

(Another way to see this: f' is linear, so on $[-1,1]$ its modulus is largest at an endpoint, and the bound at $x=\pm1$ reads $|f'(\pm1)|\le\frac{3}{2}+\frac{1}{2}+2=4$.)

A case of equality

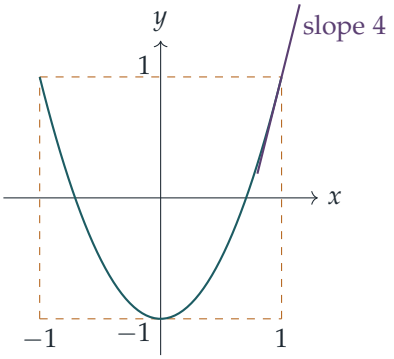
For equality at $x=1$ we need every step above to be tight there. At $x=1$ the coefficients $x+\frac{1}{2}$, $x-\frac{1}{2}$ and $-2x$ are $\frac{3}{2}$, $\frac{1}{2}$ and -2. To get $f'(1)=+4$ we therefore want

$$f(1)=1, \qquad f(-1)=1, \qquad f(0)=-1.$$

These give $c=-1$, $a+b+c=1$ and $a-b+c=1$, so $b=0$ and $a=2$:

$$a=2, \quad b=0, \quad c=-1, \qquad f(x)=2x^2-1.$$

Now check the condition. For $|x|\le 1$ we have $0\le x^2\le 1$, so $-1\le 2x^2-1\le 1$, that is, $|f(x)|\le 1$. And $f'(x)=4x$, so $f'(1)=4$ with $x=1$ in the interval. This is the example we need.



The curve $y=2x^2-1$ (teal) stays inside the square $|x|\le 1$, $|y|\le 1$, swinging from top to bottom and back, and so it is as steep as it can be at the ends.

(Any of $\pm(2x^2-1)$ gives $|f'(\pm1)|=4$. The choice $f(x)=2x^2-1$ gives $f'(1)=+4$, and so does $-(2x^2-1)$ at $x=-1$.)

Checks

- Try $f(x)=x^2+x+1$: $f(1)=3$, $f(-1)=1$, $f(0)=1$. At $x=2$ the identity gives $3\cdot\frac{5}{2}+1\cdot\frac{3}{2}-2\cdot 2=5=f'(2)$.

- A random search over 20,000 quadratics scaled so that $\max_{|x|\le 1} |f(x)| = 1$ never found $\max_{|x|\le 1} |f'(x)| = 2|a| + |b|$ above 4 (the largest was about 3.91), and $2x^2 - 1$ reaches 4 exactly.

Common slips

- Bounding $\left|x + \frac{1}{2}\right|$ and $\left|x - \frac{1}{2}\right|$ separately by $\frac{3}{2}$ each. That only gives $|f'(x)| \le 5$. The two distances must be bounded together.
- Proposing an example without checking $|f(x)| \le 1$ on the whole interval, not just at 0 and ± 1.

Beyond STEP

This is the degree-two case of Markov's inequality: if a polynomial p of degree n has $|p| \le 1$ on $[-1, 1]$, then $|p'| \le n^2$ there. Equality comes from the Chebyshev polynomial $T_n(\cos\theta) = \cos n\theta$, and $2x^2 - 1 = T_2(x)$. Writing a polynomial through its values at chosen points is Lagrange interpolation, a basic tool in numerical analysis and approximation theory. For more problems in this spirit, see Siklos, *Advanced Problems in Mathematics*.

WORKED SOLUTION

STEP I 1988, Question 8

Pure • Geometry in Three Dimensions

STEP I 1988, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

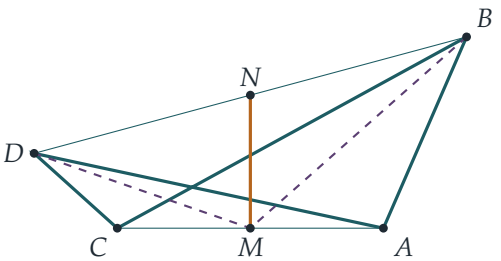
step.maths.org/questions/88-s1-q8

The key idea

Each diagonal splits the quadrilateral into two triangles, and the equal opposite sides make those two triangles congruent (SSS). Congruent triangles have equal medians, so the midpoint of one diagonal is equidistant from the two ends of the other. A point equidistant from both ends of a segment, joined to the segment's midpoint, gives a line perpendicular to that segment: this is just the isosceles triangle.

Solution

Let M be the midpoint of AC and N the midpoint of BD.



The figure is a projection of a skew quadrilateral; the dashed lines MB and MD are the medians used below, and the amber segment is MN.

First, $M \neq N$

The line MN only makes sense if M and N are different points. If $M = N$, the diagonals AC and BD would bisect each other at a common point. Two lines through a common point lie in a plane, so A, B, C, D would all lie in that plane, and $ABCD$ would be a (planar) parallelogram. Since $ABCD$ is skew, $M \neq N$. This is exactly where the word "skew" is used: for a parallelogram the opposite sides are equal but the two midpoints

coincide.

MN is perpendicular to BD

Look at the triangles ABC and CDA, which share the diagonal AC. We have

$$AB = CD, \qquad BC = DA, \qquad CA = AC,$$

so the triangles are congruent (SSS), with $A \leftrightarrow C$, $B \leftrightarrow D$, $C \leftrightarrow A$. Under this correspondence the median from B to the midpoint of AC in triangle ABC matches the median from D to the midpoint of CA in triangle CDA. Both medians end at the same point M, so

$$MB = MD.$$

Ask yourself Why must congruent triangles have equal medians?

A median is determined entirely by the side lengths. For instance, by the cosine rule (or Apollonius's theorem), the median m from B in triangle ABC satisfies $4m^2 = 2AB^2 + 2BC^2 - AC^2$. The same side lengths give the same median, so $MB^2 = \frac{1}{4}(2AB^2 + 2BC^2 - AC^2) = \frac{1}{4}(2CD^2 + 2DA^2 - CA^2) = MD^2$.

Now M is equidistant from B and D, and N is the midpoint of BD. The triangles MNB and MND have $MB = MD$, $NB = ND$ and the common side MN, so they are congruent (SSS), and the angles $\angle MNB$ and $\angle MND$ are equal. They are also supplementary, because B, N, D lie on a line. So each is a right angle: $MN \perp BD$. (The triangles are genuine, not degenerate: if M lay on the line BD with $MB = MD$, then M would be the midpoint N, which we have ruled out.)

MN is perpendicular to AC

The same argument works with the roles of the diagonals swapped. Triangles ABD and CDB share the diagonal BD, and

$$AB = CD, \qquad AD = CB, \qquad BD = DB,$$

so they are congruent, with $A \leftrightarrow C$, $B \leftrightarrow D$, $D \leftrightarrow B$. The medians from A and from C both end at the midpoint N of BD, so $NA = NC$. Since M is the midpoint of AC and $N \neq M$, the isosceles-triangle argument above gives $MN \perp AC$.

So the line joining the midpoints of the diagonals is perpendicular to each diagonal, as required.

Or with vectors

Take the origin at M, so that $C = -A$ as position vectors, and $\overrightarrow{MN} = \frac{1}{2}(\mathbf{b} + \mathbf{d})$. Squaring the two given equalities,

$$AB^2 = CD^2 \iff |\mathbf{b}|^2 - 2\mathbf{a}\cdot\mathbf{b} = |\mathbf{d}|^2 + 2\mathbf{a}\cdot\mathbf{d},$$
$$BC^2 = AD^2 \iff |\mathbf{b}|^2 + 2\mathbf{a}\cdot\mathbf{b} = |\mathbf{d}|^2 - 2\mathbf{a}\cdot\mathbf{d}.$$

Subtracting gives $4\mathbf{a}\cdot\mathbf{b} = -4\mathbf{a}\cdot\mathbf{d}$, that is $\mathbf{a}\cdot(\mathbf{b}+\mathbf{d}) = 0$. Since $\overrightarrow{AC} = -2\mathbf{a}$, this says $\overrightarrow{MN} \perp AC$. Putting the origin at N instead gives $MN \perp BD$ in the same way.

Checks

Try $A = (1,0,0)$, $B = (1,2,1)$, $C = (-1,0,0)$, $D = (-1,-2,1)$. Then $AB = CD = \sqrt{5}$ and $BC = AD = 3$, and the four points are not coplanar (the scalar triple product $\overrightarrow{AB}\cdot(\overrightarrow{AC}\times\overrightarrow{AD})$ is 8). Here $M = (0,0,0)$ and $N = (0,0,1)$, so MN is along the z-axis, while AC is along the x-axis and $\overrightarrow{BD} = (-2,-4,0)$ has no z-component: both are perpendicular to MN. We also tested 20 000 random skew quadrilaterals with equal opposite sides; the cosine of the angle between MN and either diagonal never exceeded 4×10^{-12}.

Common slips

- Not explaining why $M \neq N$. Without the skew condition the result fails, since for a parallelogram there is no line MN at all.
- Claiming $MB = MD$ "by symmetry" without naming the congruent triangles.
- Proving perpendicularity to one diagonal only. The second needs a different pair of congruent triangles.

Beyond STEP

In the example, the half-turn about the z-axis (the line MN) swaps A with C and B with D. This is true in general: every skew quadrilateral with equal opposite sides is carried onto itself by a rotation through 180° about MN. (The rotation fixes M and reverses every direction perpendicular to MN, so it sends A to C; likewise it sends B to D.) Conversely, the half-turn makes the equal sides obvious. Seeing geometric facts as consequences of a symmetry, an isometry of space, is a central idea in first-year geometry and group theory; Allenby's *Rings, Fields and Groups* develops the group side, and Siklos's *Advanced Problems in Mathematics* has more STEP-style geometry.

WORKED SOLUTION

STEP I 1988, Question 9

Pure • Integration by Substitution

STEP I 1988, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/88-s1-q9

The key idea

Both integrals yield to a standard technique once you choose the right piece to work on. In (i) the product of $\ln x$ with a power of x calls for integration by parts, differentiating the logarithm so that it disappears. In (ii) the awkward part is the square root, so we make it the new variable. The $\cos x\,\mathrm{d}x$ is exactly what the substitution needs, and what is left is a rational function that partial fractions can handle.

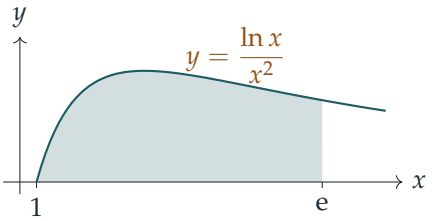
Solution

Part (i)

Integrate by parts with $u = \ln x$ and $\dfrac{\mathrm{d}v}{\mathrm{d}x} = \dfrac{1}{x^2}$, so that $\dfrac{\mathrm{d}u}{\mathrm{d}x} = \dfrac{1}{x}$ and $v = -\dfrac{1}{x}$:

$$\begin{aligned}\int_1^{\mathrm{e}} \frac{\ln x}{x^2}\,\mathrm{d}x &= \left[-\frac{\ln x}{x}\right]_1^{\mathrm{e}} + \int_1^{\mathrm{e}} \frac{1}{x^2}\,\mathrm{d}x \\ &= \left(-\frac{1}{\mathrm{e}} - 0\right) + \left[-\frac{1}{x}\right]_1^{\mathrm{e}} \\ &= -\frac{1}{\mathrm{e}} + 1 - \frac{1}{\mathrm{e}} = 1 - \frac{2}{\mathrm{e}}.\end{aligned}$$

We used $\ln \mathrm{e} = 1$ and $\ln 1 = 0$. The answer is positive, as it must be: the integrand is positive on $(1, \mathrm{e}]$.



The shaded area is $1 - 2/\mathrm{e} \approx 0.264$.

Part (ii)

The integrand is defined when $\sin x \neq 0$ and $1 + \sin x > 0$. Put

$$u = \sqrt{1+\sin x}, \qquad \text{so } u^2 = 1 + \sin x, \quad 2u\,\mathrm{d}u = \cos x\,\mathrm{d}x, \quad \sin x = u^2 - 1.$$

Here $u > 0$, and $u \neq 1$ because $\sin x \neq 0$. Then

$$\int \frac{\cos x}{\sin x \sqrt{1+\sin x}}\,\mathrm{d}x = \int \frac{2u}{(u^2-1)\,u}\,\mathrm{d}u = \int \frac{2}{u^2-1}\,\mathrm{d}u.$$

Ask yourself What happened to the $\cos x$? Do we have to write it in terms of u?

No. The whole of $\cos x\,\mathrm{d}x$ is replaced by $2u\,\mathrm{d}u$, which is the reason this substitution works so cleanly. Had the numerator been something else, we would have needed $\cos x = \pm\sqrt{1-\sin^2 x}$, with all its sign trouble.

Partial fractions give $\dfrac{2}{u^2-1} = \dfrac{1}{u-1} - \dfrac{1}{u+1}$, so

$$\int \frac{2}{u^2-1}\,\mathrm{d}u = \ln|u-1| - \ln|u+1| + C = \ln\left|\frac{u-1}{u+1}\right| + C.$$

Returning to x,

$$\int \frac{\cos x}{\sin x\sqrt{1+\sin x}}\,\mathrm{d}x = \ln\left|\frac{\sqrt{1+\sin x}-1}{\sqrt{1+\sin x}+1}\right| + C.$$

The modulus signs matter: when $-1 < \sin x < 0$ we have $0 < u < 1$, and $\dfrac{u-1}{u+1}$ is negative. (Strictly, the constant may differ on each interval where the integrand is defined, since the domain is split up by the zeros of $\sin x$.)

An equivalent form comes from multiplying the top and bottom of the fraction by $u-1$. Since $(u+1)(u-1) = u^2 - 1 = \sin x$, the answer is also

$$2\ln\left|\sqrt{1+\sin x} - 1\right| - \ln|\sin x| + C.$$

Checks

- Simpson's rule gives $\int_1^{\mathrm{e}} x^{-2}\ln x\,\mathrm{d}x \approx 0.264241$, matching $1 - 2/\mathrm{e} = 0.264241\ldots$.
- For (ii), differentiate the answer: by the chain rule $\dfrac{\mathrm{d}}{\mathrm{d}x}\ln\left|\dfrac{u-1}{u+1}\right| = \dfrac{2}{u^2-1}\cdot\dfrac{\cos x}{2u}$, which is the integrand. Numerically, $\int_{0.3}^{1.2}$ of the integrand is 0.926006 by Simpson's rule, which agrees with the difference of the antiderivative. The agreement is just as good on intervals where $\sin x < 0$, such as $[-1.2, -0.2]$, where the integral is -2.361027.

Common slips

- In (i), choosing $u = 1/x^2$ and $\mathrm{d}v/\mathrm{d}x = \ln x$ instead. This works in principle but is far messier; differentiate the logarithm.
- In (ii), dropping the modulus signs, so the answer is undefined whenever $\sin x < 0$.
- Losing the factor 2 from $2u\,\mathrm{d}u$, which halves the answer.

Beyond STEP

Part (ii) is an instance of a general principle: an integrand that is a rational function of x and $\sqrt{ax+b}$ becomes a rational function of u after the substitution $u = \sqrt{ax+b}$, and rational functions can always be integrated by partial fractions. Here $\sin x$ plays the role of x. In a first-year analysis course you meet the other side of the story: the theorems that justify substitution and integration by parts, and why the constant of integration may differ between the separate intervals of the domain. Bartle and Sherbert, *Introduction to Real Analysis*, covers both carefully.

WORKED SOLUTION

STEP I 1988, Question 10

Mechanics • Projectile Motion

STEP I 1988, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/88-s1-q10

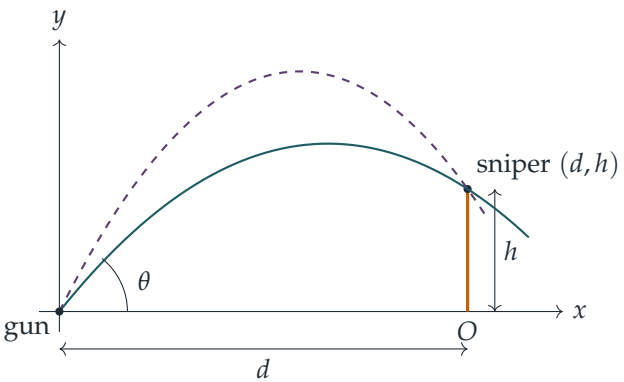
The key idea

Turn the question round. Put the gun at horizontal distance d from the base of the tree and ask: is there *any* angle of elevation for which a bullet with speed v passes through the top of the tree? The trajectory equation, written in terms of $\tan\theta$, is a quadratic in $\tan\theta$, so the answer is "yes" exactly when that quadratic has a real root. The discriminant condition then turns out to be an inequality on d alone.

Solution

The model

We treat the bullet as a particle moving under gravity alone (no air resistance), fired from ground level with speed v at an angle θ above the horizontal, where $0 < \theta \leq \frac{\pi}{2}$. Since the gun's position is unknown, let it be at distance $d \geq 0$ from the base O of the tree, and take axes with origin at the gun, x horizontal towards the tree and y vertically up. The top of the tree is then the point (d, h).



Hitting the sniper

After time t the bullet is at $x = vt\cos\theta$, $y = vt\sin\theta - \frac{1}{2}gt^2$. If $\theta < \frac{\pi}{2}$ we eliminate $t = x/(v\cos\theta)$ and use $\sec^2\theta = 1 + \tan^2\theta$ to get the trajectory

$$y = x\tan\theta - \frac{gx^2}{2v^2}(1 + \tan^2\theta).$$

The bullet hits the sniper if and only if (d, h) lies on this curve. Suppose first that $d > 0$. Writing $T = \tan\theta$ and $k = \dfrac{gd^2}{2v^2} > 0$, the condition is

$$kT^2 - dT + (h + k) = 0.$$

Every real number T is $\tan\theta$ for exactly one $\theta \in (-\frac{\pi}{2}, \frac{\pi}{2})$. Can a root be negative, though? The product of the roots is $(h + k)/k > 0$ and their sum is $d/k > 0$, so any real roots are both positive and give genuine elevations $0 < \theta < \frac{\pi}{2}$. Hence some elevation works exactly when this quadratic has a real root, that is, when its discriminant is non-negative:

$$\begin{aligned} d^2 - 4k(h+k) \geq 0 &\iff d^2 \geq \frac{2gd^2}{v^2}\left(h + \frac{gd^2}{2v^2}\right) \\ &\iff v^4 \geq 2ghv^2 + g^2d^2 \qquad \text{(dividing by } d^2 > 0\text{, multiplying by } v^4\text{)} \\ &\iff d^2 \leq \frac{v^2}{g^2}(v^2 - 2gh). \end{aligned}$$

Ask yourself What if the gun was right at the base of the tree, so that $d = 0$ and the trajectory equation above is no use?

Then the bullet must have been fired vertically ($\theta = \frac{\pi}{2}$), and it reaches height h if and only if $v^2 \geq 2gh$, by $v^2 = u^2 - 2gs$ with final speed at least zero. This is the same condition with $d = 0$, so the inequality covers every case.

Since $d \geq 0$, taking the non-negative square root gives

$$d \leq \frac{v}{g}\sqrt{v^2 - 2gh}.$$

So the gun was at a point on the horizontal ground whose distance from the base of the tree is at most $(v/g)\sqrt{v^2 - 2gh}$: it lay within (or, in the extreme case, on) the circle centred on the base of the tree with that radius, as required. The square root is real because we know the shot happened: with $d = 0$ the inequality forces $v^2 \geq 2gh$, which simply says the bullet has enough energy to climb to height h.

The condition is also sufficient. From any point inside the circle the quadratic has two real positive roots, so there are two possible elevations (a flatter and a steeper shot, as in the figure), and on the circle itself they merge into one.

Checks

- With $h = 0$ the radius is v^2/g, the familiar maximum range on level ground. As $v^2 \to 2gh$ the radius shrinks to 0: the gun can only just reach the sniper by firing straight up from the base.
- Units: $(v/g)\sqrt{v^2}$ has units $(\mathrm{m\,s^{-1}})^2/(\mathrm{m\,s^{-2}}) = \mathrm{m}$.
- Numerically, with $g = 9.8$, $v = 20$ and $h = 8$ the radius is about 31.8 m. Stepping the motion forward from $d = 31.8\,\mathrm{m}$ at the single permitted elevation, $\tan\theta = v^2/(gd)$, brings the bullet to height 8.000 m at the tree. A scan over all elevations confirms that points a few per cent outside the circle can't be reached and points inside can.

Common slips

- Using the maximum range v^2/g or the maximum height alone. The sniper is at height h, so both coordinates matter at once.
- Forgetting to check that the roots for $\tan\theta$ give elevations above the horizontal, or ignoring the vertical shot $d = 0$.

Beyond STEP

For fixed v, the points reachable from the gun fill the region below the parabola $y = \dfrac{v^2}{2g} - \dfrac{gx^2}{2v^2}$, the *envelope* of the family of trajectories (the "parabola of safety"). Our discriminant condition is exactly $h \leq$ this envelope at $x = d$. Envelopes of families of curves appear again in the calculus of variations and the theory of first-order differential equations. Kleppner and Kolenkow, *An Introduction to Mechanics*, is a good next step for projectile problems treated with this kind of care.

WORKED SOLUTION

STEP I 1988, Question 11

Mechanics • Centre of Mass

STEP I 1988, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/88-s1-q11

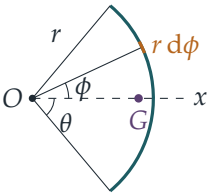
The key idea

For the arc, symmetry puts the centre of mass on the bisecting radius, and one integral over the angle gives its distance from the centre. For the framework, the striking fact is that it balances on *every* point of the semicircle. The floor's reaction always acts along a radius of the semicircle, and the only point lying on all those radii is the centre. So the centre of mass of the whole framework must be the centre of the semicircle, and taking moments about that point gives an equation for h.

Solution

The uniform arc

Let the arc have centre O and density ρ per unit length. Measure the angle ϕ from the bisecting radius, so the arc is $-\theta \le \phi \le \theta$. Put the x-axis along the bisector. By symmetry in that axis the centre of mass lies on it, so we only need $\bar{x}$.



The element between ϕ and $\phi + \mathrm{d}\phi$ has length $r\,\mathrm{d}\phi$, so mass $\rho r\,\mathrm{d}\phi$, and its x-coordinate is $r\cos\phi$. The total mass is $\rho r \cdot 2\theta$. Hence

$$\bar{x} = \frac{\displaystyle\int_{-\theta}^{\theta} r\cos\phi\;\rho r\,\mathrm{d}\phi}{2\rho r\theta} = \frac{r\left[\sin\phi\right]_{-\theta}^{\theta}}{2\theta} = \frac{r\sin\theta}{\theta}.$$

So the centre of mass lies on the bisecting radius, at distance

$$\boxed{\bar{x} = \frac{r\sin\theta}{\theta}}$$

from the centre. For a semicircular arc, $\theta = \frac{\pi}{2}$, and this is $\dfrac{2r}{\pi}$, which is the case we need next.

The framework: where must the centre of mass be?

We model the framework as rigid and made of uniform thin rods of the same density ρ per unit length, and the floor as horizontal. Let O be the centre of the semicircle, which is the midpoint of the bottom side of the rectangle.

Ask yourself Why does "equilibrium at any point of the semicircle" pin down the centre of mass completely?

Suppose the framework rests with the point P of the semicircle on the floor. The floor is tangent to the semicircle at P, so the vertical through P is perpendicular to that tangent: it is the radius line OP. Only two forces act, the weight at the centre of mass G and the reaction at P. For equilibrium they must act along the same line, so G lies on the vertical through P, which is the line OP. This holds for every P on the semicircle. Two different points P give two different lines through O, and these meet only at O. Hence $G = O$.

Conversely, if $G = O$ then for every P the weight acts along OP, straight through the point of contact, so the framework is in equilibrium wherever it touches the floor. The condition is therefore exactly "G is at O".

Moments about the bottom side

Take moments about the line of the bottom side, measuring heights upwards from O (so the semicircle is below the line). The pieces are:

piece	mass	height of its centre	moment
top rod	$2r\rho$	$2h$	$4rh\rho$
two side rods	$2 \times 2h\rho$	h	$4h^2\rho$
bottom rod	$2r\rho$	0	0
semicircle	$\pi r\rho$	$-\dfrac{2r}{\pi}$	$-2r^2\rho$

The semicircle's centre of mass comes from the arc formula with $\theta = \frac{\pi}{2}$, and it is *below* O. Setting the total moment to zero, because G is at height 0, and dividing by 2ρ:

$$2h^2 + 2rh - r^2 = 0.$$

This is a quadratic in h, with roots

$$h = \frac{-2r \pm \sqrt{4r^2 + 8r^2}}{4} = \frac{(-1 \pm \sqrt{3})\, r}{2}.$$

Since $h > 0$ we take the positive root:

$$\boxed{h = \frac{(\sqrt{3}-1)\,r}{2} \approx 0.366\,r}$$

Notice that the bottom rod has zero moment about its own line. The answer is the same whether or not the framework includes it, so a slightly different reading of the diagram doesn't matter.

Checks

- As $\theta \to 0$, $\frac{r\sin\theta}{\theta} \to r$: a very short arc has its centre of mass on the arc itself. For a full circle, $\theta = \pi$, we get 0, the centre, as symmetry demands.
- With $h = \frac{(\sqrt{3}-1)r}{2}$: $2h^2 = (2-\sqrt{3})r^2$ and $2rh = (\sqrt{3}-1)r^2$, which add to r^2 as required.
- We computed the centre of mass numerically by splitting every rod into 20 000 pieces. With $r = 1$ and $h \approx 0.36603$, it lies within 10^{-9} of O. Increasing h by 10% moves it up by about 0.03, so the framework would no longer balance.

Common slips

- Using $\frac{4r}{3\pi}$, which is for a semicircular *lamina*, instead of $\frac{2r}{\pi}$ for a semicircular *wire*.
- Weighting by the angle rather than by length: the mass of the element is $\rho r\,\mathrm{d}\phi$, and the factor r cancels only because it also appears in the total mass.
- Asserting that G is at O without a reason. The argument is that the reaction always acts along a radius.

Beyond STEP

The arc integral is the simplest example of a centre of mass as a *line integral* $\int \mathbf{r}\,\rho\,\mathrm{d}s$ along a curve. First-year courses go on to surfaces and solids using double and triple integrals, and Pappus's theorems link these centres of mass to areas and volumes of revolution. The balancing argument is also the idea behind rocking stability, where a body resting on a curved surface is stable if its centre of mass lies below the centre of curvature. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats centres of mass and rigid-body equilibrium in this spirit.

WORKED SOLUTION

STEP I 1988, Question 12

Mechanics • Variable Force Motion

STEP I 1988, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/88-s1-q12

The key idea

Work out, for each method, the furthest the skater travels towards the bank; she is safe exactly when d exceeds that distance for at least one method. Turning is classic circular motion: a constant force perpendicular to the velocity gives a circle of fixed radius, and the radius is the distance she still advances. Braking is a variable-force problem in which the force depends on speed but we want a *distance*, which is the cue to write the acceleration as $v\frac{\mathrm{d}v}{\mathrm{d}x}$.

Solution

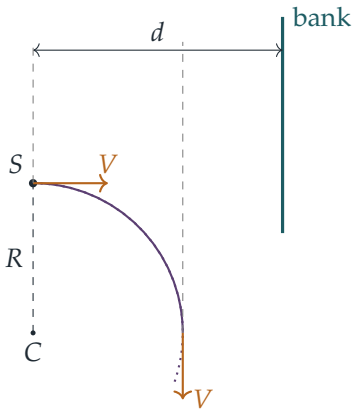
The model

We treat the skater as a particle of mass M on smooth horizontal ice. The only horizontal force on her is the one she applies, so weight and the normal reaction cancel and play no further part. We measure distance x from her starting point towards the bank, so the bank is at $x = d$.

Method 1: turning in a circle

A force perpendicular to the velocity does no work, so her speed stays V. A force of constant magnitude F always at right angles to a velocity of constant speed V provides the centripetal force for a circle of radius R, where

$$F = \frac{MV^2}{R}, \qquad \text{so} \qquad R = \frac{MV^2}{F}.$$



She starts at S moving straight at the bank, so the centre C of the circle lies on the line through S *parallel* to the bank, at distance R from S. Her distance towards the bank is then $x = R\sin\theta$ after turning through angle θ, which is greatest, equal to R, when $\theta = \frac{\pi}{2}$; there she is moving parallel to the bank, and afterwards she curves away. So she misses the bank if

$$d > \frac{MV^2}{F}.$$

(If $d = MV^2/F$ exactly she just grazes the bank, moving along it.)

Method 2: braking

Now the force acts directly backwards along her line of motion, so she moves in a straight line towards the bank and slows down. Newton's second law, with the acceleration written as $v\frac{\mathrm{d}v}{\mathrm{d}x}$ because we want distance rather than time, gives

$$Mv\frac{\mathrm{d}v}{\mathrm{d}x} = -\frac{F(V^2+v^2)}{2V^2}.$$

Separating the variables and integrating from $x = 0$, where $v = V$, to the point $x = X$ where she stops:

$$\int_V^0 \frac{v}{V^2+v^2}\,\mathrm{d}v = -\frac{F}{2MV^2}\int_0^X \mathrm{d}x,$$
$$\left[\tfrac{1}{2}\ln(V^2+v^2)\right]_V^0 = -\frac{FX}{2MV^2},$$
$$\tfrac{1}{2}\ln\frac{V^2}{2V^2} = -\tfrac{1}{2}\ln 2 = -\frac{FX}{2MV^2},$$

so her stopping distance is

$$X = \frac{MV^2}{F}\ln 2.$$

Ask yourself Does she actually come to rest, or does she only approach zero speed as she creeps forward for ever?

The braking force never drops below $\frac{1}{2}F$, so her speed falls to zero in a finite time. Explicitly, $M\frac{\mathrm{d}v}{\mathrm{d}t} = -\frac{F}{2V^2}(V^2+v^2)$ integrates to $\frac{2MV}{F}\left(\arctan 1 - \arctan\frac{v}{V}\right) = t$, so she stops at $t = \pi MV/(2F)$, having gone the finite distance X. (Once at rest she stops pushing.) So braking saves her when $d > X$; if $d = X$ she reaches the bank with zero speed, touching it without any impact.

Comparing the methods

Since $\ln 2 \approx 0.693 < 1$, the stopping distance is shorter than the turning radius: braking is the better method, and whenever turning works, braking works too. Hence she can avoid hitting the bank precisely when

$$\boxed{d > \frac{MV^2}{F}\ln 2}$$

(with $d = \frac{MV^2}{F}\ln 2$ the borderline case in which she arrives at the bank at rest).

The particle assumption

A real skater has size: her body and arms reach some way in front of her centre of mass, so in practice she needs d larger than the value above by roughly that reach. More seriously, the forces come from her skates on the ice. To turn she must lean into the curve, and to brake hard she must twist her blades across her motion (a hockey stop), which turns her body sideways. Rotation and posture like this are exactly what a particle model ignores. The model also quietly assumes the ice can supply these forces at all, although it is called smooth; the blades grip sideways even when there is no friction along them.

Checks

- Units: MV^2/F is $\mathrm{kg\,m^2\,s^{-2}/N} = \mathrm{m}$, a length, as it must be.
- Bounds: the braking force lies between $\frac{1}{2}F$ and F. A constant force F would stop her in $MV^2/(2F)$ and a constant $\frac{1}{2}F$ in MV^2/F, and indeed $\frac{1}{2} < \ln 2 < 1$.
- Numerically: stepping both motions with a Runge–Kutta integrator for random M, V, F (for example $R = MV^2/F = 7.0747$) gave furthest advance 7.0747 when turning and stopping distance $4.9038 = R\ln 2$ when braking, with stopping time $\pi MV/(2F)$.

Common slips

- Placing the centre of the circle on the line of motion (or at the bank), and getting a reach of $2R$ or 0, rather than putting the centre beside her so that the reach is R.
- Using $\frac{\mathrm{d}v}{\mathrm{d}t}$ for the braking and then struggling to get a distance; $v\,\frac{\mathrm{d}v}{\mathrm{d}x}$ gives it in one integration.
- Giving two separate conditions without comparing them. The question asks for the set of d for which she *can* escape, so we need the better of the two methods.

Beyond STEP

Writing $\ddot{x} = v\,\frac{\mathrm{d}v}{\mathrm{d}x}$ is the first step towards energy methods: for a force depending only on position it is the work–energy theorem, and in university mechanics it develops into potentials and phase-plane pictures of motion. The skater's choice between steering and braking is also a simple optimal control problem: what is the best way to use a limited force to avoid an obstacle? Kleppner and Kolenkow, *An Introduction to Mechanics*, treats velocity-dependent forces and circular motion at exactly the next level up.

WORKED SOLUTION

STEP I 1988, Question 13

Mechanics • Elastic Strings and Collisions

STEP I 1988, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/88-s1-q13

The key idea

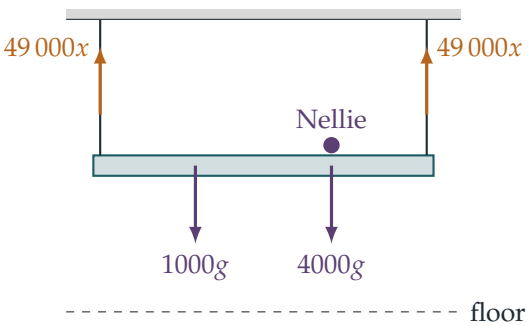
There are three stages, and each one has its own conservation law. Nellie's landing is a sudden inelastic collision, so momentum is conserved but energy is not. The descent to the floor involves only gravity and the elastic ropes, so energy is conserved. After Nellie steps off, energy is conserved again, and we have to ask whether the stored elastic energy is enough to lift the plank past the point where the ropes are at their natural length.

Solution

We model the plank as a rigid body that stays horizontal, so we can treat it as a particle of mass 1000 kg moving vertically. Nellie is a particle of mass 4000 kg, and the ropes are light elastic strings obeying Hooke's law. Both ropes always have the same extension x, so together they exert an upward force

$$2 \cdot \frac{490\,000}{10}\, x = 98\,000\, x \text{ N}.$$

So the pair of ropes acts like a single spring of stiffness $K = 98\,000 \text{ N m}^{-1}$, with elastic energy $\frac{1}{2}Kx^2 = 49\,000\, x^2$ J.



The starting position

Before Nellie arrives the plank hangs in equilibrium, so the extension e_0 satisfies

$$98\,000\,e_0 = 1000 \times 9.8 \quad \Longrightarrow \quad e_0 = 0.1 \text{ m}.$$

The landing

Nellie lands and stays on the plank, so this is a perfectly inelastic collision. The impact takes a very short time, and the ropes' tension and the weights are finite forces, so their impulse during the impact is negligible. Momentum is therefore conserved, and the common speed v just afterwards satisfies

$$4000 \times 5 = 5000\,v \quad \Longrightarrow \quad v = 4 \text{ m s}^{-1}.$$

Ask yourself Can we use conservation of energy straight from Nellie's arrival?

No. When an elephant lands on a plank and stays there, a lot of kinetic energy turns into heat and sound: here $\frac{1}{2} \cdot 4000 \cdot 5^2 = 50\,000\,\text{J}$ becomes $\frac{1}{2} \cdot 5000 \cdot 4^2 = 40\,000\,\text{J}$, so $10\,000\,\text{J}$ is lost. Energy is conserved only from the instant just after the landing.

The descent

Let h be the plank's initial height above the floor. Plank and Nellie (total mass 5000 kg) start with speed $4\,\text{m s}^{-1}$ and come to rest at the floor, a distance h lower, where the extension is $0.1 + h$. The only forces doing work are gravity and the ropes, so

$$\underbrace{\tfrac{1}{2} \cdot 5000 \cdot 4^2}_{\text{KE lost}} + \underbrace{5000 \cdot 9.8\,h}_{\text{GPE lost}} = \underbrace{49\,000\left[(0.1+h)^2 - 0.1^2\right]}_{\text{elastic energy gained}},$$

$$40\,000 + 49\,000\,h = 49\,000\,(0.2h + h^2).$$

Dividing by 49 000 and rearranging,

$$h^2 - 0.8\,h - \tfrac{40}{49} = 0, \qquad \text{so} \qquad h = 0.4 + \sqrt{0.16 + \tfrac{40}{49}} = \frac{14 + 2\sqrt{299}}{35},$$

taking the positive root because $h > 0$. Hence

$$h \approx 1.39 \text{ m}.$$

The ropes stay taut throughout this stage, because the extension only grows as the plank goes down.

After Nellie alights

Nellie steps off gently, so the plank alone (1000 kg) is at rest at the floor with extension

$$X = 0.1 + h \approx 1.488 \text{ m}.$$

The ropes go slack if the plank rises far enough for the extension to fall to zero, which means rising a height X. While the ropes are taut, energy is conserved, and so the plank reaches that level if the stored elastic energy exceeds the gain in potential energy:

$$\tfrac{1}{2}KX^2 = 49\,000\,X^2 \approx 108\,500\text{ J}, \qquad 1000 \cdot 9.8 \cdot X \approx 14\,600\text{ J}.$$

The first is far bigger, so the plank arrives at the natural-length position still moving upwards, with kinetic energy of about 93 900 J (a speed of about $13.7\,\mathrm{m\,s^{-1}}$). Above this point the ropes would have to be shorter than their natural length, so they are slack. **Yes, the ropes become slack.**

Ask yourself Couldn't we just use simple harmonic motion?

Yes, while the ropes are taut. The plank alone performs SHM about its equilibrium extension 0.1 m, and it starts $h \approx 1.39$ m below it, so the amplitude is 1.39 m. The top of the oscillation would be at extension $0.1 - 1.39 < 0$, which is impossible for a taut rope. So the ropes must go slack before the plank reaches the top. (In fact the SHM argument gives the answer as soon as $h > 0.1$.)

Once the ropes are slack the plank moves freely under gravity, rising a further $13.7^2/(2 \times 9.8) \approx 9.6$ m. This is less than 10 m, so (just!) the plank doesn't reach the ropes' fixing points before falling back.

Checks

- Substituting $h = 1.3881$ into the energy equation: both sides equal 108 017 J, to the nearest joule.
- Stepping $\ddot{x} = g - Kx/5000$ numerically from $x = 0.1$, $\dot{x} = 4$ gives the plank stopping after descending 1.3881 m.
- Stepping the plank's motion after Nellie leaves shows the extension reaching zero while the plank is moving up at $13.7\,\mathrm{m\,s^{-1}}$, and a highest point 11.07 m above the floor, which matches $X + 9.58$.
- Units: 490 000 N / 10 m gives $\mathrm{N\,m^{-1}}$, so every term in the energy equation is in joules.

Common slips

- Using energy conservation through the landing, which gives $\sqrt{20} \approx 4.47\,\mathrm{m\,s^{-1}}$ for the combined speed instead of $4\,\mathrm{m\,s^{-1}}$.
- Measuring elastic energy from the natural length but forgetting the initial extension 0.1 m: the energy gained is $49\,000[(0.1 + h)^2 - 0.1^2]$, not $49\,000\,h^2$.
- Using the stiffness of one rope instead of two.

Beyond STEP

The pair of ropes is a nonlinear spring: it pulls when stretched but cannot push when compressed. Systems like this, which switch between different equations of motion, are called piecewise-linear or “impact” oscillators, and even simple ones can behave in surprisingly complicated ways. Kleppner and Kolenkow, *An Introduction to Mechanics*, has a careful treatment of energy, collisions and oscillations that underpins all of this question.

WORKED SOLUTION

STEP I 1988, Question 14

Probability and Statistics • Censored Normal Expectation

STEP I 1988, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/88-s1-q14

The key idea

The variable X_M is just $\min(X, M)$: the normal variable with everything above M squashed down to M. Its expectation splits into an integral over $x < M$ and a constant times a tail probability, and the integral is easy because $x\phi(x)$ is, up to sign, the derivative of $\phi(x)$. The tourist story is the same idea in disguise. The number of passengers a bus fleet carries is $\min(\text{demand}, \text{capacity})$, and after a normal approximation that is exactly an X_M.

Solution

The expectation of X_M

Since $X_M = X$ on $\{X < M\}$ and $X_M = M$ on $\{X \geq M\}$,

$$\mathrm{E}(X_M) = \int_{-\infty}^{M} x\,\phi(x)\,\mathrm{d}x \;+\; M\,\mathrm{P}(X \geq M) = \int_{-\infty}^{M} x\,\phi(x)\,\mathrm{d}x \;+\; M(1-\Phi(M)).$$

? **Ask yourself** Is there a function whose derivative is $x\phi(x)$?

Yes. With $\phi(x) = \frac{1}{\sqrt{2\pi}}\mathrm{e}^{-x^2/2}$, the chain rule gives $\phi'(x) = -x\phi(x)$. So

$$\int_{-\infty}^{M} x\,\phi(x)\,\mathrm{d}x = \Big[-\phi(x)\Big]_{-\infty}^{M} = -\phi(M) + 0,$$

because $\phi(x) \to 0$ as $x \to -\infty$. Hence

$$\mathrm{E}(X_M) = -\phi(M) + M(1-\Phi(M)),$$

as required.

As a sanity check, letting $M \to \infty$ gives $\mathrm{E}(X_M) \to 0 = \mathrm{E}(X)$, since $M(1-\Phi(M)) \to 0$. Letting $M \to -\infty$ gives $\mathrm{E}(X_M) \approx M$, as it should.

The buses

Let N be the number of tourists, on one visit, who are sent to the bus station. We assume that each of the 1024 tourists is directed independently, with probability $\frac{1}{2}$ each way. Then

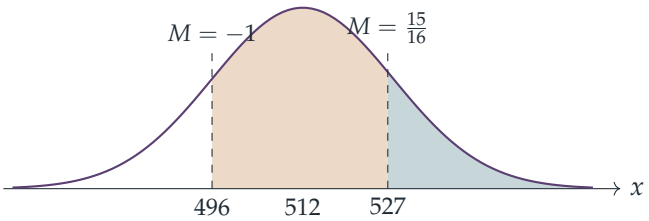
$$N \sim \mathrm{B}(1024, \tfrac{1}{2}), \qquad \mathrm{E}(N) = 512, \qquad \mathrm{Var}(N) = 1024 \cdot \tfrac{1}{2} \cdot \tfrac{1}{2} = 256 = 16^2.$$

With $n = 1024$ this large, N is very well approximated by $512 + 16X$ with X standard normal.

With 16 buses the capacity is $16 \times 31 = 496$. With 17 it is 527. A fleet of capacity C carries $\min(N, C)$ passengers, so its profit for one visit, in vloska, is

$$\begin{aligned} \min(N, C) &\approx \min(512 + 16X,\ C) \\ &= 512 + 16 \min\Big(X, \frac{C - 512}{16}\Big) = 512 + 16\, X_M, \qquad \text{where } M = \frac{C - 512}{16}. \end{aligned}$$

So $C = 496$ gives $M = -1$, and $C = 527$ gives $M = \frac{15}{16}$.



When demand lands in the amber band, the 17th bus carries the excess over 496. When demand lands in the teal tail, it carries a full 31.

Using $\phi(1) \approx 0.24197$ and $\Phi(1) \approx 0.84134$, and noting $1 - \Phi(-1) = \Phi(1)$,

$$\mathrm{E}(X_{-1}) = -\phi(1) - \big(1 - \Phi(-1)\big) = -\phi(1) - \Phi(1) \approx -0.24197 - 0.84134 = -1.08332.$$

Using $\phi(0.9375) \approx 0.25707$ and $1 - \Phi(0.9375) \approx 0.17425$,

$$\mathrm{E}(X_{15/16}) \approx -0.25707 + 0.9375 \times 0.17425 \approx -0.25707 + 0.16336 = -0.09371.$$

The extra bus therefore raises the expected profit per visit by

$$16\big(\mathrm{E}(X_{15/16}) - \mathrm{E}(X_{-1})\big) \approx 16 \times 0.98961 \approx 15.83 \text{ vloska.}$$

There are 50 visits a year, so the extra bus is worth about

$$50 \times 15.83 \approx 792 \text{ vloska a year.}$$

This is the largest annual bribe worth considering. Paying any more than the expected extra profit would leave the cooperative worse off on average.

Ask yourself Shouldn't we use a continuity correction, since N is discrete?

Here it isn't needed. A continuity correction helps when we approximate a probability such as $\mathrm{P}(N \le c)$, where the function being averaged jumps at an integer. The function $k \mapsto \min(k, C)$ has no jump, and the uncorrected normal approximation is extremely accurate for it. The exact binomial calculation gives 791.68, which agrees with the figure above to two decimal places. Shifting the capacities by $\frac{1}{2}$ would actually make things worse, giving 775.

Checks

- Midpoint quadrature of $\int_{-\infty}^{M} x\phi(x)\,\mathrm{d}x + M(1-\Phi(M))$ agrees with the formula to eight decimal places for several values of M. A Monte Carlo simulation of $\min(X, M)$ with 4×10^5 samples agrees too.
- Exact summation over $\mathrm{B}(1024, \frac{1}{2})$ gives an extra expected profit of 15.8337 per visit, which is 791.68 a year, matching the normal approximation.
- Rough plausibility: the extra bus earns something whenever $N > 496$, which happens with probability about 0.84, and it is often full. So a figure a little over half of 31 per visit is reasonable.

Common slips

- Forgetting to scale: $\min(N, C)$ is $512 + 16X_M$, so the difference in $\mathrm{E}(X_M)$ must be multiplied by the standard deviation 16, and then by 50.
- Sign errors with $M = -1$: the term $M(1-\Phi(M))$ is $-(1-\Phi(-1)) = -\Phi(1)$, not $-(1-\Phi(1))$.

Beyond STEP

The quantity $\mathrm{E}\min(X, M)$ is the expected value of a *censored* normal variable. Closely related expressions, such as $\mathrm{E}(X-M)^+ = \phi(M) - M(1-\Phi(M))$, are the backbone of inventory theory (the "newsvendor problem") and of the Black–Scholes price of an option. The identity $\mathrm{E}\min(N, C+1) - \mathrm{E}\min(N, C) = \mathrm{P}(N > C)$ is a neat way to see the value of each extra unit of capacity. For more on these ideas, see Grimmett and Welsh, *Probability: An Introduction*.

WORKED SOLUTION

STEP I 1988, Question 15

Probability and Statistics • Binomial Hypothesis Test

STEP I 1988, Question 15

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/88-s1-q15

The key idea

Turn "more than half its points from fouls" into a simple inequality between the number of fouls and the number of goals. Conditioning on the number of goals then leaves a count of equally likely foul values. For the second part, ask how many nine-foul occasions the *old* distribution would lead us to expect in 300 tries, and how surprising the observed number would be if nothing had changed.

Solution

The probability

For a particular team in a game, let G be the number of goals it scores and F the number of fouls committed against it. Then G and F are independent, with

$$\mathrm{P}(G=0),\ \ldots,\ \mathrm{P}(G=4) \;=\; \tfrac{1}{9},\ \tfrac{2}{9},\ \tfrac{3}{9},\ \tfrac{2}{9},\ \tfrac{1}{9}, \qquad \mathrm{P}(F=f)=\tfrac{1}{10} \quad (f=0,1,\ldots,9).$$

The team scores $2G$ points from goals and F from fouls, so $2G+F$ in total. It gains more than half its points from fouls exactly when

$$F > \tfrac{1}{2}(2G+F) \iff F > 2G.$$

(If the team scores no points at all, then $F=0=2G$ and it does not count, which the strict inequality handles correctly.)

Given $G=g$, the favourable values of F are $2g+1,\ldots,9$, which is $9-2g$ of the ten equally likely values. So, using the independence of F and G,

$$\mathrm{P}(F>2G \mid G=g) = \frac{9-2g}{10}, \qquad g=0,1,2,3,4,$$

giving $\frac{9}{10}, \frac{7}{10}, \frac{5}{10}, \frac{3}{10}, \frac{1}{10}$. By the law of total probability,

$$\mathrm{P}(F>2G) = \frac{1\cdot 9+2\cdot 7+3\cdot 5+2\cdot 3+1\cdot 1}{9\times 10} = \frac{45}{90} = \frac{1}{2}.$$

The neat answer has a reason: the goal distribution is symmetric about 2, and $(9-2g)/10$ is linear in g, so its average equals its value at $g=2$, namely $\frac{1}{2}$.

Has the foul distribution changed?

The number of fouls a team commits is the number committed against its opponents, so under the old rules each team commits 9 fouls in a game with probability $\frac{1}{10}$. Let X be the number of the 300 team-games in which a team committed 9 fouls. We test

$$H_0:\ p=\tfrac{1}{10}\quad\text{(no change)}\qquad\text{against}\qquad H_1:\ p<\tfrac{1}{10},$$

where p is the probability of a team committing 9 fouls in a game. The alternative is one-sided because the rule change was meant to make large numbers of fouls *less* likely.

? **Ask yourself** Is a binomial model justified here?

We need the 300 team-games to be independent, each with the same probability p. The question gives independence within a game, and it is reasonable to assume that different games are independent too. So under H_0, $X\sim \mathrm{B}(300,\frac{1}{10})$, with

$$\mathrm{E}(X)=30,\qquad \mathrm{Var}(X)=300\times\tfrac{1}{10}\times\tfrac{9}{10}=27,\qquad \text{standard deviation } \sqrt{27}\approx 5.20.$$

We observed $X=12$, which is 18 below the expected 30, about 3.5 standard deviations. With the normal approximation and a continuity correction,

$$\mathrm{P}(X\le 12)\approx\Phi\left(\frac{12.5-30}{\sqrt{27}}\right)=\Phi(-3.37)\approx 0.0004.$$

The exact binomial probability is even smaller, about 0.0001. Either way, a result this extreme would happen less than once in a thousand seasons if nothing had changed. It is far beyond the usual 5% or even 1% significance levels, so we reject H_0. This *is* good evidence that the distribution of the number of fouls has changed, and in the hoped-for direction: nine-foul games have become rarer.

Two caveats are worth stating. The test only shows that $\mathrm{P}(9\text{ fouls})$ has gone down, not how the rest of the distribution has changed (although that is the natural explanation). Statistics also cannot prove that the rule change *caused* the decrease, since other things may have changed between seasons.

Checks

- The goal probabilities $\frac{1}{9}(1+2+3+2+1)$ sum to 1, as they must.
- Enumerating all 5×10 pairs (g,f) with exact fractions gives $\mathrm{P}(F>2G)=\frac{1}{2}$, and a simulation of 200,000 games gave 0.4997.
- Summing the binomial probabilities directly gives $\mathrm{P}(X\le 12)=9.2\times10^{-5}$ for $X\sim\mathrm{B}(300,0.1)$. The normal approximation (3.8×10^{-4}) and the Poisson approximation with mean 30 (1.7×10^{-4}) agree that it is tiny.

Common slips

- Comparing F with G instead of with $2G$: a goal is worth two points.
- Counting $F \geq 2G$ instead of $F > 2G$. "More than half" is strict, and a team with no points gains none of them from fouls.
- Using the 150 games as the sample size instead of the 300 team-games, or forgetting to state the null hypothesis and the binomial model before computing.

Beyond STEP

The second part is a hypothesis test for a binomial proportion, and the tail probability we found is a p-value. At university you'll meet the likelihood ratio and chi-squared tests, which compare a whole observed distribution of foul counts with the old one, not just one value. Blitzstein and Hwang, *Introduction to Probability*, covers the probability side (conditioning, binomial and normal approximations) very clearly.

WORKED SOLUTION

STEP I 1988, Question 16

Probability and Statistics • Continuous Distributions and Conditioning

STEP I 1988, Question 16

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/88-s1-q16

The key idea

The first part is a normalisation-and-integration exercise: the density is only given up to a constant, so we must find that constant by insisting the total probability is 1, and the integral of $t^2/(1+t^2)^2$ yields to splitting it into $1/(1+t^2)$ and the derivative of $t/(1+t^2)$. After that the testing scheme is a short tree of cases. Only the tests at ages 4 and 5 can matter (no car is rusty before age 3), so we split according to whether the car was already rusty at its fourth test, and the last part is Bayes' theorem on the same tree.

Solution

The distribution of the onset of rust

Let the car become rusty at age $3+T$, where $T \geq 0$ has density $f(t) = k\,t^2/(1+t^2)^2$ for some constant k. The key integral comes from noticing that

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{t}{1+t^2}\right) = \frac{1-t^2}{(1+t^2)^2}, \qquad \text{so} \qquad \frac{t^2}{(1+t^2)^2} = \frac{1}{2}\left(\frac{1}{1+t^2} - \frac{1-t^2}{(1+t^2)^2}\right),$$

which you can confirm by putting the bracket over the common denominator $(1+t^2)^2$: the numerator is $(1+t^2)-(1-t^2) = 2t^2$. Hence

$$\int_0^t \frac{s^2}{(1+s^2)^2}\,\mathrm{d}s = \frac{1}{2}\left(\arctan t - \frac{t}{1+t^2}\right).$$

As $t \to \infty$, $\arctan t \to \frac{\pi}{2}$ and $t/(1+t^2) \to 0$, so the integral over $[0,\infty)$ is $\frac{\pi}{4}$. For f to be a density we need $k \cdot \frac{\pi}{4} = 1$, so $k = 4/\pi$. Therefore

$$F(t) = \mathrm{P}(\text{rusty before age } 3+t) = \mathrm{P}(T < t) = \frac{2}{\pi}\left(\arctan t - \frac{t}{1+t^2}\right), \quad t \geq 0.$$

(Notice $F(t) \to 1$: every car rusts eventually.)

Two values we need are

$$F(1) = \frac{2}{\pi}\left(\frac{\pi}{4} - \frac{1}{2}\right) = \frac{1}{2} - \frac{1}{\pi} \approx 0.1817,$$
$$F(2) = \frac{2}{\pi}\left(\arctan 2 - \frac{2}{5}\right) \approx 0.4502.$$

Failing the fifth-anniversary test

Before age 3 no car is rusty, so the tests at ages 1, 2 and 3 are passed for certain (the chance of rust starting at the exact moment of the third test is zero). To fail at age 5 a car must pass at age 4 and fail at age 5, and failing requires being rusty at age 5. Rust never goes away, so there are two disjoint ways this can happen.

- *Rusty already at age* 4 (probability $F(1)$). It passes the fourth test with probability $\frac{1}{2}$ and then, still rusty, fails the fifth with probability $\frac{1}{2}$. Contribution: $\frac{1}{4}F(1)$.
- *Rust begins between ages* 4 *and* 5 (probability $F(2) - F(1)$). It passes the fourth test for certain and fails the fifth with probability $\frac{1}{2}$. Contribution: $\frac{1}{2}\big(F(2) - F(1)\big)$.

Here we are assuming, as the question intends, that the outcomes of different tests on a rusty car are independent of each other and of the time rust began.

> **Ask yourself** Why don't we multiply by the probability of passing the tests at ages 1, 2 and 3?

Because those tests happen before any car can be rusty, so every car passes them; that factor is 1.

Adding the two cases,

$$\begin{aligned}\mathrm{P}(\text{fails at age 5}) &= \tfrac{1}{4}F(1) + \tfrac{1}{2}\big(F(2) - F(1)\big) = \tfrac{1}{2}F(2) - \tfrac{1}{4}F(1)\\ &= \frac{1}{\pi}\left(\arctan 2 - \frac{2}{5}\right) - \frac{1}{8} + \frac{1}{4\pi}\\ &= \boxed{\frac{1}{\pi}\left(\arctan 2 - \frac{3}{20}\right) - \frac{1}{8} \approx 0.1797.}\end{aligned}$$

Rusty at the fourth test, given destruction at the fifth

A car destroyed straight after its fifth test is exactly a car that failed that test (having passed all earlier ones), which is the event just found. Let A be the event that it was rusty at its fourth test and D the event that it failed the fifth. The first bullet above is precisely $A \cap D$, so

$$\mathrm{P}(A \cap D) = \tfrac{1}{4}F(1) = \frac{1}{8} - \frac{1}{4\pi} = \frac{\pi - 2}{8\pi}.$$

By the definition of conditional probability,

$$\mathrm{P}(A \mid D) = \frac{\mathrm{P}(A \cap D)}{\mathrm{P}(D)} = \frac{\dfrac{\pi - 2}{8\pi}}{\dfrac{1}{\pi}\left(\arctan 2 - \dfrac{3}{20}\right) - \dfrac{1}{8}}.$$

Multiplying top and bottom by 8π,

$$\boxed{\mathrm{P}(A \mid D) = \frac{\pi - 2}{8\arctan 2 - \frac{6}{5} - \pi} = \frac{5(\pi - 2)}{40\arctan 2 - 6 - 5\pi} \approx 0.253.}$$

So only about a quarter of the cars destroyed at five years had slipped through their fourth test while already rusty; most had rusted during the fifth year.

Checks

- $F(0) = 0$ and $F(t) \to 1$ as $t \to \infty$, as a distribution function must. Simpson's rule on $\frac{4}{\pi}\int_0^2 t^2/(1+t^2)^2\,\mathrm{d}t$ gives 0.45018, matching $F(2)$.
- A simulation of 400 000 cars (sampling the onset time and running the annual tests) gave 0.1789 for failing at five years and 0.255 for the conditional probability, against the exact 0.1797 and 0.2528.
- Sanity: $\mathrm{P}(D)$ must lie below $\frac{1}{2}F(2) \approx 0.225$ (the chance of being rusty at five and then failing, ignoring earlier removal), and it does.

Common slips

- Forgetting to normalise: the density is only *proportional* to $t^2/(1+t^2)^2$, and missing the factor $4/\pi$ throws off every later answer.
- Treating "rusty at age 5" as a single case with probability $F(2)$ and multiplying by $\frac{1}{2}$. That ignores the cars which were rusty at age 4 and were removed at that test.
- In the last part, conditioning on "rusty at age 5" rather than on "failed at age 5".

Beyond STEP

The function F here is a distribution function, and the ratio $f(t)/(1-F(t))$ is the *hazard rate*, the instantaneous chance of rusting given the car has survived so far; it is the central object of survival analysis and reliability theory. The final part is a small example of filtering: inferring a hidden state (rusty or not) from imperfect observations (the tests). Grimmett and Welsh, *Probability: An Introduction*, covers continuous distributions and conditioning in this spirit.

WORKED SOLUTION

STEP II 1988, Question 1

Pure • Partial Fractions and Series

STEP II 1988, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/88-s2-q1

The key idea

Split f into partial fractions, so that everything comes down to expanding two simple fractions, $\frac{1}{x-1}$ and $\frac{1}{x-2}$, as geometric series. The only real decision is *which way round* to expand each one. A geometric series $\sum r^n$ converges only when $|r| < 1$, so for each fraction we pull out whichever of the two terms in the denominator is larger in size. That choice changes as $|x|$ passes 1 and then 2, and this is why there are three different expansions.

Solution

Partial fractions

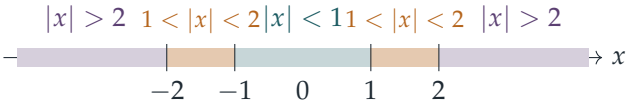
Write $\dfrac{1}{(x-1)(x-2)} = \dfrac{A}{x-1} + \dfrac{B}{x-2}$. The cover-up rule gives $A = \frac{1}{1-2} = -1$ and $B = \frac{1}{2-1} = 1$, so

$$\mathrm{f}(x) = \frac{1}{x-2} - \frac{1}{x-1}.$$

The two building blocks we need are the geometric series

$$\frac{1}{1-r} = \sum_{n=0}^{\infty} r^n \qquad (|r| < 1),$$

used with r equal to one of x, $\frac{x}{2}$, $\frac{1}{x}$ or $\frac{2}{x}$, whichever has modulus less than 1 in the range we are working in.



The range $|x| < 1$

Here $|x| < 1$ and also $|\frac{x}{2}| < \frac{1}{2} < 1$, so both fractions can be expanded in powers of x:

$$-\frac{1}{x-1} = \frac{1}{1-x} = \sum_{n=0}^{\infty} x^n,$$
$$\frac{1}{x-2} = -\frac{1}{2}\cdot\frac{1}{1-\frac{x}{2}} = -\frac{1}{2}\sum_{n=0}^{\infty}\left(\frac{x}{2}\right)^n.$$

Adding,

$$f(x) = \sum_{n=0}^{\infty} x^n - \frac{1}{2}\sum_{n=0}^{\infty}\left(\frac{x}{2}\right)^n \qquad (|x| < 1),$$

as required.

The range $1 < |x| < 2$

The second fraction is unchanged: we still have $|\frac{x}{2}| < 1$, so $\frac{1}{x-2} = -\frac{1}{2}\sum_{n\geq 0}(\frac{x}{2})^n$ exactly as before.

Ask yourself Why can't we keep using $\sum x^n$ for the first fraction?

Because $|x| > 1$ now, so the terms x^n do not tend to zero and $\sum x^n$ diverges. Instead we take out the larger term, x, from the denominator, which leaves a ratio $\frac{1}{x}$ with $|\frac{1}{x}| < 1$:

$$-\frac{1}{x-1} = -\frac{1}{x}\cdot\frac{1}{1-\frac{1}{x}} = -\frac{1}{x}\sum_{n=0}^{\infty} x^{-n} = -\sum_{n=0}^{\infty} x^{-(n+1)} = -\sum_{n=1}^{\infty} x^{-n},$$

where the last step just relabels $n+1$ as n. Adding the two pieces,

$$f(x) = -\sum_{n=1}^{\infty} x^{-n} - \frac{1}{2}\sum_{n=0}^{\infty}\left(\frac{x}{2}\right)^n \qquad (1 < |x| < 2),$$

as required.

The range $|x| > 2$

The first fraction keeps its expansion from the previous range, since $|\frac{1}{x}| < \frac{1}{2} < 1$. But now $|\frac{x}{2}| > 1$, so the second fraction must be turned round in the same way, taking out x and leaving the ratio $\frac{2}{x}$, which has $|\frac{2}{x}| < 1$:

$$\frac{1}{x-2} = \frac{1}{x}\cdot\frac{1}{1-\frac{2}{x}} = \frac{1}{x}\sum_{n=0}^{\infty}\left(\frac{2}{x}\right)^n = \sum_{n=0}^{\infty} 2^n x^{-(n+1)} = \sum_{n=1}^{\infty} 2^{n-1}x^{-n}.$$

Hence, for $|x| > 2$,

$$f(x) = \sum_{n=1}^{\infty} 2^{n-1}x^{-n} - \sum_{n=1}^{\infty} x^{-n} = \sum_{n=1}^{\infty}(2^{n-1}-1)x^{-n}.$$

The $n = 1$ coefficient is $2^0 - 1 = 0$, so the series really starts at x^{-2}:

$$\boxed{\mathrm{f}(x) = \sum_{n=2}^{\infty} (2^{n-1} - 1)x^{-n} = \frac{1}{x^2} + \frac{3}{x^3} + \frac{7}{x^4} + \frac{15}{x^5} + \cdots \qquad (|x| > 2).}$$

(Combining the two series term by term is legitimate because each converges.) Either the two-series form or the combined form is a correct answer.

Checks

- *Leading behaviour.* For large $|x|$, $\mathrm{f}(x) = \frac{1}{x^2 - 3x + 2} \approx \frac{1}{x^2}$, which matches the first term of the last expansion. The vanishing x^{-1} coefficient is no accident: the two partial fractions have equal and opposite $\frac{1}{x}$ behaviour.
- *A value.* At $x = 3$, $\mathrm{f}(3) = \frac{1}{2 \cdot 1} = \frac{1}{2}$, and $\sum_{n \geq 2} (2^{n-1} - 1)3^{-n} = \frac{1}{2} \sum_{n \geq 2} (\frac{2}{3})^n - \sum_{n \geq 2} (\frac{1}{3})^n = \frac{1}{2} \cdot \frac{4}{3} - \frac{1}{6} = \frac{1}{2}$. Similarly $\mathrm{f}(\frac{1}{2}) = \frac{4}{3}$ and $\mathrm{f}(\frac{3}{2}) = -4$ agree with the first two expansions.
- Numerically, all three series (summed to 4000 terms) agree with f to about 10^{-14} at hundreds of random points in their ranges, positive and negative.

Common slips

- Getting the sign wrong in $\frac{1}{x-2} = -\frac{1}{2} \cdot \frac{1}{1 - x/2}$: the factor pulled out is -2, not 2.
- Being out by one in the index after multiplying by $\frac{1}{x}$: $\frac{1}{x} \sum_{n \geq 0} x^{-n}$ starts at x^{-1}, so the relabelled sum starts at $n = 1$.
- Using the same expansion in every range without checking that the ratio has modulus less than 1.

Beyond STEP

These three expansions are the real-variable shadow of *Laurent series* in complex analysis. For complex z, the function $\frac{1}{(z-1)(z-2)}$ has a different Laurent expansion in each of the regions $|z| < 1$, $1 < |z| < 2$ and $|z| > 2$: a disc and two annuli bounded by circles through the poles, which is exactly what the circles $|x| = 1$ and $|x| = 2$ become. The coefficients of z^{-1} in such expansions (residues) turn out to control contour integrals. Needham's *Visual Complex Analysis* gives a beautifully geometric account of power and Laurent series.

WORKED SOLUTION

STEP II 1988, Question 2

Pure • Symmetric Algebraic Equations

STEP II 1988, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/88-s2-q2

The key idea

Clear the fractions, then subtract one equation from the other. The difference of two squares on the right produces a factor $y - z$, matching the $y - z$ that appears on the left, and the hypothesis $y \neq z$ is exactly what lets us cancel it. After that, everything comes down to one symmetric expression in x, y and z, and symmetry does the rest.

Solution

Proving $x + y + z = a$

Since $y \neq 0$ and $z \neq 0$ we may multiply through:

$$2ay - 3y^2 = (z - x)^2, \tag{1}$$
$$2az - 3z^2 = (x - y)^2. \tag{2}$$

Subtracting (2) from (1), and using a difference of two squares on the right,

$$\begin{aligned} 2a(y - z) - 3(y^2 - z^2) &= (z - x)^2 - (x - y)^2 \\ &= \big[(z - x) - (x - y)\big]\big[(z - x) + (x - y)\big] \\ &= (y + z - 2x)(z - y). \end{aligned}$$

The left-hand side is $(y - z)\big[2a - 3(y + z)\big]$, so

$$(y - z)\big[2a - 3(y + z)\big] = -(y - z)(y + z - 2x).$$

Ask yourself Where exactly is $y \neq z$ needed?

Right here: because $y - z \neq 0$ we can divide by it, giving

$$2a - 3y - 3z = 2x - y - z \quad \Longrightarrow \quad 2a = 2x + 2y + 2z,$$

so $x + y + z = a$, as required.

Proving the third equation

Substitute $a = x + y + z$ back into (1):

$$y(2x + 2y + 2z - 3y) = (z - x)^2 \implies 2xy - y^2 + 2yz = z^2 - 2zx + x^2,$$

which rearranges to

$$x^2 + y^2 + z^2 - 2xy - 2yz - 2zx = 0. \tag{3}$$

The striking thing is that (3) is *symmetric* in x, y and z. Now look at what we want. Since $x \neq 0$, the equation $2a - 3x = \dfrac{(y-z)^2}{x}$ is equivalent to

$$x(2a - 3x) = (y - z)^2,$$

and with $a = x + y + z$ this reads

$$x(2y + 2z - x) = y^2 - 2yz + z^2 \iff x^2 + y^2 + z^2 - 2xy - 2yz - 2zx = 0,$$

which is exactly (3). Hence

$$2a - 3x = \frac{(y-z)^2}{x}.$$

(In fact all three equations are just (3) together with $a = x + y + z$, each written from the point of view of one variable.)

Can the last equation hold when $y = z$?

Yes. If $y = z$ the two given equations become the same equation, $2a - 3y = \dfrac{(y-x)^2}{y}$, and the last equation becomes $2a - 3x = 0$, i.e. $a = \frac{3}{2}x$. Putting $a = \frac{3}{2}x$ into the first,

$$3x - 3y = \frac{(x-y)^2}{y} \iff 3y(x - y) = (x - y)^2 \iff (x - y)(x - 4y) = 0.$$

So, for instance, $x = y = z = 1$ with $a = \frac{3}{2}$ satisfies all three equations:

$$2a - 3y = 0 = \frac{(z-x)^2}{y}, \qquad 2a - 3z = 0 = \frac{(x-y)^2}{z}, \qquad 2a - 3x = 0 = \frac{(y-z)^2}{x}.$$

Another example is $x = 4$, $y = z = 1$, $a = 6$: then $2a - 3y = 9 = \frac{(1-4)^2}{1}$ for each of the first two, and $2a - 3x = 0 = \frac{0^2}{4}$.

So the last equation does *not* hold only if $y \neq z$: the condition $y \neq z$ guarantees it, but it can also hold when $y = z$. What fails when $y = z$ is the *guarantee*. For example $x = 1$, $y = z = 2$, $a = \frac{13}{4}$ satisfies both given equations (each side equals $\frac{1}{2}$), but $2a - 3x = \frac{7}{2} \neq 0$. Notice too that the first example has $x + y + z = 3 \neq a$, so with $y = z$ even the first conclusion can fail.

Checks

- Take $x = 9$, $y = 4$, $z = 1$, which satisfy (3): $81 + 16 + 1 - 72 - 8 - 18 = 0$. Then $a = 14$ and $2a - 3y = 16 = \frac{(1-9)^2}{4}$, $2a - 3z = 25 = \frac{(9-4)^2}{1}$ and $2a - 3x = 1 = \frac{(4-1)^2}{9}$. All three hold.
- A numerical check verified the identity $(1) - (2) = (y - z)\big(2a - 2(x + y + z)\big)$ exactly on 10,000 random rational triples, and confirmed all three equations on about 1,900 random points of (3) with $y \neq z$, plus the three $y = z$ examples above.

Common slips

- Dividing by $y - z$ without saying why it's allowed. That step is the whole reason the hypothesis is there.
- Answering the final part with "no" or "yes" and no example. A single explicit triple with $y = z$ settles it.
- Thinking that $y = z$ makes the last equation false. It only removes the guarantee.

Beyond STEP

Equation (3) is a quadratic form, and its zero set is a double cone in (x, y, z)-space. If $x, y, z > 0$, write $x = p^2$, $y = q^2$, $z = r^2$: then (3) factorises as

$$-(p + q + r)(-p + q + r)(p - q + r)(p + q - r) = 0,$$

which is 16 times the squared area of a triangle with sides p, q, r (Heron's formula). So (3) says that $\sqrt{x}$, $\sqrt{y}$, $\sqrt{z}$ form a degenerate triangle, for example $\sqrt{9} = \sqrt{4} + \sqrt{1}$ in the check above. Quadratic forms and their classification are a standard topic in first-year linear algebra; see Strang, *Introduction to Linear Algebra*.

WORKED SOLUTION

STEP II 1988, Question 3

Pure • Symmetries of Roots

STEP II 1988, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/88-s2-q3

The key idea

Don't work with the coefficients first; work with the set of roots. The property says the set of roots is unchanged by the operation $k \mapsto 1/k$, so either the two roots swap or each is fixed. Adding the second operation $k \mapsto 1 - k$ does the same job again. For the cubic, applying the two operations over and over produces up to six values from one root, and a cubic has room for only three: the roots must sit at special points where some of the six values coincide.

Solution

The reciprocal property

Let the roots be α and β (possibly equal, possibly complex), so $\alpha + \beta = -b$ and $\alpha\beta = c$. First, 0 can't be a root, because 0^{-1} doesn't exist; so $c \neq 0$.

The map $k \mapsto 1/k$ must send each root to a root, so $1/\alpha \in \{\alpha, \beta\}$.

- If $1/\alpha = \beta$, then $\alpha\beta = 1$, that is, $c = 1$. Conversely, if $c = 1$ the roots are nonzero with $\beta = 1/\alpha$, so the reciprocal of each root is the other root, and the property holds for *every* real b. (This includes repeated roots ± 1 when $b = \mp 2$, and complex roots on the unit circle when $|b| < 2$.)
- If $1/\alpha = \alpha$ but $1/\alpha \neq \beta$, then $\alpha = \pm 1$ and $\alpha\beta \neq 1$. Applying the property to β, its reciprocal is not α (since $\alpha\beta \neq 1$), so $1/\beta = \beta$ and $\beta = \pm 1$ too. With $\alpha\beta \neq 1$ the roots must be 1 and -1, giving $x^2 - 1 = 0$: $b = 0$, $c = -1$. This equation does have the property.

So the property holds exactly when

$$c = 1 \ \ (b \text{ any real number}), \qquad \text{or} \qquad b = 0,\ c = -1.$$

Adding the property $k \mapsto 1-k$

The same argument with $1-k$ in place of $1/k$: either $1-\alpha = \beta$, so $\alpha+\beta = 1$; or both roots are fixed, $1-\alpha = \alpha$ and $1-\beta = \beta$, so $\alpha = \beta = \frac{1}{2}$ and again $\alpha+\beta = 1$. Either way $b = -1$, and conversely $b = -1$ makes the roots α and $1-\alpha$, so the property holds.

The case $b = 0$, $c = -1$ fails this, so we need $c = 1$ and $b = -1$:

$$b = -1, \qquad c = 1,$$

giving $x^2 - x + 1 = 0$ with roots $\frac{1}{2} \pm \frac{\sqrt{3}}{2}\mathrm{i}$. Each has modulus 1 and real part $\frac{1}{2}$, so $1/k$ and $1-k$ are both the complex conjugate $\bar{k}$, the other root.

The cubic

Now let $x^3 + px^2 + qx + r = 0$ have real coefficients and both properties. As before 0 is not a root, and nor is 1 (else $1 - 1 = 0$ would be one).

A cubic with real coefficients has at least one real root; call it a. Applying the two operations repeatedly, each of the following is a root:

$$a, \quad \frac{1}{a}, \quad 1-a, \quad \frac{1}{1-a}, \quad 1-\frac{1}{a} = \frac{a-1}{a}, \quad \frac{1}{(a-1)/a} = \frac{a}{a-1}.$$

(The fourth is the reciprocal of the third, the fifth is $1-$ the second, the sixth the reciprocal of the fifth.) These are six real numbers, but the cubic has at most three distinct roots, so two of them coincide.

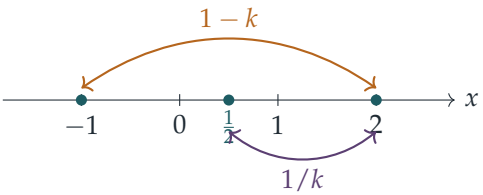
? **Ask yourself** How does "two of these are equal" pin down a?

Call the six maps $f_1, \ldots, f_6$, with f_1 the identity. Each one has an inverse in the list: $\frac{1}{k}$, $1-k$ and $\frac{k}{k-1}$ undo themselves, while $\frac{1}{1-k}$ and $\frac{k-1}{k}$ undo each other. One can also check that composing any two of the maps gives another in the list. So if $f_i(a) = f_j(a)$ with $i \neq j$, applying f_i^{-1} gives $a = g(a)$, where $g = f_i^{-1} \circ f_j$ is one of the six and is not the identity. So a is a fixed point of one of the five non-identity maps:

$$\begin{aligned}
\tfrac{1}{a} = a &\iff a = \pm 1, & \tfrac{1}{1-a} = a &\iff a^2 - a + 1 = 0,\\
1-a = a &\iff a = \tfrac{1}{2}, & \tfrac{a-1}{a} = a &\iff a^2 - a + 1 = 0,\\
\tfrac{a}{a-1} = a &\iff a(a-2) = 0.
\end{aligned}$$

The quadratic $a^2 - a + 1$ has no real roots (it is the quadratic from before), and $a \neq 0, 1$. Hence $a \in \{-1, \frac{1}{2}, 2\}$.

Whichever of these a is, the six values above run through exactly -1, 2 and $\frac{1}{2}$: the map $1-k$ swaps -1 and 2 and fixes $\frac{1}{2}$, while $1/k$ swaps 2 and $\frac{1}{2}$ and fixes -1.



So -1, 2 and $\frac{1}{2}$ are three distinct roots of the cubic. A monic cubic is determined by its three roots, so it must be

$$(x+1)(x-2)\left(x-\tfrac{1}{2}\right) = (x^2-x-2)\left(x-\tfrac{1}{2}\right) = x^3 - \tfrac{3}{2}x^2 - \tfrac{3}{2}x + 1.$$

Conversely, this cubic does have both properties, since its root set $\{-1, \frac{1}{2}, 2\}$ is sent to itself by $k \mapsto 1/k$ and by $k \mapsto 1-k$. So

$$x^3 - \tfrac{3}{2}x^2 - \tfrac{3}{2}x + 1 = 0$$

is the only such cubic, as required.

Checks

Substitute $x = 2$: $8 - 6 - 3 + 1 = 0$; $x = -1$: $-1 - \frac{3}{2} + \frac{3}{2} + 1 = 0$. The product of the roots is $-1 \cdot 2 \cdot \frac{1}{2} = -1 = -r$, and their sum is $\frac{3}{2} = -p$. Notice the cubic's coefficients read the same forwards and backwards up to sign $(1, -\frac{3}{2}, -\frac{3}{2}, 1)$: that palindromic shape is exactly what $k \mapsto 1/k$ forces. A computer test of 20,000 random (b, c) confirmed the reciprocal condition, and random real a away from $0, \pm 1, \frac{1}{2}, 2$ always gave six distinct values.

Common slips

- Answering only "$c = 1$" in the first part: the case where each root is its own reciprocal adds $b = 0$, $c = -1$.
- Overlooking the pair $\frac{1}{2} \pm \frac{\sqrt{3}}{2}\mathrm{i}$, which is closed under both maps. Starting from a *real* root is what rules it out: a real cubic has a real root, and that root's six values can't include these.
- Proving only that the given cubic works; the question asks you to show it is the *only* one.

Beyond STEP

The six maps $k \mapsto k$, $1/k$, $1-k$, $\frac{1}{1-k}$, $\frac{k-1}{k}$, $\frac{k}{k-1}$ form a group under composition, a copy of the symmetric group S_3; it permutes the three points $0, 1, \infty$, and the six values are the possible cross-ratios of four points. The counting step is the orbit–stabiliser

theorem: an orbit is shorter than six only at points fixed by some non-identity element. Allenby's *Rings, Fields and Groups* develops groups acting on sets, and Needham's *Visual Complex Analysis* shows these maps as Möbius transformations of the plane.

WORKED SOLUTION

STEP II 1988, Question 4

Pure • Loci in the Argand Diagram

STEP II 1988, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/88-s2-q4

The key idea

Write the complex number as $a + ib$ and turn "real" or "imaginary" into one real equation in a and b. That gives a familiar curve in the Argand diagram. Then squaring is a map of the plane, and we find where it sends that curve by writing x and y (or p and q) in terms of a and b and eliminating.

A word on the printed text: it says $w^2 - 2x$, but x has not been introduced at that point, and the final part (with $t^2 - 2t$) shows the intended condition. We read it as $w^2 - 2w$ *is real*.

Solution

Part (i)

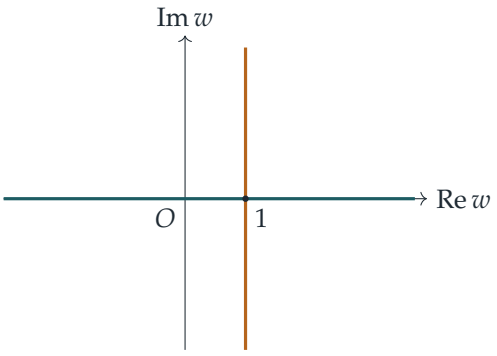
Let $w = a + ib$ with a, b real. Then

$$w^2 - 2w = (a^2 - b^2 - 2a) + i\,(2ab - 2b) = (a^2 - b^2 - 2a) + 2i\,b(a - 1).$$

This is real exactly when its imaginary part vanishes, so when

$$b(a - 1) = 0, \qquad \text{that is,} \qquad b = 0 \ \text{ or } \ a = 1.$$

The locus of w is therefore the real axis together with the vertical line $\operatorname{Re} w = 1$. The two lines cross at $w = 1$.



Part (ii)

We square each of the two lines separately.

The real axis. If $b = 0$ then $w = a$ is real and $w^2 = a^2$. So $y = 0$ and $x = a^2 \geq 0$, and as a runs through the reals, a^2 takes every value in $[0, \infty)$. This piece of the locus is the non-negative x-axis: the half-line $y = 0$, $x \geq 0$.

The line $a = 1$. If $w = 1 + ib$ then

$$w^2 = 1 - b^2 + 2ib, \qquad \text{so} \qquad x = 1 - b^2, \quad y = 2b.$$

Eliminating $b = \frac{1}{2}y$ gives

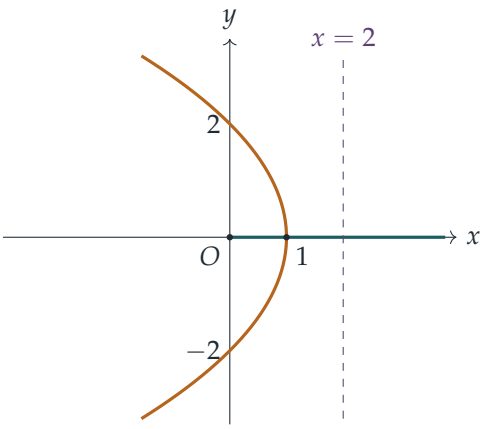
$$x = 1 - \tfrac{1}{4}y^2, \qquad \text{or} \qquad y^2 = 4(1 - x) = -4(x - 1).$$

Every point of this curve is reached, by taking $b = \frac{1}{2}y$. It is a parabola with its axis along the x-axis, vertex at $(1, 0)$, opening towards negative x. Comparing with the standard form $Y^2 = 4cX$, here $c = -1$ with $X = x - 1$. So the focus is at $X = -1$, which is the origin, and the directrix is $X = 1$, which is the line $x = 2$. Indeed $x^2 + y^2 = x^2 + 4 - 4x = (2 - x)^2$, so every point is as far from O as from the line $x = 2$. The parabola cuts the y-axis at $(0, \pm 2)$.

? **Ask yourself** Why doesn't the half-line extend to negative x?

The real axis in the w-plane is covered twice (by a and $-a$), but squaring a real number never gives a negative result. So the whole real axis folds onto the non-negative half. Similarly, the line $\mathrm{Re}\, w = 1$ goes onto the parabola once, because $b \mapsto 2b$ is one-to-one.

So the locus of (x, y) is the parabola $y^2 = 4(1 - x)$, with focus at the origin and directrix $x = 2$, together with the half-line $y = 0$, $x \geq 0$. The half-line runs from the focus through the vertex $(1, 0)$, which is the image of $w = 1$, where the two lines of part (i) meet.



The final part

Let $t = a + \mathrm{i}b$. "Imaginary" here means purely imaginary, so we need the real part of $t^2 - 2t$ to vanish:

$$a^2 - b^2 - 2a = 0, \qquad \text{that is,} \qquad (a-1)^2 - b^2 = 1.$$

So t lies on a rectangular hyperbola centred at 1, with vertices at $t = 0$ and $t = 2$. On it, $b^2 = a^2 - 2a = a(a-2) \geq 0$, so $a \leq 0$ or $a \geq 2$.

Now $t^2 = p + \mathrm{i}q$ gives

$$p = a^2 - b^2 = 2a, \qquad q = 2ab,$$

where we used the hyperbola's equation $a^2 - b^2 = 2a$ to simplify p. So $a = \frac{1}{2}p$, and then

$$q^2 = 4a^2b^2 = 4a^2(a^2 - 2a) = p^2\left(\tfrac{1}{4}p^2 - p\right), \qquad \text{so} \qquad 4q^2 = p^3(p-4).$$

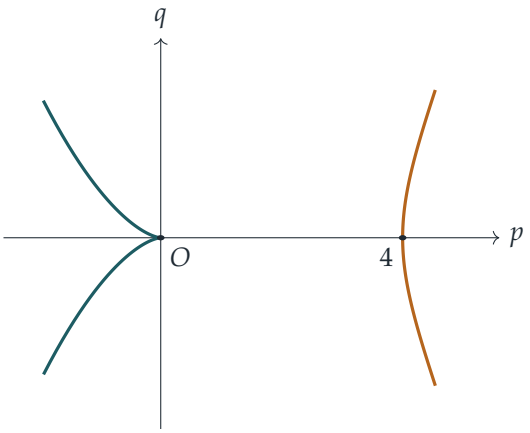
The conditions $a \leq 0$ or $a \geq 2$ become $p \leq 0$ or $p \geq 4$, which are exactly where $p^3(p-4) \geq 0$. Conversely, take a point on this curve. If $p \neq 0$, put $a = \frac{1}{2}p$ and $b = q/p$. Then $b^2 = q^2/p^2 = \frac{1}{4}p(p-4) = a^2 - 2a$, so t is on the hyperbola, and $t^2 = (a^2 - b^2) + 2\mathrm{i}ab = p + \mathrm{i}q$. If $p = 0$ then $q = 0$, and this point comes from $t = 0$. So the locus is the whole curve

$$4q^2 = p^3(p-4).$$

Its shape:

- It is symmetric in the p-axis, because only q^2 appears.
- There are no points with $0 < p < 4$. The curve has two pieces, one coming from each branch of the hyperbola.
- The right piece starts at $(4, 0)$, the image of $t = 2$. Near there $q^2 \approx 16(p-4)$, so the tangent is vertical.
- The left piece passes through the origin, the image of $t = 0$. Near there $q^2 \approx -p^3$, so $|q| \approx |p|^{3/2}$, a cusp pointing right, with both arms tangent to the p-axis.

- For large $|p|$, $|q| \approx \frac{1}{2}p^2$, so both pieces open out like parabolas.



(The vertical scale is compressed. The teal piece comes from the branch of the hyperbola with $\operatorname{Re} t \leq 0$, the amber piece from the branch with $\operatorname{Re} t \geq 2$.)

Checks

- Part (i): $w = 1 + 2\mathrm{i}$ gives $w^2 - 2w = (-3 + 4\mathrm{i}) - (2 + 4\mathrm{i}) = -5$, which is real. Part (ii): its square $-3 + 4\mathrm{i}$ satisfies $16 = 4(1 - (-3))$.
- Final part: $t = -1 + \mathrm{i}\sqrt{3}$ satisfies $1 - 3 + 2 = 0$. Its square is $-2 - 2\sqrt{3}\,\mathrm{i}$, and $4q^2 = 48 = (-8)(-6) = p^3(p - 4)$.
- We checked numerically at thousands of random points that each locus maps onto the stated curve, and that every point of each curve is reached.

Common slips

- Forgetting the real axis in part (i), and so missing the half-line in part (ii). Or drawing the whole x-axis instead of only $x \geq 0$.
- Not using $a^2 - b^2 = 2a$ to get $p = 2a$. Without it, the elimination is much messier.
- Missing the gap $0 < p < 4$, or the cusp at the origin, in the final sketch.

Beyond STEP

Squaring, $z \mapsto z^2$, is the simplest conformal map. It doubles angles, which is why the lines and hyperbolas here turn into parabolas and cusped curves. At $z = 0$ the derivative vanishes, so angles are no longer preserved, and that is where the cusp comes from. A standard trick in fluid flow and electrostatics is to use maps like this to

turn a hard region into an easy one. Needham's *Visual Complex Analysis* explains these mappings beautifully with pictures.

WORKED SOLUTION

STEP II 1988, Question 5

Pure • Roots of Unity and Tangents

STEP II 1988, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/88-s2-q5

The key idea

Write $z = \cos\theta + \mathrm{i}\sin\theta$ and expand z^7 with the binomial theorem. The imaginary part is $\sin 7\theta$, and dividing it by $\cos^7\theta$ turns it into a polynomial in $t = \tan\theta$. The seventh roots of unity make $\sin 7\theta$ vanish, so they hand us seven roots of a degree-7 polynomial for free. After that the tangent product is just the product of the roots, read off from the constant term.

Solution

The roots of the sextic

By de Moivre's theorem, $(\cos\theta + \mathrm{i}\sin\theta)^7 = \cos 7\theta + \mathrm{i}\sin 7\theta$. Write $c = \cos\theta$, $s = \sin\theta$. Expanding the left-hand side with the binomial theorem and keeping the terms with an odd power of i gives

$$\sin 7\theta = 7c^6 s - 35c^4 s^3 + 21c^2 s^5 - s^7.$$

Whenever $\cos\theta \neq 0$ we may divide by c^7 and put $t = \tan\theta$:

$$\frac{\sin 7\theta}{\cos^7\theta} = 7t - 35t^3 + 21t^5 - t^7 = -t\,(t^6 - 21t^4 + 35t^2 - 7). \qquad (*)$$

The roots of $z^7 = 1$ are $z = \cos\theta + \mathrm{i}\sin\theta$ with $\theta = \frac{2k\pi}{7}$, and for these $\sin 7\theta = \sin 2k\pi = 0$, the imaginary part of $z^7 = 1$. It is a little tidier to use $\theta = \frac{k\pi}{7}$, $k = 0, 1, \ldots, 6$, which also gives $\sin 7\theta = \sin k\pi = 0$ (and gives the same set of tangents, because tan has period π).

For each of these θ:

- $\cos\theta \neq 0$, since $\frac{k\pi}{7} = \frac{\pi}{2}$ would need $2k = 7$; so $(*)$ is valid;
- $\sin 7\theta = 0$, so by $(*)$ the number $t = \tan\frac{k\pi}{7}$ is a root of $p(t) = t^7 - 21t^5 + 35t^3 - 7t$.

The seven values $\tan\frac{k\pi}{7}$ ($k = 0, \ldots, 6$) are distinct: the angles $0, \frac{\pi}{7}, \ldots, \frac{6\pi}{7}$ are distinct points of $[0, \pi)$, none equal to $\frac{\pi}{2}$, and tan is one-to-one on $[0, \pi) \setminus \{\frac{\pi}{2}\}$ (positive and increasing on $[0, \frac{\pi}{2})$, negative on $(\frac{\pi}{2}, \pi)$, also increasing there, with the two pieces taking values in $[0, \infty)$ and $(-\infty, 0)$).

Ask yourself How do we know we have found *all* the roots, not just some?

A polynomial of degree 7 has at most 7 roots. We have found 7 distinct ones, so they are all of them. The value $k = 0$ gives $t = 0$, the root coming from the factor t. Hence the roots of $t^6 - 21t^4 + 35t^2 - 7 = 0$ are exactly

$$t = \tan\frac{k\pi}{7}, \qquad k = 1, 2, 3, 4, 5, 6,$$

that is, $\pm\tan\frac{\pi}{7}$, $\pm\tan\frac{2\pi}{7}$, $\pm\tan\frac{3\pi}{7}$, since $\tan\frac{(7-k)\pi}{7} = -\tan\frac{k\pi}{7}$.

The product $\tan\frac{2\pi}{7}\tan\frac{4\pi}{7}\tan\frac{6\pi}{7}$

For a monic polynomial of even degree, the product of the roots equals the constant term. So

$$\left(\tan\tfrac{\pi}{7}\right)\left(-\tan\tfrac{\pi}{7}\right)\left(\tan\tfrac{2\pi}{7}\right)\left(-\tan\tfrac{2\pi}{7}\right)\left(\tan\tfrac{3\pi}{7}\right)\left(-\tan\tfrac{3\pi}{7}\right) = -7,$$

which says

$$\left(\tan\tfrac{\pi}{7}\tan\tfrac{2\pi}{7}\tan\tfrac{3\pi}{7}\right)^2 = 7.$$

All three angles lie in $(0, \frac{\pi}{2})$, so each tangent is positive, and the product is the positive square root: $\tan\frac{\pi}{7}\tan\frac{2\pi}{7}\tan\frac{3\pi}{7} = \sqrt{7}$.

Now use $\tan(\pi - x) = -\tan x$: $\tan\frac{4\pi}{7} = -\tan\frac{3\pi}{7}$ and $\tan\frac{6\pi}{7} = -\tan\frac{\pi}{7}$. The two minus signs cancel, so

$$\tan\frac{2\pi}{7}\tan\frac{4\pi}{7}\tan\frac{6\pi}{7} = \tan\frac{2\pi}{7}\tan\frac{3\pi}{7}\tan\frac{\pi}{7} = \sqrt{7},$$

as required.

The cases $n = 9$ and $n = 11$

The same argument works for any odd $n = 2m + 1$. Taking the imaginary part of $(\cos\theta + \mathrm{i}\sin\theta)^n$ and dividing by $\cos^n\theta$,

$$\frac{\sin n\theta}{\cos^n\theta} = \sum_{r=0}^{m}(-1)^r\binom{n}{2r+1}t^{2r+1} = t\,q_n(t),$$

where q_n has degree $2m = n - 1$, leading coefficient $(-1)^m$ and constant term $\binom{n}{1} = n$. Explicitly,

$$q_9(t) = t^8 - 36t^6 + 126t^4 - 84t^2 + 9,$$
$$q_{11}(t) = -t^{10} + 55t^8 - 330t^6 + 462t^4 - 165t^2 + 11.$$

Exactly as before (n is odd, so no $\frac{k\pi}{n}$ equals $\frac{\pi}{2}$), the roots of q_n are the $n-1$ distinct numbers $\tan\frac{k\pi}{n}$, $k = 1, \ldots, n-1$, which pair up as $\pm\tan\frac{k\pi}{n}$ for $k = 1, \ldots, m$. The product of the roots of q_n is (constant term)/(leading coefficient), because the degree is even:

$$(-1)^m \prod_{k=1}^{m} \tan^2\frac{k\pi}{n} = \frac{n}{(-1)^m} \quad \Longrightarrow \quad \prod_{k=1}^{m} \tan\frac{k\pi}{n} = \sqrt{n},$$

taking the positive root since all these angles are in $(0, \frac{\pi}{2})$. For $n = 9$ this is q_9's constant term 9 over leading coefficient 1; for $n = 11$ it is $11/(-1) = -11 = (-1)^5 \prod \tan^2$. Either way, the product of $\tan\frac{k\pi}{n}$ over $k = 1, \ldots, m$ is 3 for $n = 9$ and $\sqrt{11}$ for $n = 11$.

Ask yourself The required product has angles $\frac{2\pi}{n}, \frac{4\pi}{n}, \ldots$, not $\frac{\pi}{n}, \frac{2\pi}{n}, \ldots$. How do we get from one to the other, and what happens to the sign?

Each even multiple $2j \le m$ is already in the list $1, \ldots, m$. Each even $2j > m$ (that is, $2j > \frac{n}{2}$) gives $\tan\frac{2j\pi}{n} = -\tan\frac{(n-2j)\pi}{n}$, and $n - 2j$ is odd and at most m. Every number from 1 to m arises exactly once in this way, because the even ones come from the first kind and the odd ones from the second. So the required product is $\pm\sqrt{n}$, with one minus sign for each even number between $\frac{n}{2}$ and n.

- $n = 9$: the even numbers $2, 4, 6, 8$; those above 4.5 are $6, 8$, two minus signs. So
$$\tan\frac{2\pi}{9}\tan\frac{4\pi}{9}\tan\frac{6\pi}{9}\tan\frac{8\pi}{9} = \sqrt{9} = 3.$$
- $n = 11$: the even numbers $2, 4, 6, 8, 10$; those above 5.5 are $6, 8, 10$, three minus signs. So
$$\tan\frac{2\pi}{11}\tan\frac{4\pi}{11}\tan\frac{6\pi}{11}\tan\frac{8\pi}{11}\tan\frac{10\pi}{11} = -\sqrt{11}.$$

(For $n = 7$ the even numbers above 3.5 are $4, 6$, two signs, which is why that product came out positive.)

Checks

- $n = 9$ has an easy partial check: $\tan\frac{6\pi}{9} = \tan\frac{2\pi}{3} = -\sqrt{3}$, and numerically $\tan\frac{2\pi}{9}\tan\frac{4\pi}{9}\tan\frac{8\pi}{9} \approx 0.8391 \times 5.6713 \times (-0.3640) \approx -1.732 = -\sqrt{3}$, giving a total of 3.
- On a calculator, $\tan\frac{2\pi}{7}\tan\frac{4\pi}{7}\tan\frac{6\pi}{7} \approx 2.645751$ and $\sqrt{7} \approx 2.645751$; the $n = 11$ product is $\approx -3.316625 = -\sqrt{11}$.
- $\tan\frac{\pi}{7} \approx 0.4816$ gives $t^6 - 21t^4 + 35t^2 - 7 \approx 0$, as it should.

Common slips

- Dividing by $\cos^7\theta$ without checking that it is non-zero, or not arguing that the seven tangents are distinct. Without these points you have not shown that you have found

all the roots.

- Getting the sign wrong for $n = 11$: the square of the product is easy, but the sign needs the count of negative tangents.
- Forgetting that the product of the roots of $a_d t^d + \cdots + a_0$ is $(-1)^d a_0/a_d$, which matters for $n = 11$, where the leading coefficient is -1.

Beyond STEP

The numbers $\tan \frac{k\pi}{n}$ are algebraic, and the polynomials q_n are their minimal polynomials (when n is prime) over the rationals. Identities like $\prod \tan \frac{k\pi}{n} = \sqrt{n}$ are shadows of Gauss sums, which show that $\sqrt{\pm p}$ lies in the field generated by the pth roots of unity, a cornerstone of algebraic number theory. Davenport's *The Higher Arithmetic* is a gentle way into this circle of ideas, and Needham's *Visual Complex Analysis* gives a geometric feel for roots of unity.

WORKED SOLUTION

STEP II 1988, Question 6

Pure • Hyperbolic Function Inequalities

STEP II 1988, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/88-s2-q6

The key idea

Each of the three functions is zero at $x = 0$, so it is enough to show that it is increasing for $x > 0$. The derivatives are chosen so that each part feeds the next: the derivative of (ii) contains the function from (i), and the derivative of (iii) is a rescaled copy of (ii). For f, logarithmic differentiation turns the sign of f′ into the sign of the function in (iii), which is clearly what that part was set up for.

Solution

Throughout, we use the principle that if $F(0) = 0$ and $F'(x) > 0$ for all $x > 0$, then $F(x) > 0$ for all $x > 0$. (By the mean value theorem, $F(x) = F(x) - F(0) = xF'(\xi)$ for some $\xi \in (0, x)$.)

Part (i)

Let $g(x) = x - \tanh x$. Then $g(0) = 0$ and

$$g'(x) = 1 - \operatorname{sech}^2 x = \tanh^2 x,$$

using $1 - \tanh^2 x = \operatorname{sech}^2 x$. For $x > 0$ we have $\tanh x > 0$, so $g'(x) > 0$. Hence $x - \tanh x > 0$ for $x > 0$.

Part (ii)

Let $h(x) = x \sinh x - 2 \cosh x + 2$. Then $h(0) = 0 - 2 + 2 = 0$ and

$$h'(x) = \sinh x + x \cosh x - 2 \sinh x = x \cosh x - \sinh x = \cosh x \, (x - \tanh x).$$

For $x > 0$, $\cosh x > 0$ and $x - \tanh x > 0$ by part (i), so $h'(x) > 0$. Hence $h(x) > 0$ for $x > 0$.

Part (iii)

Let $k(x) = 2x\cosh 2x - 3\sinh 2x + 4x$. Then $k(0) = 0$ and

$$\begin{aligned} k'(x) &= 2\cosh 2x + 4x\sinh 2x - 6\cosh 2x + 4 \\ &= 4x\sinh 2x - 4\cosh 2x + 4. \end{aligned}$$

? **Ask yourself** Where have we seen this combination before?

Writing $u = 2x$, this is $2u\sinh u - 4\cosh u + 4 = 2\left(u\sinh u - 2\cosh u + 2\right) = 2h(u)$. So

$$k'(x) = 2h(2x) > 0 \quad \text{for } x > 0,$$

by part (ii), since $2x > 0$. Hence $k(x) > 0$ for $x > 0$.

No turning points of f

For $x > 0$ every factor in $\mathrm{f}(x)$ is positive, so $\mathrm{f}(x) > 0$ and we may take logarithms:

$$\ln \mathrm{f}(x) = \ln x + \tfrac{1}{3}\ln\cosh x - \ln\sinh x.$$

Differentiating,

$$\frac{\mathrm{f}'(x)}{\mathrm{f}(x)} = \frac{1}{x} + \frac{\sinh x}{3\cosh x} - \frac{\cosh x}{\sinh x} = \frac{3\sinh x\cosh x + x\sinh^2 x - 3x\cosh^2 x}{3x\sinh x\cosh x}.$$

Now we put the numerator in terms of double angles, using $\sinh x\cosh x = \frac{1}{2}\sinh 2x$, $\sinh^2 x = \frac{1}{2}(\cosh 2x - 1)$ and $\cosh^2 x = \frac{1}{2}(\cosh 2x + 1)$:

$$\begin{aligned} 3\sinh x\cosh x + x\sinh^2 x - 3x\cosh^2 x &= \tfrac{3}{2}\sinh 2x + \tfrac{x}{2}(\cosh 2x - 1) - \tfrac{3x}{2}(\cosh 2x + 1) \\ &= \tfrac{3}{2}\sinh 2x - x\cosh 2x - 2x \\ &= -\tfrac{1}{2}\left(2x\cosh 2x - 3\sinh 2x + 4x\right). \end{aligned}$$

Therefore

$$\mathrm{f}'(x) = -\frac{\mathrm{f}(x)\left(2x\cosh 2x - 3\sinh 2x + 4x\right)}{6x\sinh x\cosh x}.$$

For $x > 0$, the bracket is positive by part (iii), and $\mathrm{f}(x)$, x, $\sinh x$ and $\cosh x$ are all positive. So $\mathrm{f}'(x) < 0$ for every $x > 0$: f' never vanishes, so f has no turning points, and in fact f is strictly decreasing on $x > 0$.

The sketch

We need the behaviour at the two ends.

As $x \to 0^+$: $x/\sinh x \to 1$ and $(\cosh x)^{1/3} \to 1$, so $\mathrm{f}(x) \to 1$. In more detail, $(\cosh x)^{1/3} = 1 + \frac{1}{6}x^2 + O(x^4)$ and $x/\sinh x = 1 - \frac{1}{6}x^2 + O(x^4)$, so the x^2 terms cancel; one more term of each series gives

$$\mathrm{f}(x) = 1 - \tfrac{1}{45}x^4 + O(x^6).$$

So the curve leaves the (excluded) point $(0, 1)$ very flat, with zero gradient.

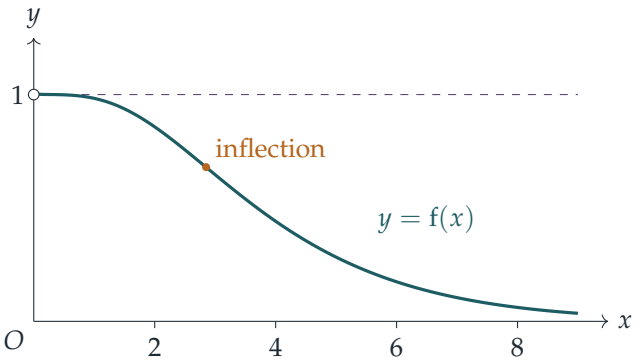
As $x \to \infty$: $\cosh x \sim \frac{1}{2}e^x$ and $\sinh x \sim \frac{1}{2}e^x$, so

$$\mathrm{f}(x) \sim \frac{x\,(\frac{1}{2}e^x)^{1/3}}{\frac{1}{2}e^x} = 2^{2/3}\,x\,e^{-2x/3} \to 0,$$

since the exponential beats the power of x. The x-axis is an asymptote, and the curve stays above it because $\mathrm{f} > 0$.

Ask yourself The curve starts flat and ends flat, yet always decreases. What does that force?

Near 0 the curve bends downwards (it is $1 - x^4/45$), and for large x it is a decaying exponential, which bends upwards. So there must be at least one point of inflection in between; numerically there is one, near $x \approx 2.85$, where $\mathrm{f} \approx 0.68$. Some values: $\mathrm{f}(1) \approx 0.983$, $\mathrm{f}(2) \approx 0.858$, $\mathrm{f}(4) \approx 0.441$, $\mathrm{f}(6) \approx 0.174$.



Checks

- At $x = 1$: $1 - \tanh 1 \approx 0.238$, $\sinh 1 - 2\cosh 1 + 2 \approx 0.089$ and $2\cosh 2 - 3\sinh 2 + 4 \approx 0.643$, all positive.
- The identity $\mathrm{f}'/\mathrm{f} = -k(x)/(6x\sinh x\cosh x)$ can be tested at any point: at $x = 1$ both sides are about -0.0592, and a numerical derivative of f agrees.
- Numerically $(1 - \mathrm{f}(x))/x^4 \approx 0.0222 \approx \frac{1}{45}$ for small x, and $\mathrm{f}(x)/(2^{2/3}xe^{-2x/3}) \to 1$ for large x.

Common slips

- Showing only that the derivative is positive, and forgetting to check that the function is 0 at $x = 0$; both are needed.
- Trying to differentiate f with the quotient and product rules directly. It works, but the logarithmic derivative is far cleaner and makes the link to (iii) visible.

- Drawing the curve crossing the x-axis, or starting at the origin. It starts at height 1 (not attained) and stays positive.

Beyond STEP

The technique of proving $F(x) > 0$ from $F(0) = 0$ and $F' > 0$ is the mean value theorem in action, and chaining it (as here, through k', h', g') is a standard way to prove inequalities between functions in analysis. The expansion $\mathrm{f}(x) = 1 - x^4/45 + \cdots$ is a taste of manipulating power series, which first-year analysis puts on a rigorous footing. A good place to read about both is Abbott, *Understanding Analysis*.

WORKED SOLUTION

STEP II 1988, Question 7

Pure • Integration by Substitution

STEP II 1988, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/88-s2-q7

The key idea

Every equality here is a substitution, and the hard part is finding the right one. The first step needs a substitution that sends $[1,2]$ to $[0,1]$ and turns $2-2x+x^2$ into something with $1+t^2$ in it: a Möbius map $x = 2/(1+t)$ does exactly that. Once $1+t^2$ appears, $t=\tan\theta$ is the obvious move. The θ-integrand is symmetric about $\theta = \frac{1}{8}\pi$, and undoing $t=\tan\theta$ on the half-range brings in $\tan\frac{1}{8}\pi = \sqrt{2}-1$. The final deduction uses the same substitutions in reverse.

Solution

Throughout, k is any real constant. All the bases we raise to powers are positive on the ranges involved, so non-integer powers cause no trouble.

From $[1,2]$ to $[0,1]$

Put $x = \dfrac{2}{1+t}$, so that $t=1$ when $x=1$, $t=0$ when $x=2$, and $\mathrm{d}x = -\dfrac{2}{(1+t)^2}\,\mathrm{d}t$. Then

$$2-2x+x^2 = \frac{2(1+t)^2-4(1+t)+4}{(1+t)^2} = \frac{2(1+t^2)}{(1+t)^2}, \qquad x^{k+1} = \frac{2^{k+1}}{(1+t)^{k+1}}.$$

So the integrand becomes

$$\frac{2^k(1+t^2)^k}{(1+t)^{2k}} \cdot \frac{(1+t)^{k+1}}{2^{k+1}} \cdot \frac{2}{(1+t)^2} = \frac{(1+t^2)^k}{(1+t)^{k+1}},$$

and the minus sign in $\mathrm{d}x$ swaps the limits from $\int_1^0$ to $\int_0^1$:

$$I = \int_0^1 \frac{(1+t^2)^k}{(1+t)^{k+1}}\,\mathrm{d}t.$$

The trigonometric form

Now put $t = \tan\theta$, with θ running from 0 to $\frac{1}{4}\pi$. Then $1 + t^2 = \sec^2\theta$, $\mathrm{d}t = \sec^2\theta\,\mathrm{d}\theta$, and

$$1 + t = \frac{\cos\theta + \sin\theta}{\cos\theta} = \frac{\sqrt{2}\cos\left(\frac{1}{4}\pi - \theta\right)}{\cos\theta},$$

using $\cos\theta + \sin\theta = \sqrt{2}\left(\cos\frac{1}{4}\pi\cos\theta + \sin\frac{1}{4}\pi\sin\theta\right)$. The integrand becomes

$$\sec^{2k}\theta \cdot \frac{\cos^{k+1}\theta}{\left[\sqrt{2}\cos\left(\frac{1}{4}\pi - \theta\right)\right]^{k+1}} \cdot \sec^2\theta = \frac{\cos^{-(k+1)}\theta}{\left[\sqrt{2}\cos\left(\frac{1}{4}\pi - \theta\right)\right]^{k+1}} = \frac{1}{\left[\sqrt{2}\cos\theta\cos\left(\frac{1}{4}\pi - \theta\right)\right]^{k+1}},$$

since the powers of $\cos\theta$ total $-2k + (k+1) - 2 = -(k+1)$. Both cosines are positive on $[0, \frac{1}{4}\pi]$, so this is valid for every real k. That gives the second form of I.

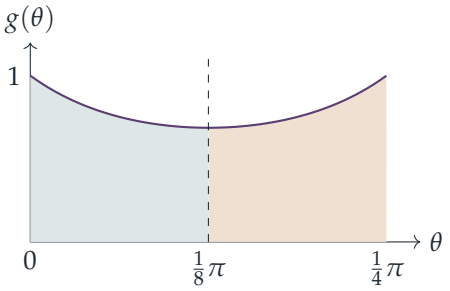
Ask yourself Why should the integral over $[0, \frac{1}{4}\pi]$ be twice the integral over $[0, \frac{1}{8}\pi]$?

Write $g(\theta)$ for the θ-integrand. Replacing θ by $\frac{1}{4}\pi - \theta$ just swaps the two factors $\cos\theta$ and $\cos(\frac{1}{4}\pi - \theta)$, so $g(\frac{1}{4}\pi - \theta) = g(\theta)$: the graph of g is symmetric about $\theta = \frac{1}{8}\pi$. (You can see this directly too: $\sqrt{2}\cos\theta\cos(\frac{1}{4}\pi - \theta) = \frac{1}{2}\left(1 + \sqrt{2}\cos(2\theta - \frac{1}{4}\pi)\right)$, which depends only on the distance of θ from $\frac{1}{8}\pi$.) Formally, substituting $\phi = \frac{1}{4}\pi - \theta$,

$$\int_{\pi/8}^{\pi/4} g(\theta)\,\mathrm{d}\theta = \int_0^{\pi/8} g(\tfrac{1}{4}\pi - \phi)\,\mathrm{d}\phi = \int_0^{\pi/8} g(\phi)\,\mathrm{d}\phi,$$

so the two halves of $[0, \frac{1}{4}\pi]$ contribute equally and

$$I = 2\int_0^{\pi/8} \frac{\mathrm{d}\theta}{\left[\sqrt{2}\cos\theta\cos\left(\frac{1}{4}\pi - \theta\right)\right]^{k+1}}.$$



The figure shows g for $k = 1$: the two shaded halves are mirror images.

Hence the integral up to $\sqrt{2}-1$

Now run the substitution $t=\tan\theta$ backwards on $[0,\frac{1}{8}\pi]$. Every step of the computation above is unchanged; only the upper limit moves, to $t=\tan\frac{1}{8}\pi$. From $\tan\frac{1}{4}\pi = \dfrac{2\tau}{1-\tau^2}=1$ with $\tau=\tan\frac{1}{8}\pi$ we get $\tau^2+2\tau-1=0$, so $\tau=-1\pm\sqrt{2}$, and the positive root is $\tan\frac{1}{8}\pi=\sqrt{2}-1$. Hence

$$I=2\int_0^{\sqrt{2}-1}\frac{(1+t^2)^k}{(1+t)^{k+1}}\,\mathrm{d}t,$$

as required. Comparing this with $I=\int_0^1$ of the same integrand, we have found that the integral over $[0,\sqrt{2}-1]$ and the integral over $[\sqrt{2}-1,1]$ are *both* equal to $\frac{1}{2}I$. This is the fact we use next.

The deduction

Left-hand side. Write $\dfrac{2-2x^2+x^4}{x^2}=\dfrac{2-2u+u^2}{u}$ with $u=x^2$. Then $\mathrm{d}u=2x\,\mathrm{d}x$, so $\dfrac{\mathrm{d}x}{x}=\dfrac{\mathrm{d}u}{2u}$, and u runs from 1 to 2:

$$\int_1^{\sqrt{2}}\left(\frac{2-2x^2+x^4}{x^2}\right)^k\frac{1}{x}\,\mathrm{d}x=\frac{1}{2}\int_1^2\frac{(2-2u+u^2)^k}{u^k}\cdot\frac{1}{u}\,\mathrm{d}u=\frac{1}{2}I.$$

Right-hand side. The integrand is $\dfrac{(2-2x+x^2)^k}{x^{k+1}}$, the same as in the definition of I, but over $[1,\sqrt{2}]$. Use the first substitution $x=2/(1+t)$ again. Now $x=\sqrt{2}$ corresponds to $1+t=\sqrt{2}$, that is $t=\sqrt{2}-1$, so

$$\begin{aligned}\int_1^{\sqrt{2}}\left(\frac{2-2x+x^2}{x}\right)^k\frac{1}{x}\,\mathrm{d}x&=\int_{\sqrt{2}-1}^1\frac{(1+t^2)^k}{(1+t)^{k+1}}\,\mathrm{d}t\\&=I-\int_0^{\sqrt{2}-1}\frac{(1+t^2)^k}{(1+t)^{k+1}}\,\mathrm{d}t=I-\tfrac{1}{2}I=\tfrac{1}{2}I.\end{aligned}$$

Both sides equal $\frac{1}{2}I$, so they are equal, which is the required result.

Checks

- $k=0$: $I=\int_1^2\frac{1}{x}\,\mathrm{d}x=\ln 2$, and both sides of the final identity are $\int_1^{\sqrt{2}}\frac{1}{x}\,\mathrm{d}x=\frac{1}{2}\ln 2$, as they should be.
- $k=1$: $I=\int_1^2\left(\frac{2}{x^2}-\frac{2}{x}+1\right)\mathrm{d}x=2-2\ln 2\approx 0.61371$. Simpson's rule on all five forms of I, and on both sides of the final identity, agrees to ten decimal places (each side is ≈ 0.30685). The same agreement holds for non-integer and negative k, such as $k=0.5$, $k=-0.7$ and $k=4.3$.

Common slips

- Forgetting the factor $\sqrt{2}$ when writing $\cos\theta + \sin\theta$ as a single cosine, which leaves a stray $2^{(k+1)/2}$.
- Trying to evaluate I. For general k there is no closed form, and none is needed: every step is a change of variable or a symmetry.
- In the deduction, missing that the right-hand side is only *part* of I and needs the "hence" result to finish.

Beyond STEP

The map $x \mapsto 2/(1+t)$ is a Möbius (fractional linear) transformation. These maps send circles and lines to circles and lines, and they are the natural changes of variable for integrands built from quadratics. Spotting a hidden symmetry of an integrand, as with $\theta \mapsto \frac{1}{4}\pi - \theta$ here, is a standard trick for evaluating definite integrals that have no elementary antiderivative. For more on Möbius transformations, read Needham, *Visual Complex Analysis*. For many more integrals of this style, see Siklos, *Advanced Problems in Mathematics*.

WORKED SOLUTION

STEP II 1988, Question 8

Pure • Matrices and Eigenvectors

STEP II 1988, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/88-s2-q8

The key idea

One year of the model is a linear map, so it is multiplication by a 2×2 matrix M. A population whose ratio never changes is one that M just rescales: an *eigenvector*. Once we have the two eigenvectors, any starting population splits into a piece along each, and each piece simply gets multiplied by its own eigenvalue every year. The long-term fate is then decided by which eigenvalue is larger and by the sign of the coefficient in front of it.

Solution

Ratios that stay the same

Writing the populations as a column vector, one year of the model is

$$\begin{pmatrix} A' \\ B' \end{pmatrix} = M \begin{pmatrix} A \\ B \end{pmatrix}, \qquad M = \begin{pmatrix} \frac{1}{4} & \frac{3}{4} \\ -\frac{1}{2} & \frac{3}{2} \end{pmatrix}.$$

The ratio $A : B$ is unchanged exactly when $(A', B') = \lambda(A, B)$ for some number λ, that is, when (A, B) is an eigenvector of M. The eigenvalues satisfy

$$\det(M - \lambda I) = \left(\tfrac{1}{4} - \lambda\right)\left(\tfrac{3}{2} - \lambda\right) + \tfrac{3}{8} = \lambda^2 - \tfrac{7}{4}\lambda + \tfrac{3}{4} = 0,$$

so $4\lambda^2 - 7\lambda + 3 = (\lambda - 1)(4\lambda - 3) = 0$, giving $\lambda = 1$ or $\lambda = \frac{3}{4}$.

- $\lambda = 1$: the first row gives $\frac{1}{4}A + \frac{3}{4}B = A$, so $A = B$. The ratio is 1, and such a population stays exactly the same each year.
- $\lambda = \frac{3}{4}$: the first row gives $\frac{1}{4}A + \frac{3}{4}B = \frac{3}{4}A$, so $A = \frac{3}{2}B$. The ratio is $\frac{3}{2}$; check the second row: $\frac{3}{2}B - \frac{1}{2} \cdot \frac{3}{2}B = \frac{3}{4}B$. Both species shrink by a factor $\frac{3}{4}$ each year.

Equivalently, without matrices: if $A = rB$ then the new ratio is $\dfrac{\frac{1}{4}r + \frac{3}{4}}{\frac{3}{2} - \frac{1}{2}r}$, and setting this

equal to r gives $2r^2 - 5r + 3 = 0$, so $r = 1$ or $r = \frac{3}{2}$. Either way, the ratio can stay fixed only if it is 1 or $\frac{3}{2}$, as required. We take $x = 1$ and $y = \frac{3}{2}$.

Subsequent years

Write

$$\begin{pmatrix} a \\ b \end{pmatrix} = p \begin{pmatrix} 1 \\ 1 \end{pmatrix} + q \begin{pmatrix} \frac{3}{2} \\ 1 \end{pmatrix}.$$

Comparing components, $p + \frac{3}{2}q = a$ and $p + q = b$. Subtracting, $\frac{1}{2}q = a - b$, so

$$q = 2(a - b), \qquad p = b - q = 3b - 2a.$$

Because M is linear, after one year the first piece is multiplied by 1 and the second by $\frac{3}{4}$; after n years (as long as the model still applies) we have

$$\begin{pmatrix} a_n \\ b_n \end{pmatrix} = (3b - 2a) \begin{pmatrix} 1 \\ 1 \end{pmatrix} + 2(a - b)\left(\frac{3}{4}\right)^n \begin{pmatrix} \frac{3}{2} \\ 1 \end{pmatrix},$$

that is,

$$a_n = (3b - 2a) + 3(a - b)\left(\tfrac{3}{4}\right)^n,$$
$$b_n = (3b - 2a) + 2(a - b)\left(\tfrac{3}{4}\right)^n.$$

Since $(\frac{3}{4})^n \to 0$, the populations head towards $(3b - 2a,\ 3b - 2a)$, a point on the line of equilibria $a = b$.

One reassurance before classifying: the aardvarks never cause trouble. If $a_n \geq 0$ and $b_n > 0$ then $a_{n+1} = \frac{1}{4}a_n + \frac{3}{4}b_n > 0$, so while the buffaloes survive, the aardvarks stay positive too. So the only thing to watch is the sign of b_n, and we work in the region $a \geq 0$, $b > 0$ of the plane.

Ask yourself Could b_n dip below zero for a while and then recover towards a positive limit?

No. In the formula for b_n, only $(\frac{3}{4})^n$ depends on n, and it decreases steadily from 1 to 0. So b_n moves monotonically from $b_0 = b$ towards its limit $3b - 2a$, and every b_n lies between b and $3b - 2a$. That makes the classification clean.

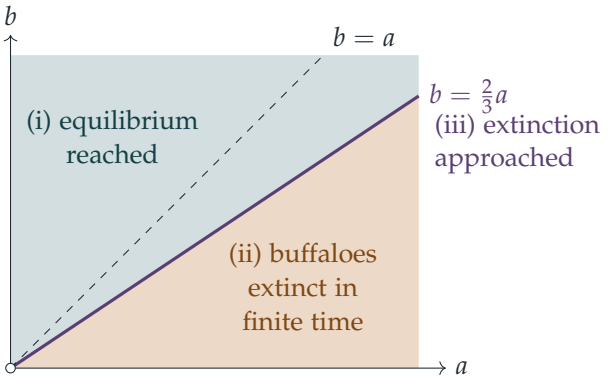
The three regions

(i) If $3b - 2a > 0$, i.e. $b > \frac{2}{3}a$, then b and $3b - 2a$ are both positive, so every b_n is positive and the model applies for ever. The population tends to the equilibrium $(3b - 2a,\ 3b - 2a)$, which is non-zero. This is the region above the line $b = \frac{2}{3}a$ (including the b-axis, $a = 0$).

(ii) If $3b - 2a < 0$, i.e. $b < \frac{2}{3}a$, then $a > \frac{3}{2}b > b$, so $a - b > 0$ and b_n decreases strictly towards the negative number $3b - 2a$. Hence $b_n \leq 0$ for some finite n: the

buffaloes become extinct in finite time. This is the region between the line $b = \frac{2}{3}a$ and the a-axis.

(iii) If $3b - 2a = 0$, i.e. $b = \frac{2}{3}a$ with $a > 0$, then $p = 0$ and the population is a pure multiple of $(\frac{3}{2}, 1)$: $b_n = b\,(\frac{3}{4})^n$, which is positive for every n but tends to 0. The buffaloes (and the aardvarks) approach extinction but never reach it. This is just the half-line $b = \frac{2}{3}a$, $a > 0$.



The dashed line $b = a$ consists of populations that are already in equilibrium; every starting point in region (i) moves towards it, sliding parallel to the line $b = \frac{2}{3}a$ (the direction $(\frac{3}{2}, 1)$ shrinks away while the $(1, 1)$ component stays fixed).

Checks

- With $n = 0$ the formulas give $a_0 = 3b - 2a + 3a - 3b = a$ and $b_0 = 3b - 2a + 2a - 2b = b$, as they must.
- Start at $(a, b) = (1, 3)$, in region (i). One year of the model gives $(\frac{1}{4} + \frac{9}{4},\ \frac{9}{2} - \frac{1}{2}) = (\frac{5}{2}, 4)$, and the formula gives $7 - 6(\frac{3}{4}) = \frac{5}{2}$ and $7 - 4(\frac{3}{4}) = 4$. The limit is $(7, 7)$.
- A computer iterated the model exactly (with fractions) from 3000 random starting points, confirmed the closed form for 40 years, and confirmed that the sign of $3b - 2a$ correctly predicts the outcome in every case, including points on $b = \frac{2}{3}a$, where e.g. $(3, 2)$ becomes $(\frac{729}{1024}, \frac{243}{512})$ after five years, a multiple $(\frac{3}{4})^5$ of the start.

Common slips

- Getting the ratio upside down: the question asks for aardvarks to buffaloes, so the values are 1 and $\frac{3}{2}$, not 1 and $\frac{2}{3}$.
- Forgetting that the model stops when $b_n \leq 0$, and then claiming that every population tends to $(3b - 2a, 3b - 2a)$, including ones with a negative "limit".
- Describing region (iii) as an area: it is only the half-line $b = \frac{2}{3}a$, the boundary between (i) and (ii).

Beyond STEP

This is a discrete linear dynamical system, and the method (diagonalise, then raise each eigenvalue to the nth power) is exactly how first-year linear algebra computes M^n. The eigenvalue 1 gives a line of fixed points and the eigenvalue $\frac{3}{4} < 1$ gives a direction that decays; the same picture, with continuous time and $e^{\lambda t}$ in place of λ^n, governs systems of linear differential equations. Strang's *Introduction to Linear Algebra* has a good chapter on difference equations and powers of matrices, and Boyce and DiPrima's *Elementary Differential Equations* does the continuous version.

WORKED SOLUTION

STEP II 1988, Question 9

Pure • Subgroups and Unions

STEP II 1988, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/88-s2-q9

This question has a full guided treatment in *Beyond STEP*, Section 6.6, *Unions of Subgroups*: https://beyondstep.keithmcnulty.org/ch06/subgroup-unions.html

The key idea

Argue by contradiction. If neither subgroup is the whole group, pick an element that only G_2 contains and an element that only G_1 contains, then ask where their product lives. Closure under multiplication and inverses gives a contradiction whichever subgroup we put it in. With three subgroups the product has somewhere else to go. The natural group to try is one in which every element is its own inverse.

Solution

Two subgroups

Suppose that every element of G lies in G_1 or in G_2, and suppose, for a contradiction, that $G_1 \neq G$ and $G_2 \neq G$.

Since $G_1 \neq G$ and $G_1 \subseteq G$, there is an element $a \in G$ with $a \notin G_1$. Every element lies in G_1 or G_2, so $a \in G_2$.

In the same way, since $G_2 \neq G$ there is an element $b \in G$ with $b \notin G_2$, and then $b \in G_1$.

So we have

$$a \in G_2, \quad a \notin G_1, \qquad\qquad b \in G_1, \quad b \notin G_2.$$

Now look at the product $ab \in G$. By hypothesis it lies in G_1 or in G_2.

- If $ab \in G_1$: since $b \in G_1$ and G_1 is a subgroup, $b^{-1} \in G_1$, and so by closure

$$a = (ab)\, b^{-1} \in G_1,$$

  which contradicts $a \notin G_1$.

- If $ab \in G_2$: since $a \in G_2$, also $a^{-1} \in G_2$, and so
$$b = a^{-1}(ab) \in G_2,$$
which contradicts $b \notin G_2$.

Both cases are impossible, so our assumption was false: $G_1 = G$ or $G_2 = G$. ■

Ask yourself Where did we use that G is finite?

We didn't. The argument uses only the subgroup properties (closure and inverses) and works for any group, finite or infinite. Equivalently: a group is never the union of two proper subgroups.

Three subgroups

Take H to be the Klein four-group

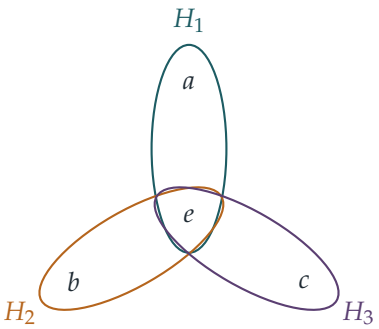
$$H = \{e,\ a,\ b,\ c\}, \qquad a^2 = b^2 = c^2 = e, \quad ab = ba = c,\ bc = cb = a,\ ca = ac = b.$$

Concretely, H is $\{(0,0),(1,0),(0,1),(1,1)\}$ under addition modulo 2 in each coordinate, or the symmetry group of a non-square rectangle (identity, two reflections and the half-turn).

Every non-identity element has order 2, so each of

$$H_1 = \{e,a\}, \qquad H_2 = \{e,b\}, \qquad H_3 = \{e,c\}$$

is a subgroup: it contains e, is closed under multiplication (the only non-trivial product is $a \cdot a = e$, and similarly for b and c), and each element is its own inverse. Each has 2 elements, so none of them is H. But between them they contain e, a, b and c, which is every element of H.



So three proper subgroups can cover a group. Another example is the quaternion group $Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}$, which is the union of the three cyclic subgroups $\{\pm 1, \pm i\}$, $\{\pm 1, \pm j\}$ and $\{\pm 1, \pm k\}$, each of order 4.

Notice why the first argument breaks down here: with $a \notin H_2$ and $b \notin H_1$, the product $ab = c$ is free to escape into the third subgroup H_3.

Checks

- A brute-force search over every subgroup of the cyclic groups of order 1 to 12, of S_3, S_4, the dihedral group of order 8, Q_8, $\mathbb{Z}_2^2$ and $\mathbb{Z}_2^3$ (1278 ordered pairs of proper subgroups) found no pair whose union is the whole group.
- The same search confirms that $\{e, a\} \cup \{e, b\} \cup \{e, ab\}$ is all of the Klein four-group, and Q_8 is the union of $\langle i\rangle$, $\langle j\rangle$, $\langle k\rangle$. Interestingly, none of the cyclic groups tested, nor S_3 or S_4, is the union of three proper subgroups.
- A quick sanity check on the first part: in a cyclic group, a generator lies in no proper subgroup at all, so a cyclic group can't even be covered by any number of proper subgroups.

Common slips

- Arguing by counting: two proper subgroups each have at most $\frac{1}{2}|G|$ elements, but they share e, so their union has at most $|G| - 1$ elements. This counting proof is correct (by Lagrange's theorem) and short, but it does need finiteness and Lagrange; make sure you state both.
- Picking $a \notin G_1$ and $b \notin G_2$ but forgetting to use that $a \in G_2$ and $b \in G_1$. Without those two facts the contradiction doesn't follow.
- Offering three subgroups that are not all proper, or not subgroups at all (for example $\{e, a, b\}$, which is not closed since $ab = c$).

Beyond STEP

Which groups are the union of three proper subgroups? A theorem of Scorza (1926) gives the answer: exactly those with a quotient isomorphic to the Klein four-group, which is why the Klein group is the basic example. The same "no union of two" fact appears in linear algebra, where a vector space over an infinite field is never a finite union of proper subspaces. For a first course in groups, try Allenby, *Rings, Fields and Groups*.

WORKED SOLUTION

STEP II 1988, Question 10

Pure • Vector Geometry of Paraboloids

STEP II 1988, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/88-s2-q10

The key idea

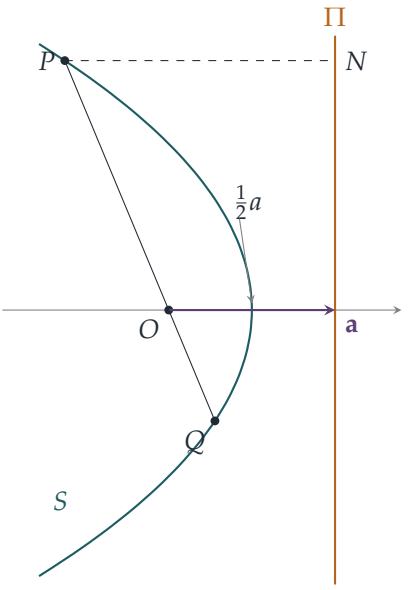
The equation has two pieces with clear geometric meanings: r is the distance from O, and $(a^2 - \mathbf{a}\cdot\mathbf{r})/a$ is the signed distance from the plane $\mathbf{r}\cdot\mathbf{a} = a^2$. So S is a paraboloid of revolution with focus O and that plane as its directrix. For the rest, work along lines through O: write a point on line OP as $\lambda\mathbf{p}$, and a point on the line through O perpendicular to $\mathbf{a}$ and $\mathbf{p}$ as $\mu\,\mathbf{a}\times\mathbf{p}$. Then substitute into the equation of S. The sign of λ matters, because $|\lambda\mathbf{p}| = |\lambda|p$.

Solution

The locus

Let Π be the plane $\mathbf{r}\cdot\mathbf{a} = a^2$. It has unit normal $\hat{\mathbf{a}} = \mathbf{a}/a$, and it lies at distance a from O, on the side towards $\mathbf{a}$. The perpendicular distance from the point $\mathbf{r}$ to Π is

$$d(\mathbf{r}) = |\,a - \hat{\mathbf{a}}\cdot\mathbf{r}\,| = \frac{|a^2 - \mathbf{a}\cdot\mathbf{r}|}{a}.$$



The diagram shows the cross-section of S by a plane containing $\mathbf{a}$. The point P is equally far from O and from its foot N on Π. The surface crosses the axis at $\frac{1}{2}\mathbf{a}$, halfway between O and Π.

Points of S are equidistant. If $\mathbf{a}\cdot\mathbf{r} + ar = a^2$, then $a^2 - \mathbf{a}\cdot\mathbf{r} = ar \geq 0$. So

$$d(\mathbf{r}) = \frac{a^2 - \mathbf{a}\cdot\mathbf{r}}{a} = \frac{ar}{a} = r = |OR|.$$

Equidistant points lie on S. Suppose $r = d(\mathbf{r})$, that is, $ar = |a^2 - \mathbf{a}\cdot\mathbf{r}|$. If $a^2 - \mathbf{a}\cdot\mathbf{r} \geq 0$, this is exactly the equation of S. Otherwise $\mathbf{a}\cdot\mathbf{r} > a^2$, and the condition says $\mathbf{a}\cdot\mathbf{r} = ar + a^2 > ar$. That contradicts $\mathbf{a}\cdot\mathbf{r} \leq ar$ (Cauchy–Schwarz). So no point beyond Π is equidistant, and S is exactly the set of points equidistant from O and Π. This is a paraboloid of revolution about the line through O in the direction of $\mathbf{a}$.

Two consequences are useful later. For P on S,

$$\mathbf{a}\cdot\mathbf{p} = a^2 - ap. \qquad (*)$$

Also $a^2 = \mathbf{a}\cdot\mathbf{p} + ap \leq 2ap$, so $p \geq \frac{1}{2}a$, and hence $2p - a \geq 0$. Equality holds only when $\mathbf{p}$ is a positive multiple of $\mathbf{a}$, that is, when P is the vertex $\frac{1}{2}\mathbf{a}$.

The second intersection Q

Points on line OP are $\lambda\mathbf{p}$, with $|\lambda\mathbf{p}| = |\lambda|p$. Substituting into the equation of S gives

$$\lambda\,\mathbf{a}\cdot\mathbf{p} + a|\lambda|p = a^2.$$

Ask yourself Why can't Q be on the same side of O as P?

If $\lambda > 0$, the left-hand side is $\lambda(\mathbf{a}\cdot\mathbf{p} + ap) = \lambda a^2$, so $\lambda = 1$, which is P itself. (Geometrically, a ray from the focus meets the paraboloid at most once.) So Q has $\lambda < 0$, and then $|\lambda| = -\lambda$. Using $(*)$,

$$\lambda(\mathbf{a}\cdot\mathbf{p} - ap) = a^2 \quad\Longrightarrow\quad \lambda(a^2 - 2ap) = a^2 \quad\Longrightarrow\quad \lambda = -\frac{a}{2p-a}.$$

This is negative, as required, provided $2p > a$, that is, provided P is not the vertex. (At the vertex, the line OP is the axis, and it meets S only once.) Hence

$$\boxed{\mathbf{q} = -\frac{a}{2p-a}\,\mathbf{p}} \qquad \text{and} \qquad |OQ| = \frac{ap}{2p-a}.$$

The points T and T'

The line through O perpendicular to both $\mathbf{a}$ and $\mathbf{p}$ is $\mathbf{t} = \mu\,\mathbf{a}\times\mathbf{p}$. (We need $\mathbf{a}\times\mathbf{p} \neq \mathbf{0}$, so P is not on the axis.) Since $\mathbf{a}\cdot\mathbf{t} = 0$, the equation of S reduces to $at = a^2$, that is, $|\mathbf{t}| = a$. So T and T' lie on the circle where S meets the plane through O parallel to Π, and

$$|\mu|\,|\mathbf{a}\times\mathbf{p}| = a.$$

Ask yourself How do we find $|\mathbf{a}\times\mathbf{p}|$ without knowing the angle between $\mathbf{a}$ and $\mathbf{p}$?

Use the identity $|\mathbf{a}\times\mathbf{p}|^2 = a^2p^2 - (\mathbf{a}\cdot\mathbf{p})^2$, which is $\sin^2\theta = 1 - \cos^2\theta$ in disguise. Then substitute $(*)$:

$$\begin{aligned}|\mathbf{a}\times\mathbf{p}|^2 &= a^2p^2 - (a^2 - ap)^2\\ &= a^2p^2 - a^4 + 2a^3p - a^2p^2\\ &= a^3(2p-a).\end{aligned}$$

Therefore

$$|\mu| = \frac{a}{a^{3/2}\sqrt{2p-a}} = \frac{1}{\sqrt{a(2p-a)}} = \frac{1}{\sqrt{2ap-a^2}},$$

and the position vectors of T and T' are $\pm\dfrac{1}{\sqrt{2ap-a^2}}\,\mathbf{a}\times\mathbf{p}$, as required.

The area of triangle PQT

The points P, O and Q are collinear, with O between P and Q. The vector $\mathbf{t}$ is parallel to $\mathbf{a}\times\mathbf{p}$, so it is perpendicular to $\mathbf{p}$ and hence to the line PQ. So OT is the altitude of triangle PQT from T, and its length is a. The base is

$$|PQ| = p + \frac{ap}{2p-a} = \frac{p(2p-a)+ap}{2p-a} = \frac{2p^2}{2p-a}.$$

Hence

$$\text{area} = \tfrac{1}{2}\cdot|PQ|\cdot|OT| = \tfrac{1}{2}\cdot\frac{2p^2}{2p-a}\cdot a = \frac{ap^2}{2p-a},$$

as required. The same holds with T' in place of T.

Checks

- Take $\mathbf{a} = (2, 0, 0)$ and $P = (-1.25, 3, 0)$, which is the point in the diagram. Then $\mathbf{a}\cdot\mathbf{p} + ap = -2.5 + 2(3.25) = 4 = a^2$, so P is on S. The formula gives $\mathbf{q} = -\frac{2}{4.5}\mathbf{p} = (0.556, -1.333, 0)$, and $\mathbf{a}\cdot\mathbf{q} + aq = 1.111 + 2.889 = 4$, so Q is on S too.
- Take P at an end of the latus rectum, so $\mathbf{p} \perp \mathbf{a}$ and $p = a$. Then $\mathbf{q} = -\mathbf{p}$, which is right by symmetry. The area is $\frac{1}{2}(2a)(a) = a^2$, and the formula gives $a \cdot a^2/a = a^2$.
- The two focal distances satisfy $\dfrac{1}{OP} + \dfrac{1}{OQ} = \dfrac{1}{p} + \dfrac{2p - a}{ap} = \dfrac{2}{a}$, the classical focal-chord property of a parabola.
- Numerically, we checked Q, T, T' and the area for 20 000 random choices of $\mathbf{a}$ and P. The largest error was about 10^{-11}.

Common slips

- Writing $|\lambda\mathbf{p}| = \lambda p$ for every λ. Then the only solution is $\lambda = 1$, and Q can't be found.
- Forgetting the converse in the first part. A point beyond Π has $|a^2 - \mathbf{a}\cdot\mathbf{r}| = \mathbf{a}\cdot\mathbf{r} - a^2$, and we have to rule out such points.
- Computing the area with a cross product of $\overrightarrow{PQ}$ and $\overrightarrow{PT}$. It works, but it is much longer than noticing that $OT \perp PQ$.

Beyond STEP

This is the focus–directrix definition of a conic, in three dimensions and coordinate-free. With e the eccentricity, the surface $e\,\mathbf{a}\cdot\mathbf{r} + ar = e\,a^2$ (distance from O equal to e times distance from Π) gives the other quadrics of revolution with a focus at O. The relation $1/OP + 1/OQ = 2/a$ is the vector form of the polar equation $\ell/r = 1 + e\cos\theta$, which is central to orbital mechanics. Strang's *Introduction to Linear Algebra* is a good next step for the vector algebra, and Siklos's *Advanced Problems in Mathematics* has more questions in this style.

WORKED SOLUTION

STEP II 1988, Question 11

Mechanics • String Winding Round Post

> **STEP II 1988, Question 11**
>
> Read the question before you go on. It is in the STEP Support Programme's question database:
>
> step.maths.org/questions/88-s2-q11

The key idea

The free part of the string always leaves the post along a *tangent*. So the particle's position is "contact point plus a tangent vector of length (unwound string)", and that is easy to write down in terms of the angle θ. Differentiating, two terms cancel and the velocity turns out to be perpendicular to the string. The tension therefore does no work, the speed stays at its initial value, and the speed formula becomes a separable differential equation for θ.

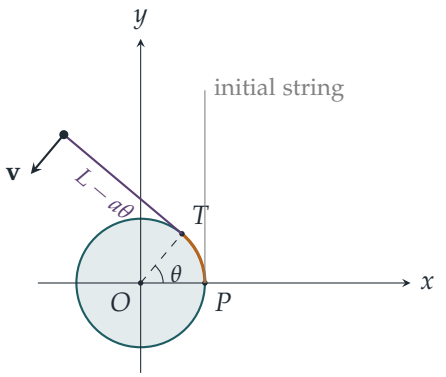
Solution

Setting up

We model the particle as a point mass on a smooth table, with a light inextensible string that stays taut. Gravity and the normal reaction are vertical and cancel, so the only horizontal force is the tension.

Take axes with origin O and put P at $(a, 0)$. The string starts perpendicular to OP, so it is tangent to the post at P, and we take the particle to start at (a, L). (The mirror-image choice just flips the sign of the y-coordinate.) As the string winds on anticlockwise, after it has wrapped through angle θ the length $a\theta$ lies along the arc from P, and the string leaves the post at

$$T = a(\cos\theta,\ \sin\theta).$$



The free part of the string, of length $L - a\theta$, runs along the tangent at T. The unit tangent pointing the way the string unwinds is $(-\sin\theta,\ \cos\theta)$: it is perpendicular to OT, and at $\theta = 0$ it is $(0, 1)$, matching the starting position.

Position and velocity

Hence the position vector of the particle is

$$\mathbf{r} = a\begin{pmatrix}\cos\theta\\ \sin\theta\end{pmatrix} + (L - a\theta)\begin{pmatrix}-\sin\theta\\ \cos\theta\end{pmatrix} = \begin{pmatrix}a\cos\theta - (L - a\theta)\sin\theta\\ a\sin\theta + (L - a\theta)\cos\theta\end{pmatrix}.$$

Differentiating with respect to t by the chain rule and the product rule,

$$\begin{aligned}\dot{\mathbf{r}} &= a\dot{\theta}\begin{pmatrix}-\sin\theta\\ \cos\theta\end{pmatrix} - a\dot{\theta}\begin{pmatrix}-\sin\theta\\ \cos\theta\end{pmatrix} + (L - a\theta)\dot{\theta}\begin{pmatrix}-\cos\theta\\ -\sin\theta\end{pmatrix}\\ &= -(L - a\theta)\dot{\theta}\begin{pmatrix}\cos\theta\\ \sin\theta\end{pmatrix}.\end{aligned}$$

The first two terms cancel: moving the contact point along the post is exactly undone by the shortening of the free string.

Since $(\cos\theta, \sin\theta)$ is a unit vector, $L - a\theta > 0$, and $\dot{\theta} > 0$ while the string winds on, the speed is

$$|\dot{\mathbf{r}}| = (L - a\theta)\dot{\theta},$$

as required.

Time to hit the post

Ask yourself The string is pulling on the particle, so why should its speed stay equal to v?

The velocity is parallel to $(\cos\theta, \sin\theta)$, while the string, and hence the tension, lies along $(-\sin\theta, \cos\theta)$. These are perpendicular, so the tension does no work. It is the only horizontal force, as the table is smooth. So the kinetic energy, and with it the speed, stays constant at v. The tension only changes the direction of the velocity.

Therefore, for all t until the particle reaches the post,

$$(L - a\theta)\,\frac{\mathrm{d}\theta}{\mathrm{d}t} = v, \qquad \theta(0) = 0.$$

Separating variables and integrating from 0 to t,

$$L\theta - \tfrac{1}{2}a\theta^2 = vt.$$

The particle hits the post when the whole string is wound on, that is, when $a\theta = L$, so $\theta = L/a$. Then

$$vt = \frac{L^2}{a} - \frac{L^2}{2a} = \frac{L^2}{2a}, \qquad \text{so} \qquad t = \frac{L^2}{2av},$$

which is the required time.

Checks

- At $\theta = 0$ the formula gives $\mathbf{r} = (a, L)$ and $\dot{\mathbf{r}} = (-L\dot{\theta}, 0)$, perpendicular to the initial string, as it should be.
- Units: $L^2/(av)$ is $\mathrm{m}^2/(\mathrm{m}\cdot\mathrm{m\,s^{-1}}) = \mathrm{s}$.
- Another way to see the answer: the particle's whole path has length

  $$\int_0^{L/a} (L - a\theta)\,\mathrm{d}\theta = \frac{L^2}{2a},$$

  travelled at constant speed v.
- We simulated the motion numerically, with a very stiff string wrapping round a post with $a = 1$, $L = 3$, $v = 2$. The simulated time matched $L\theta - \frac{1}{2}a\theta^2 = vt$ to within 10^{-5}, and integrating $\dot{\theta} = v/(L - a\theta)$ to $a\theta = L$ gave 2.25, which is $L^2/(2av)$.

Common slips

- Using conservation of angular momentum about O. The tension does not pass through O (it acts along a tangent), so angular momentum about O is *not* conserved. Energy is the right tool.
- Writing the free string along $(\cos\theta, \sin\theta)$, i.e. radially, instead of along the tangent.
- Forgetting that the speed formula needs $\dot{\theta} > 0$. The absolute value of the vector is $(L - a\theta)|\dot{\theta}|$.

Beyond STEP

The path traced out by the particle is an *involute* of the circle, the curve traced by the end of a taut thread unwound from a spool. Involutes are why gear teeth are shaped as they are, and they were central to Huygens' work on pendulum clocks. Its arc length

$\int (L - a\theta)\, \mathrm{d}\theta$ is a first example of the differential geometry of plane curves (evolutes, involutes and curvature). For the mechanics side, including when energy and angular momentum are or are not conserved, read Kleppner and Kolenkow, *An Introduction to Mechanics*.

WORKED SOLUTION

STEP II 1988, Question 12

Mechanics • Falling Chains

STEP II 1988, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

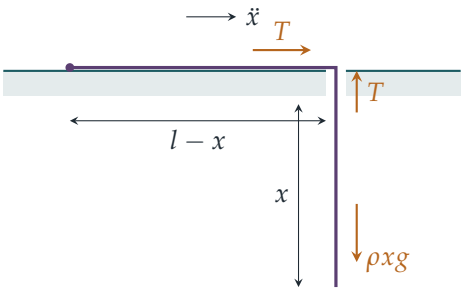
step.maths.org/questions/88-s2-q12

The key idea

Because the string is inextensible, every bit of it moves with the same speed, so the whole string behaves like one body whose driving force is the weight of the part that hangs below the table. That weight grows as more string goes through the hole, which gives a simple linear differential equation in the hanging length. Its solution is a hyperbolic cosine, and the time we want comes from inverting it.

Solution

There is no friction anywhere: the table is smooth, the hole is small and smooth (so it changes the string's direction without exerting any force along it), and the string has mass ρ per unit length. Let x be the length hanging below the table at time t, so that the length lying on the table is $l - x$. Initially $x = l - d$ and $\dot{x} = 0$, and the end reaches the top of the hole when $x = l$. Since the table is higher than l, the hanging part never touches the floor before then.



Let T be the tension in the string at the hole. As the hole is smooth, the tension has the same value T just above and just below it.

The part on the table. It has mass $\rho(l - x)$ and moves horizontally towards the hole with acceleration $\ddot{x}$ (inextensibility: it moves as fast as the hanging part). The only

horizontal force on it is the tension T, since the table is smooth. So

$$\rho(l-x)\,\ddot{x} = T.$$

The hanging part. It has mass ρx and accelerates downwards at $\ddot{x}$, under its weight $\rho x g$ downwards and the tension T upwards:

$$\rho x\,\ddot{x} = \rho x g - T.$$

Adding the two equations eliminates T:

$$\rho l\,\ddot{x} = \rho g x, \qquad \text{that is} \qquad \ddot{x} = n^2 x, \quad n = \sqrt{\frac{g}{l}}.$$

Ask yourself The amount of string on each side of the hole keeps changing. Is it really all right to use $F = ma$ for each piece?

Yes. Each equation is applied at one instant to the particular piece of string that is on that side of the hole at that instant; the string moves as a whole with a single speed $\dot{x}$, so the rate of change of momentum of the whole string is $\rho l\ddot{x}$, and the only external force with a component along the direction of motion is the weight $\rho x g$ of the hanging part. (The reaction of the table balances the weight of the horizontal part, and the hole only turns the string.) So the combined equation $\rho l\ddot{x} = \rho g x$ is just Newton's second law for the whole string. An energy check below confirms it.

The general solution of $\ddot{x} = n^2 x$ is $x = A\cosh nt + B\sinh nt$. From $\dot{x} = 0$ at $t = 0$ we get $B = 0$, and from $x = l - d$ at $t = 0$ we get $A = l - d$:

$$x = (l-d)\cosh nt.$$

The end reaches the top of the hole when $x = l$, so at the time t_1 with

$$\cosh nt_1 = \frac{l}{l-d}.$$

Since $l/(l-d) > 1$ this has exactly one positive solution, and using $\operatorname{arcosh} u = \ln\bigl(u + \sqrt{u^2-1}\bigr)$,

$$\sqrt{u^2-1} = \frac{\sqrt{l^2-(l-d)^2}}{l-d} = \frac{\sqrt{2ld-d^2}}{l-d},$$

so the time taken is

$$\boxed{t_1 = \sqrt{\frac{l}{g}}\,\operatorname{arcosh}\!\left(\frac{l}{l-d}\right) = \sqrt{\frac{l}{g}}\,\ln\left(\frac{l+\sqrt{2ld-d^2}}{l-d}\right).}$$

Checks

- **Energy.** Take the table top as the zero of potential energy. The hanging part has its centre of mass at depth $x/2$, so its potential energy is $-\frac{1}{2}\rho g x^2$. Conservation of

energy from rest gives $\frac{1}{2}\rho l\dot{x}^2 = \frac{1}{2}\rho g\left(x^2 - (l-d)^2\right)$, and differentiating with respect to t gives $l\ddot{x} = gx$ again.

- **Small d.** If d is tiny, almost all the string hangs and it should fall with acceleration close to g, taking time about $\sqrt{2d/g}$. Indeed, for small d, $\cosh nt_1 = 1 + \frac{d}{l} + \cdots$ and $\cosh nt_1 \approx 1 + \frac{1}{2}n^2t_1^2$ give $t_1^2 \approx 2d/(n^2l) = 2d/g$.
- **$d \to l$.** If almost nothing hangs, the string barely starts, and $t_1 \to \infty$, as the logarithm shows.
- **Numerically.** With $l = 2\,\text{m}$, $d = 1.5\,\text{m}$ and $g = 9.8\,\text{m}\,\text{s}^{-2}$ the formula gives $t_1 \approx 0.932\,\text{s}$, which matches a step-by-step numerical integration of the motion.

Common slips

- Writing $\rho x\ddot{x} = \rho xg$ for the hanging part alone, forgetting that it has to drag the string on the table (equivalently, using the wrong mass: the mass being accelerated is the whole string, ρl).
- Trying a solution of the form $x = A\cos nt$: the equation is $\ddot{x} = +n^2x$, so the motion grows exponentially rather than oscillating.
- Setting $x = d$ instead of $x = l$ for the finishing condition: the end reaches the hole when the whole length l is hanging.

Beyond STEP

This is a variable-mass-distribution problem: the mass is fixed, but the part feeling gravity changes with position, leading to a linear equation whose solutions are exponentials. If the string fell off a frictionless table edge with a pile of string being picked up from rest, momentum would be lost in inelastic "jerks" and energy would not be conserved; telling those cases apart is a classic theme in undergraduate mechanics. Kleppner and Kolenkow, *An Introduction to Mechanics*, has a careful treatment of systems with moving mass and of chains and ropes.

WORKED SOLUTION

STEP II 1988, Question 13

Mechanics • Friction and Equilibrium

STEP II 1988, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

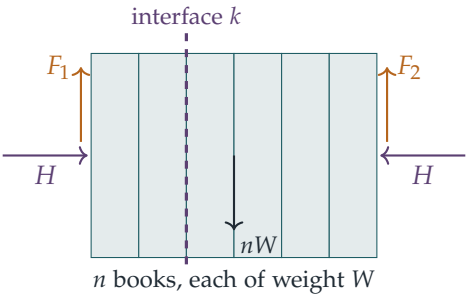
step.maths.org/questions/88-s2-q13

The key idea

Every vertical force that holds the books up is friction, and friction at any contact is capped at μ times the normal reaction there. So we need two things: the normal reaction at every contact (horizontal equilibrium makes this easy) and the friction each contact must supply (vertical equilibrium of a suitable *block* of books). Then we find which contact is closest to slipping. Treating a group of neighbouring books as one body keeps the algebra short.

Solution

Model. Suppose there are n identical books, each of weight W, with their covers vertical. Each hand presses horizontally with force $H \leq Q$ and supplies an upward frictional force; call these F_1 (left hand) and F_2 (right hand). Because each hand's vertical force is exactly this friction, we need $F_1, F_2 \leq P$. The question tells us to ignore turning effects, so we only need force balance on each book or block of books.



Normal reactions. Take the books numbered $1, 2, \ldots, n$ from left to right, and let interface k be the contact between book k and book $k+1$ ($1 \leq k \leq n-1$). The only horizontal forces on the block of books $1, \ldots, k$ are the left hand's push H and the

normal reaction from book $k+1$. So that reaction is also H: *every* contact, hand or book, has normal reaction H, and so every contact can supply friction of size at most μH.

The hands. Vertically, for the whole row,

$$F_1 + F_2 = nW.$$

Each of F_1, F_2 is at most P (arm strength) and at most $\mu H \le \mu Q$ (friction). Write $L = \min(P, \mu Q)$. Then $nW = F_1 + F_2 \le 2L$, so

$$n \le \frac{2L}{W} = \frac{2\min(P, \mu Q)}{W}.$$

This is necessary. If it holds, the best plan is to press as hard as possible, $H = Q$ (a larger H only raises every friction limit), and to share the load equally, $F_1 = F_2 = \frac{1}{2}nW \le L$. Then both hands are within their limits.

Ask yourself Could the books slip past one another somewhere in the middle, even though the hands hold?

The interfaces between books. We check this directly. Apply vertical equilibrium to the block of books $1, \ldots, k$. Its weight kW is held up by $F_1 = \frac{1}{2}nW$ from the hand and by a friction force f_k (upwards positive) from book $k+1$:

$$f_k = kW - \tfrac{1}{2}nW = \left(k - \tfrac{n}{2}\right) W.$$

For $1 \le k \le n-1$ we have $\left|k - \frac{n}{2}\right| \le \frac{n}{2} - 1$, with equality at $k = 1$ and $k = n-1$, the interfaces next to the outer books. So the largest friction any interface needs is $\left(\frac{n}{2} - 1\right)W$, which is W less than the $\frac{1}{2}nW$ each hand supplies. Since every interface has the same limit μQ as a hand, if the hands do not slip then no pair of books slips. (Physically: each outer book is held by its hand, and the friction it passes inwards is the hand's friction minus its own weight.)

Answer. The row can be lifted exactly when $nW \le 2\min(P, \mu Q)$. So the maximum number of books is

$$\boxed{n_{\max} = \left\lfloor \frac{2\min(P,\, \mu Q)}{W} \right\rfloor}$$

that is, the largest integer not exceeding both $2P/W$ and $2\mu Q/W$. Which bound applies tells us what limits the librarian: arm strength in lifting when $P < \mu Q$, and grip when $\mu Q < P$.

Checks

- Take $W = 10\,\text{N}$, $P = 60\,\text{N}$, $Q = 100\,\text{N}$ and $\mu = 0.4$. Then $\min(60, 40) = 40$ and $n_{\max} = 8$. With 8 books each hand gives $40\,\text{N}$ of friction, exactly its limit, and the worst interface needs $30\,\text{N} < 40\,\text{N}$. A ninth book would need $45\,\text{N}$ from each hand.

- With one book ($n = 1$) there are no interfaces, and we just need $W \le 2\min(P, \mu Q)$, which is clearly right.
- By computer we solved the full set of inequalities (any $H \le Q$, any split F_1, F_2, every interface) for 3000 random choices of W, P, Q, μ, and the largest feasible n always matched the formula.

Common slips

- Forgetting that the arm's vertical force *is* the friction, so both limits P and μQ apply to the same force.
- Assuming the middle books are the weakest point without checking. The forces between books are smaller than at the hands, not larger.
- Giving $2\min(P, \mu Q)/W$ without the floor: the answer must be a whole number of books.

Beyond STEP

Here friction is *statically indeterminate*: only $F_1 + F_2$ is fixed by equilibrium, and we asked whether *some* split keeps every contact within its limit. Questions like this are small linear programmes, and the same idea of checking whether some equilibrium is admissible underlies the limit analysis of structures. Kleppner and Kolenkow, *An Introduction to Mechanics*, is a good place to see friction and equilibrium treated carefully.

WORKED SOLUTION

STEP II 1988, Question 14

Mechanics • Rotating Rigid Bodies

STEP II 1988, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/88-s2-q14

The key idea

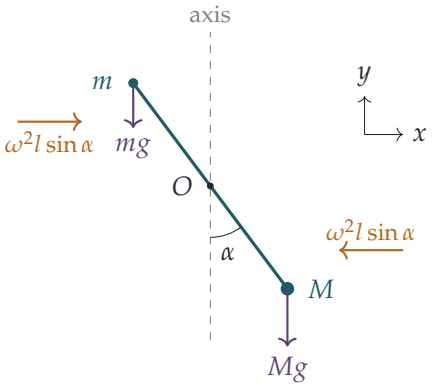
In the steady state nothing moves relative to the rotating vertical plane that contains the rod, so each particle simply goes round a horizontal circle at constant speed. That fixes each particle's acceleration completely. Newton's second law then tells us the force the rod exerts on each particle, and two facts about a *light* rod finish the job: the forces on it must balance, and (because it swings freely about the axle) their moments about the axle must balance too.

Solution

Model. Treat the masses as particles on a light rigid rod, pivoted smoothly at O so that the axle exerts no couple about its own direction. Work in the vertical plane containing the rod, which turns with angular speed ω about the vertical line through O. Take x horizontal in this plane and y vertically up, with origin O.

Suppose M is at $(l\sin\alpha,\ -l\cos\alpha)$ and m is at $(-l\sin\alpha,\ l\cos\alpha)$; we shall see in a moment that the heavier particle must be the lower one. Each particle moves on a horizontal circle of radius $l\sin\alpha$ about the vertical axis at constant speed, so its acceleration is $\omega^2 l\sin\alpha$ horizontally towards the axis:

$$\mathbf{a}_M = (-\omega^2 l\sin\alpha,\ 0), \qquad \mathbf{a}_m = (\omega^2 l\sin\alpha,\ 0).$$



(Amber arrows are accelerations, plum arrows are weights.)

Forces from the rod. Let $\mathbf{R}_M$ and $\mathbf{R}_m$ be the forces the rod exerts on M and m. These need not lie along the rod: a rigid rod can also push sideways. Newton's second law gives $\mathbf{R}_M - Mg\,\mathbf{j} = M\mathbf{a}_M$, so

$$\mathbf{R}_M = \big(-M\omega^2 l\sin\alpha,\ Mg\big), \qquad \mathbf{R}_m = \big(m\omega^2 l\sin\alpha,\ mg\big).$$

By Newton's third law the particles push on the rod with $-\mathbf{R}_M$ at M's end and $-\mathbf{R}_m$ at m's end.

> **Ask yourself** The axle exerts an unknown force on the rod at O. How can we take moments without knowing it?

Moments about the axle. The axle's force acts at O, so it has no moment about O; and the rod swings freely, so the axle supplies no couple about its own direction (the line through O perpendicular to our plane). A light rod has no angular momentum of its own, so the moments of the two end forces about this line must cancel. The moment of a force (F_x, F_y) acting at (x, y) is $xF_y - yF_x$ (anticlockwise). For $-\mathbf{R}_M$ at $(l\sin\alpha, -l\cos\alpha)$:

$$l\sin\alpha\,(-Mg) - (-l\cos\alpha)(M\omega^2 l\sin\alpha) = -Mgl\sin\alpha + M\omega^2 l^2\sin\alpha\cos\alpha,$$

and for $-\mathbf{R}_m$ at $(-l\sin\alpha,\ l\cos\alpha)$:

$$(-l\sin\alpha)(-mg) - l\cos\alpha\,(-m\omega^2 l\sin\alpha) = mgl\sin\alpha + m\omega^2 l^2\sin\alpha\cos\alpha.$$

Setting the sum to zero and dividing by $l\sin\alpha \neq 0$ (as $\alpha > 0$):

$$(M+m)\,\omega^2 l\cos\alpha = (M-m)\,g \quad\Longrightarrow\quad \omega^2 = \left(\frac{M-m}{M+m}\right)\frac{g}{l\cos\alpha},$$

dividing by $\cos\alpha > 0$, as required.

If instead M were the *upper* particle, the same calculation (swap M and m) would give $(M+m)\omega^2 l\cos\alpha = (m-M)g < 0$, which is impossible. So the heavier particle is indeed below O, as assumed.

The reaction on the axle. The rod is light, so the total force on it is zero: the axle pushes on it with $\mathbf{R}_M + \mathbf{R}_m$, and hence the rod pushes on the axle with

$$\mathbf{F} = -(\mathbf{R}_M + \mathbf{R}_m) = \Big((M-m)\,\omega^2 l \sin\alpha,\ -(M+m)\,g \Big).$$

This points downwards, with horizontal part directed outwards on M's side. If ϕ is its angle with the downward vertical, then, using the value of ω^2,

$$\begin{aligned} \tan\phi &= \frac{(M-m)\,\omega^2 l \sin\alpha}{(M+m)\,g} = \frac{M-m}{M+m} \cdot \frac{l\sin\alpha}{g} \cdot \frac{M-m}{M+m} \cdot \frac{g}{l\cos\alpha} \\ &= \left(\frac{M-m}{M+m}\right)^2 \tan\alpha. \end{aligned}$$

Since $0 \le \phi < \frac{1}{2}\pi$, $\phi = \tan^{-1}\left[\left(\frac{M-m}{M+m}\right)^2 \tan\alpha\right]$, which is the required result.

Checks

- If $m = 0$ the system is a conical pendulum of length l, and the formula gives the familiar $\omega^2 = g/(l\cos\alpha)$. The reaction angle is then α: the rod pulls along itself, like a string.
- If $M = m$ the rod is balanced at O, so $\omega = 0$ and the reaction is vertical, with size $2Mg$; both formulae agree.
- Numerically, with $M = 3$, $m = 1$, $l = 0.7$, $\alpha = 0.6$ and $g = 9.8$ we get $\omega^2 \approx 8.48$ and a reaction angle of about 9.71°. Stepping the equation of motion for the rod's angle from rest at α keeps it at α, and computing the particles' accelerations from their positions in three dimensions reproduces the same angle.

Common slips

- Assuming the rod's forces on the particles act along the rod. They don't: a rigid rod transmits sideways forces, and that is why we take moments.
- Forgetting that m's acceleration is towards the axis too, so on the *opposite* side from M's. The horizontal parts of the reaction partly cancel, giving $M - m$ rather than $M + m$.
- Putting the heavier particle on top. The torque equation shows this cannot happen.

Beyond STEP

The rod's angular momentum about O is not parallel to the vertical spin axis, so it turns as the system rotates. Changing it needs a couple, which the axle supplies in a direction perpendicular to the one we used. This is the start of rigid-body dynamics with the inertia tensor, and it explains why badly balanced wheels wobble on their bearings. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats this carefully.

WORKED SOLUTION

STEP II 1988, Question 15

Probability and Statistics • Sums of Discrete Uniforms

STEP II 1988, Question 15

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/88-s2-q15

The key idea

Each paper's mark is uniform on $\{0, 1, \ldots, m\}$, so every possible list of marks is equally likely, and a probability is just a count of lists divided by the total number of lists. The count for two papers comes from a simple listing, and the count for three papers comes from conditioning on the third mark and adding up two-paper counts (a *convolution*). Symmetry ($k \mapsto m - k$ on each paper) then gives the top range almost for free.

A word of warning before we start: the three-paper formula printed in the paper contains a misprint. We derive the correct one below and point out where the printed version differs.

Solution

Let X_1, X_2, X_3 be the marks on the papers. They are independent, since the papers are marked independently, and each is uniform on $\{0, 1, \ldots, m\}$. So each ordered pair (X_1, X_2) has probability $(m+1)^{-2}$, and each ordered triple has probability $(m+1)^{-3}$. Write $N_2(n)$ and $N_3(n)$ for the number of ordered pairs (triples) of marks in $\{0, \ldots, m\}$ with sum n.

Two papers

We need $X_1 = k$ and $X_2 = n - k$, with both in $[0, m]$, so

$$\max(0,\, n - m) \leq k \leq \min(m,\, n).$$

- If $0 \leq n \leq m$: k runs from 0 to n, so $N_2(n) = n + 1$.
- If $m < n \leq 2m$: k runs from $n - m$ to m, so $N_2(n) = m - (n - m) + 1 = 2m - n + 1$.

Hence

$$P(X_1+X_2=n)=\begin{cases}\dfrac{n+1}{(m+1)^2}, & 0\le n\le m,\\[2ex] \dfrac{2m-n+1}{(m+1)^2}, & m<n\le 2m,\end{cases}$$

which is the required result for $m<n\le 2m$, together with the result for $0\le n\le m$. The distribution is a "tent", rising by one each step up to $n=m$ and then falling.

Three papers

Condition on the third mark: if $X_3=k$ then we need $X_1+X_2=n-k$, so

$$N_3(n)=\sum_{k=0}^{m}N_2(n-k)=\sum_{j=n-m}^{n}N_2(j),$$

where $j=n-k$ and $N_2(j)=0$ when $j<0$ or $j>2m$.

Case $0\le n\le m$**.** Only $j=0,1,\dots,n$ contribute (negative j give zero), and all of them lie in the rising part of the tent:

$$N_3(n)=\sum_{j=0}^{n}(j+1)=\frac{(n+1)(n+2)}{2},\qquad P(X_1+X_2+X_3=n)=\frac{(n+1)(n+2)}{2(m+1)^3}.$$

Case $m<n\le 2m$**.** Now $1\le n-m\le m<n\le 2m$, so the range $j=n-m,\dots,n$ straddles the peak of the tent. We split it there:

$$N_3(n)=\sum_{j=n-m}^{m}(j+1)+\sum_{j=m+1}^{n}(2m-j+1).$$

The first sum adds the integers from $n-m+1$ to $m+1$:

$$\sum_{j=n-m}^{m}(j+1)=\frac{(m+1)(m+2)-(n-m)(n-m+1)}{2}.$$

In the second sum, put $i=2m-j+1$, which runs from $2m-n+1$ up to m:

$$\sum_{j=m+1}^{n}(2m-j+1)=\frac{m(m+1)-(2m-n)(2m-n+1)}{2}.$$

Expanding the four products,

$$\begin{aligned}2N_3(n)&=(m^2+3m+2)-(n^2-2mn+m^2+n-m)\\&\quad+(m^2+m)-(4m^2-4mn+n^2+2m-n)\\&=6mn-3m^2-2n^2+3m+2.\end{aligned}$$

Dividing by the $(m+1)^3$ equally likely triples,

$$P(X_1+X_2+X_3=n)=\frac{6mn-3m^2-2n^2+3m+2}{2(m+1)^3},\qquad m<n\le 2m.$$

Ask yourself This isn't the formula printed in the question. Who is right?

The printed version has $-4m^2$ in place of $-3m^2$ and $(m+1)^2$ in place of $(m+1)^3$; both are misprints. A single case settles it: with $m=1$ and $n=2$, three marks from $\{0,1\}$ add up to 2 in exactly 3 of the 8 equally likely ways, so the probability is $\frac{3}{8}$. Our formula gives $\frac{12-3-8+3+2}{16}=\frac{3}{8}$, while the printed one gives $\frac{12-4-8+3+2}{8}=\frac{5}{8}$. The denominator must be $(m+1)^3$ in any case, since that is the number of equally likely outcomes. As a further check, at $n=m$ our middle formula gives $\frac{(m+1)(m+2)}{2(m+1)^3}$, agreeing with the first case, as it should, since the two sums then join up.

Case $2m < n \le 3m$. Replace each mark X_i by $m - X_i$. This is again uniform on $\{0,\dots,m\}$, and the three new marks are still independent, so

$$\begin{aligned} P(X_1+X_2+X_3=n) &= P\big((m-X_1)+(m-X_2)+(m-X_3)=n\big) \\ &= P(X_1+X_2+X_3=3m-n). \end{aligned}$$

For $2m < n \le 3m$ we have $0 \le 3m-n < m$, so the first case applies with n replaced by $3m-n$:

$$P(X_1+X_2+X_3=n)=\frac{(3m-n+1)(3m-n+2)}{2(m+1)^3}, \qquad 2m<n\le 3m.$$

(The same symmetry, with $2m-n$ in place of $3m-n$, turns the two-paper result for $n \le m$ into the result for $n > m$.)

Checks

- Small case: for $m=1$ the three formulae give $\frac{1}{8},\frac{3}{8},\frac{3}{8},\frac{1}{8}$ for $n=0,1,2,3$, which add to 1 and match the binomial distribution of three fair coin tosses.
- The middle formula is symmetric under $n \mapsto 3m-n$: replacing n by $3m-n$ in $6mn-3m^2-2n^2$ leaves it unchanged, as the symmetry argument says it must.
- By complete enumeration for every m from 1 to 8, all five formulae agree exactly with the counts; for example, $m=10$ and $n=15$ give $\frac{91}{1331}$ both ways.

Common slips

- Forgetting that $N_2(j)$ has two different formulae, and using $j+1$ all the way through the sum in the middle range.
- Trying to reach the printed middle formula and "forcing" the algebra. If a careful calculation disagrees with a given answer, test a tiny case before doubting yourself.

Beyond STEP

The distribution of a sum of independent random variables is the *convolution* of their distributions, and that is exactly what our sum $\sum_k N_2(n-k)$ is. Probability generating functions make this mechanical: the count $N_3(n)$ is the coefficient of t^n in $(1+t+\cdots+t^m)^3$. With more and more papers, the shape goes from a tent to a smooth bump and approaches the normal curve (the central limit theorem). Grimmett and Welsh, *Probability: An Introduction*, covers generating functions and the central limit theorem at the right level.

WORKED SOLUTION

STEP II 1988, Question 16

Probability and Statistics • Normal Distribution

STEP II 1988, Question 16

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/88-s2-q16

The key idea

Everything about the roots is controlled by the single random variable B, so each question about X_1 and X_2 should be turned into a question about which interval B lies in. The discriminant tells us when the roots are real, and Vieta's formulas (sum $-2B$, product 1) make the conditions in (i) and (ii) easy to translate. After that, each answer is a conditional probability or conditional expectation for a standard normal variable, and the expectation needs one short integral.

Solution

Throughout, $B \sim \mathrm{N}(0,1)$, with density $\phi(z) = \frac{1}{\sqrt{2\pi}}\mathrm{e}^{-z^2/2}$ and distribution function Φ. Let R be the event that the roots are real.

Real roots

The discriminant of $X^2 + 2BX + 1$ is $4B^2 - 4$, so the roots are real exactly when $B^2 \geq 1$, that is, $|B| \geq 1$. By symmetry of the normal distribution,

$$\mathrm{P}(R) = \mathrm{P}(|B| \geq 1) = 2\big(1 - \Phi(1)\big) = 2(1 - 0.84134) = 0.31731\ldots \approx 0.317.$$

(Whether we include the equal-root case $|B| = 1$ makes no difference, since it has probability zero.)

When the roots are real they are

$$X_{1,2} = -B \pm \sqrt{B^2 - 1}, \qquad X_1 + X_2 = -2B, \qquad X_1 X_2 = 1.$$

Part (i)

Ask yourself How do we turn "both roots exceed $\frac{1}{5}$" into a condition on B?

First, both roots must be positive. Their product is $1 > 0$, so they have the same sign, and that sign is the sign of their sum $-2B$. So we need $B < 0$, and with R this means $B \leq -1$. Write $b = -B \geq 1$; the roots are then $b \pm \sqrt{b^2 - 1}$, both positive.

Since the product is 1, the smaller root is the reciprocal of the larger one:

$$b - \sqrt{b^2 - 1} = \frac{1}{b + \sqrt{b^2 - 1}}.$$

So the smaller root exceeds $\frac{1}{5}$ if and only if the larger root is less than 5, and then the larger root automatically exceeds $\frac{1}{5}$ as well. The condition is therefore

$$\sqrt{b^2 - 1} < 5 - b.$$

This needs $b < 5$, and for $1 \leq b < 5$ both sides are non-negative, so we may square without changing the inequality:

$$b^2 - 1 < 25 - 10b + b^2 \iff 10b < 26 \iff b < 2.6.$$

(Values $b \geq 5$ fail anyway, as $2.6 < 5$.) Hence, given real roots, both roots exceed $\frac{1}{5}$ exactly when $-2.6 < B \leq -1$, and

$$\begin{aligned} \mathrm{P}\left(X_1, X_2 > \tfrac{1}{5} \,\middle|\, R\right) &= \frac{\mathrm{P}(-2.6 < B \leq -1)}{\mathrm{P}(|B| \geq 1)} = \frac{\Phi(2.6) - \Phi(1)}{2\left(1 - \Phi(1)\right)} \\ &= \frac{0.99534 - 0.84134}{0.31731} = 0.4853\ldots \approx 0.485. \end{aligned}$$

Part (ii)

Since $|X_1 + X_2| = 2|B|$, we want $\mathrm{E}\left(2|B| \,\middle|\, |B| \geq 1\right)$. For a conditional expectation given an event, we integrate over the event and divide by its probability:

$$\mathrm{E}\left(|B| \,\middle|\, |B| \geq 1\right) = \frac{1}{\mathrm{P}(|B| \geq 1)} \int_{|z| \geq 1} |z|\, \phi(z)\, \mathrm{d}z = \frac{2\int_1^\infty z\, \phi(z)\, \mathrm{d}z}{2\left(1 - \Phi(1)\right)},$$

using the symmetry of ϕ. The integral is easy because $z\,\phi(z)$ is minus the derivative of $\phi(z)$:

$$\int_1^\infty z\, \phi(z)\, \mathrm{d}z = \Big[-\phi(z)\Big]_1^\infty = \phi(1) = \frac{\mathrm{e}^{-1/2}}{\sqrt{2\pi}} = 0.24197\ldots$$

Therefore

$$\mathrm{E}\left(|X_1 + X_2| \,\middle|\, R\right) = \frac{2\phi(1)}{1 - \Phi(1)} = \frac{2 \times 0.24197}{0.15866} = 3.050\ldots \approx 3.05.$$

Checks

- Try $B = -2.6$: the equation is $X^2 - 5.2X + 1 = 0$, with roots 5 and $\frac{1}{5}$ exactly, the boundary case in (i).
- The answer to (ii) must exceed 2, because $2|B| \geq 2$ on R. It does, and it is plausible: the normal tail beyond 1 is thin, so $|B|$ is usually only a little above 1 (its conditional mean is about 1.53).
- A simulation of 2×10^6 standard normal values of B gave 0.3177, 0.4855 and 3.049, agreeing with 0.317, 0.485 and 3.05.

Common slips

- Forgetting to condition: in (i) and (ii) we must divide by $\mathrm{P}(R) = 0.317$. The unconditional probability $\Phi(2.6) - \Phi(1) = 0.154$ is not the answer.
- Including $B \geq 1$ in (i). Both roots are then negative, so neither exceeds $\frac{1}{5}$.
- Computing $\mathrm{E}|X_1 + X_2|$ as $|\mathrm{E}(X_1 + X_2)|$, which is 0 by symmetry.

Beyond STEP

The quantity $\phi(1)/(1 - \Phi(1))$ is the mean of a *truncated* normal distribution, and the ratio $\phi(a)/(1 - \Phi(a))$ is known as the inverse Mills ratio; it appears in statistics (for example, in models of censored data) and in reliability theory as the hazard rate of the normal distribution. Questions about random polynomials, such as how many real roots a polynomial with random coefficients has on average, form a surprisingly rich area of probability. For conditional expectation and continuous distributions, see Grimmett and Welsh, *Probability: An Introduction*, or Blitzstein and Hwang, *Introduction to Probability*.

WORKED SOLUTION

STEP III 1988, Question 1

Pure • Curve Sketching and Bounds

STEP III 1988, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/88-s3-q1

The key idea

For the sketch, the factor x^2e^{-x} is never negative, so the sign of y is decided entirely by $1+x$. That, together with one derivative and the behaviour at the ends and near $x=-1$, gives the whole shape. For the integral, the cleanest move is to divide x^2 by $1+x$: most of the integrand then integrates to nothing, and what is left is easy to trap between two simple bounds.

Solution

The graph

Write $f(x)=\dfrac{x^2e^{-x}}{1+x}$, defined for all $x\neq -1$.

Sign and zeros. Since $x^2\geq 0$ and $e^{-x}>0$, we have $f(x)<0$ for $x<-1$ and $f(x)\geq 0$ for $x>-1$, with $f(x)=0$ only at $x=0$. The factor x^2 means the curve touches the x-axis at the origin rather than crossing it.

Asymptotes and ends.

- As $x\to -1$ the numerator tends to $e>0$ and the denominator to 0, so $x=-1$ is a vertical asymptote, with $f\to -\infty$ as $x\to -1^-$ and $f\to +\infty$ as $x\to -1^+$.
- As $x\to +\infty$, $f(x)\approx xe^{-x}\to 0^+$, because exponentials beat powers. So $y=0$ is an asymptote on the right, approached from above.
- As $x\to -\infty$, $\dfrac{x^2}{1+x}\approx x\to -\infty$ and $e^{-x}\to +\infty$, so $f(x)\to -\infty$, very fast.

Stationary points. By the quotient rule,

$$f'(x) = \frac{(2x - x^2)\mathrm{e}^{-x}(1+x) - x^2\mathrm{e}^{-x}}{(1+x)^2} = \frac{\mathrm{e}^{-x}\left(2x + x^2 - x^3 - x^2\right)}{(1+x)^2}$$
$$= \frac{x\left(2 - x^2\right)\mathrm{e}^{-x}}{(1+x)^2}.$$

So $f'(x) = 0$ at $x = 0$ and $x = \pm\sqrt{2}$. The factors e^{-x} and $(1+x)^2$ are positive, so the sign of f' is the sign of $x(2 - x^2)$:

interval	$x < -\sqrt{2}$	$-\sqrt{2} < x < -1$	$-1 < x < 0$	$0 < x < \sqrt{2}$	$x > \sqrt{2}$
$f'(x)$	$+$	$-$	$-$	$+$	$-$

This gives:

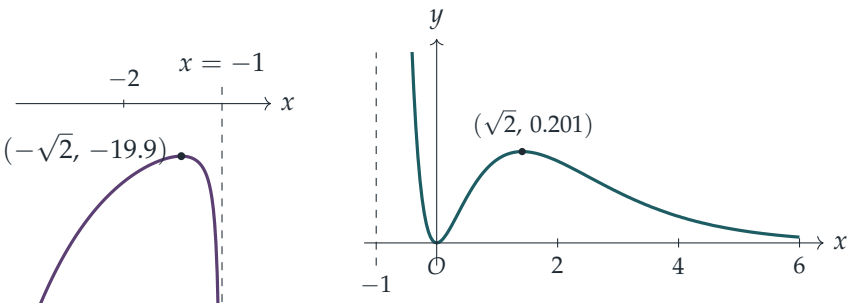
- a local maximum at $x = \sqrt{2}$, where $y = \dfrac{2\mathrm{e}^{-\sqrt{2}}}{1+\sqrt{2}} = 2(\sqrt{2}-1)\mathrm{e}^{-\sqrt{2}} \approx 0.201$;
- a local minimum at the origin;
- a local maximum at $x = -\sqrt{2}$ on the left-hand branch, where $y = \dfrac{2\mathrm{e}^{\sqrt{2}}}{1-\sqrt{2}} = -2(\sqrt{2}+1)\mathrm{e}^{\sqrt{2}} \approx -19.9$.

Ask yourself How can a branch lying entirely below the axis have a maximum?

On $x < -1$ the curve comes up from $-\infty$ as $x \to -\infty$ and goes back down to $-\infty$ at the asymptote $x = -1$, so it must turn somewhere in between. The highest point is at $x = -\sqrt{2}$, and it is still well below the axis. The branch is an upside-down "hump" hanging beneath the axis, with height about -19.9.

On the right, the curve falls from $+\infty$ just to the right of $x = -1$ down to 0 at the origin, rises to its small maximum at $x = \sqrt{2}$, then decays to 0. It is concave up near the origin and concave down at the maximum, so there are points of inflection on either side of $x = \sqrt{2}$ (numerically near $x \approx 0.35$ and $x \approx 2.5$). We don't need their exact positions for a sketch.

The two branches have very different vertical scales, so we draw them separately. On the left, y runs down to about -70. On the right, the y-scale is stretched so that the maximum of 0.2 is visible.



The integral

Let $I = \displaystyle\int_0^\infty \frac{x^2 e^{-x}}{1+x}\,dx$. Dividing x^2 by $1+x$,

$$\frac{x^2}{1+x} = x - 1 + \frac{1}{1+x}.$$

Now $\displaystyle\int_0^\infty x e^{-x}\,dx = 1$ (integrating by parts with $u = x$) and $\displaystyle\int_0^\infty e^{-x}\,dx = 1$, so the first two terms contribute $1 - 1 = 0$, and

$$I = \int_0^\infty \frac{e^{-x}}{1+x}\,dx.$$

(All three integrals converge, so splitting the integral up is legitimate.) For $x > 0$ we have $0 < \dfrac{1}{1+x} < 1$, so

$$0 < \frac{e^{-x}}{1+x} < e^{-x} \qquad (x > 0).$$

The inequalities are strict on the whole interval, and the functions are continuous, so integrating keeps them strict:

$$0 < I < \int_0^\infty e^{-x}\,dx = 1.$$

So I lies between 0 and 1, as required.

Another way. We can skip the division. For $x > 0$, $\dfrac{x^2}{1+x} < \dfrac{x^2}{x} = x$, so $0 < f(x) < x e^{-x}$ and $0 < I < \int_0^\infty x e^{-x}\,dx = 1$. The division is worth knowing anyway, because it shows exactly where I comes from.

Checks

- Differentiating f numerically at a thousand random points agrees with $f'(x) = x(2 - x^2)e^{-x}/(1+x)^2$.
- Simpson's rule on $[0, 60]$ gives $I \approx 0.59635$, and $\int_0^\infty \frac{e^{-x}}{1+x}\,dx$ gives the same value, comfortably between 0 and 1.

- A rough estimate from the sketch: on $x > 0$ the curve is a hump of height about 0.2 that has mostly died away by $x = 6$. A triangle of that height and base has area about 0.6, consistent with 0.596.

Common slips

- Missing the stationary point at $x = -\sqrt{2}$ because it lies on the "other" branch, or calling it a minimum because y is negative there.
- Saying the curve crosses the axis at the origin. The double factor x^2 makes it touch.
- Claiming $I < 1$ from $\mathrm{e}^{-x} \leq 1$ alone. That bound gives $\int_0^\infty \frac{x^2}{1+x}\,\mathrm{d}x$, which diverges.

Beyond STEP

The value of I is $\mathrm{e}\,E_1(1) \approx 0.5963$, where $E_1(t) = \int_t^\infty \mathrm{e}^{-u}u^{-1}\,\mathrm{d}u$ is the exponential integral. This number is known as the Gompertz constant. It cannot be written in terms of elementary functions, which is why the question only asks for bounds. Estimating integrals you cannot evaluate, by comparing them with ones you can, is a central skill in analysis (the "comparison test" for improper integrals). Abbott's *Understanding Analysis* develops it carefully.

WORKED SOLUTION

STEP III 1988, Question 2

Pure • Linear Recurrence Relations

STEP III 1988, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/88-s3-q2

The key idea

A linear recurrence with constant coefficients behaves like a quadratic: powers x^n solve it exactly when x is a root of the auxiliary equation. Since the recurrence fixes every term once the first two are known, matching u_0 and u_1 is all it takes to get the only solution. The last part has variable coefficients, but the factors $(n+1)$, $(n+2)$, $(n+3)$ suggest dividing through by their product, and then a simple substitution turns it into a constant-coefficient recurrence of the first kind.

Solution

(The paper prints the leading coefficient of the recurrence as α. This is clearly a misprint for a, matching the quadratic ax^2+bx+c, and we read it that way.) Since the quadratic has two roots, it really is a quadratic, so $a \neq 0$.

The formula satisfies the recurrence

Put $u_n = A\alpha^n + B\beta^n$ for any constants A, B. Then

$$\begin{aligned} au_{n+2} + bu_{n+1} + cu_n &= A\alpha^n\left(a\alpha^2 + b\alpha + c\right) + B\beta^n\left(a\beta^2 + b\beta + c\right) \\ &= A\alpha^n \cdot 0 + B\beta^n \cdot 0 = 0, \end{aligned}$$

because α and β are roots of $ax^2+bx+c=0$. So the recurrence holds for every $n \geq 0$.

With the particular A and B given, the formula also reproduces the starting values. For $n=0$,

$$A + B = \frac{(u_1 - \beta u_0) - (u_1 - \alpha u_0)}{\alpha - \beta} = \frac{(\alpha - \beta)u_0}{\alpha - \beta} = u_0,$$

and for $n=1$,

$$A\alpha + B\beta = \frac{\alpha u_1 - \alpha\beta u_0 - \beta u_1 + \alpha\beta u_0}{\alpha - \beta} = \frac{(\alpha - \beta)u_1}{\alpha - \beta} = u_1.$$

Dividing by $\alpha - \beta$ is allowed because the roots are distinct.

It is the only solution

Ask yourself What exactly are we inducting on, and why do we need two terms at a time?

Suppose $u_0, u_1, u_2, \ldots$ is any sequence satisfying the recurrence, and let A, B be built from its u_0 and u_1 as above. Write $w_n = A\alpha^n + B\beta^n$. We prove by induction on n the statement

$$P(n): \quad u_n = w_n \text{ and } u_{n+1} = w_{n+1}.$$

Each new term depends on the two before it, so the statement has to carry two consecutive terms.

Base case. $P(0)$ says $u_0 = w_0$ and $u_1 = w_1$, which is what we checked above.

Inductive step. Assume $P(n)$. Both (u_n) and (w_n) satisfy the recurrence, and $a \neq 0$, so

$$u_{n+2} = -\frac{bu_{n+1} + cu_n}{a} = -\frac{bw_{n+1} + cw_n}{a} = w_{n+2}.$$

Together with $u_{n+1} = w_{n+1}$ this is $P(n+1)$.

So $P(n)$ holds for all $n \geq 0$, and $u_n = A\alpha^n + B\beta^n$ for every n. Any solution of the recurrence is of this form, with A and B fixed by u_0 and u_1. The formula gives the only solution.

The sequence v_n

Every coefficient is a product of two of the three factors $(n+1)$, $(n+2)$, $(n+3)$, each missing a different one. Dividing by $(n+1)(n+2)(n+3)$, which is never zero for $n \geq 0$, gives

$$8\,\frac{v_{n+2}}{n+3} - 2\,\frac{v_{n+1}}{n+2} - \frac{v_n}{n+1} = 0.$$

Each fraction has the form $v_k/(k+1)$. So put

$$w_n = \frac{v_n}{n+1},$$

and the recurrence becomes

$$8w_{n+2} - 2w_{n+1} - w_n = 0, \qquad w_0 = 0, \quad w_1 = \tfrac{1}{2}.$$

The auxiliary equation is $8x^2 - 2x - 1 = 0$, that is, $(2x-1)(4x+1) = 0$. Its roots $\alpha = \frac{1}{2}$ and $\beta = -\frac{1}{4}$ are real and distinct, so the first part applies, with $\alpha - \beta = \frac{3}{4}$:

$$A = \frac{w_1 - \beta w_0}{\alpha - \beta} = \frac{1/2}{3/4} = \frac{2}{3}, \qquad B = \frac{w_1 - \alpha w_0}{\beta - \alpha} = \frac{1/2}{-3/4} = -\frac{2}{3}.$$

Hence $w_n = \frac{2}{3}\left(\left(\frac{1}{2}\right)^n - \left(-\frac{1}{4}\right)^n\right)$. The first part also tells us this is the only possibility for w_n, and so for v_n. Multiplying back by $n+1$,

$$\boxed{v_n = \frac{2(n+1)}{3}\left(\frac{1}{2^n} - \left(-\frac{1}{4}\right)^n\right)} \qquad (n = 0, 1, 2, \ldots).$$

Checks

The formula gives $v_0 = \frac{2}{3}(1-1) = 0$ and $v_1 = \frac{4}{3}\left(\frac{1}{2}+\frac{1}{4}\right) = 1$, as required. Running the original recurrence by hand with $n = 0$ and $n = 1$ gives $16v_2 = 6v_1 + 6v_0$, so $v_2 = \frac{3}{8}$, and then $48v_3 = 16v_2 + 12v_1$, so $v_3 = \frac{3}{8}$. The formula agrees: $v_2 = 2\left(\frac{1}{4}-\frac{1}{16}\right) = \frac{3}{8}$ and $v_3 = \frac{8}{3}\left(\frac{1}{8}+\frac{1}{64}\right) = \frac{3}{8}$. We checked the formula exactly against the recurrence, in rational arithmetic, for n up to 41, and the first part on hundreds of random quadratics and starting values.

Common slips

- Doing the induction on one term at a time. Knowing only $u_n = w_n$ doesn't let you deduce anything about u_{n+1}: the hypothesis must cover two consecutive terms.
- Forgetting to say why we may divide by a (it is non-zero because the equation has two roots) and by $\alpha - \beta$ (the roots are distinct).
- Applying the first part to v_n directly. Its coefficients depend on n, so it is the rescaled sequence $w_n = v_n/(n+1)$ that the first part applies to.

Beyond STEP

The solutions of a second-order linear recurrence form a two-dimensional vector space, and α^n, β^n are a basis for it. This is exactly parallel to the differential equation $ay'' + by' + cy = 0$ with solutions $e^{\alpha x}$ and $e^{\beta x}$. Writing the recurrence as $\mathbf{u}_{n+1} = M\mathbf{u}_n$ for a 2×2 matrix M shows that α and β are the eigenvalues of M, and repeated roots correspond to a matrix that can't be diagonalised. Strang's *Introduction to Linear Algebra* treats difference equations through eigenvalues in just this way, and Boyce and DiPrima's *Elementary Differential Equations* develops the parallel theory for differential equations.

WORKED SOLUTION

STEP III 1988, Question 3

Pure • Complex Loci

STEP III 1988, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/88-s3-q3

The key idea

The map $w = (z+\mathrm{i})/(z-\mathrm{i})$ can be turned round: solving for z gives $z = \mathrm{i}(w+1)/(w-1)$, so any condition on z becomes a condition on w. For the circle, the parametrisation does the work instead, because $z-\mathrm{i}$ is then simply $2\mathrm{e}^{\mathrm{i}\theta}$. For the real and imaginary cases, think of moduli and arguments: a real z is equidistant from i and $-\mathrm{i}$.

Solution

The parametric form

The curve $|z-\mathrm{i}| = 2$ is the circle with centre i and radius 2, so

$$z = \mathrm{i} + 2\mathrm{e}^{\mathrm{i}\theta}, \qquad -\pi < \theta \leq \pi,$$

or, with $z = x + \mathrm{i}y$, $x = 2\cos\theta$ and $y = 1 + 2\sin\theta$.

We'll also use the inverse map. From $w(z-\mathrm{i}) = z+\mathrm{i}$ we get $z(w-1) = \mathrm{i}(w+1)$, so for $w \neq 1$

$$z = \frac{\mathrm{i}(w+1)}{w-1}.$$

No z gives $w = 1$, since that would need $z+\mathrm{i} = z-\mathrm{i}$. Every $w \neq 1$ comes from exactly one $z \neq \mathrm{i}$.

Part (i)

Substituting the parametric form, $z+\mathrm{i} = 2\mathrm{i} + 2\mathrm{e}^{\mathrm{i}\theta}$ and $z-\mathrm{i} = 2\mathrm{e}^{\mathrm{i}\theta}$, so

$$w = \frac{2\mathrm{i} + 2\mathrm{e}^{\mathrm{i}\theta}}{2\mathrm{e}^{\mathrm{i}\theta}} = 1 + \mathrm{i}\,\mathrm{e}^{-\mathrm{i}\theta} = 1 + \mathrm{e}^{\mathrm{i}(\pi/2-\theta)}.$$

As θ runs once round, $\pi/2 - \theta$ also takes every angle once. So w traces the *whole* circle

$$|w-1| = 1,$$

with centre 1 and radius 1. For example, $z = 3\mathrm{i}$ gives $w = 2$, and $z = -\mathrm{i}$ gives $w = 0$. The point $w = 1$ isn't on this circle, which is consistent with the fact that it's never attained.

Part (ii)

If $z = x$ is real, it's the same distance from i as from $-\mathrm{i}$, so $|x + \mathrm{i}| = |x - \mathrm{i}|$ and

$$|w| = \frac{|x + \mathrm{i}|}{|x - \mathrm{i}|} = 1.$$

Explicitly, multiplying top and bottom by $x + \mathrm{i}$,

$$w = \frac{(x + \mathrm{i})^2}{x^2 + 1} = \frac{x^2 - 1}{x^2 + 1} + \mathrm{i}\,\frac{2x}{x^2 + 1}.$$

Ask yourself Is every point of the unit circle reached?

No, not quite. Take any w with $|w| = 1$ and $w \neq 1$, and write $w = \mathrm{e}^{\mathrm{i}\phi}$. Then

$$z = \mathrm{i}\,\frac{\mathrm{e}^{\mathrm{i}\phi} + 1}{\mathrm{e}^{\mathrm{i}\phi} - 1} = \mathrm{i}\,\frac{2\cos(\phi/2)}{2\mathrm{i}\sin(\phi/2)} = \cot\tfrac{\phi}{2},$$

which is real. So every such point is attained, but $w = 1$ never is: it's the limit as $x \to \pm\infty$. The locus is the unit circle $|w| = 1$ with the single point $w = 1$ removed. For instance, $x = 0$ gives $w = -1$, and $x = 1$ gives $w = \mathrm{i}$.

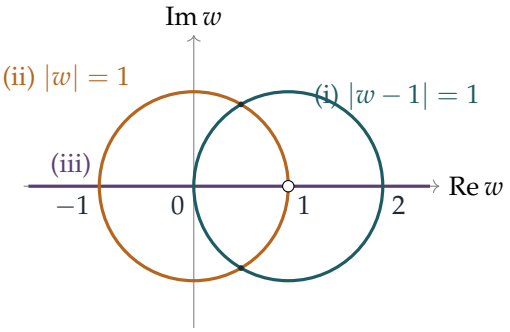
Part (iii)

If $z = \mathrm{i}y$ with y real and $y \neq 1$, then

$$w = \frac{\mathrm{i}(y + 1)}{\mathrm{i}(y - 1)} = \frac{y + 1}{y - 1} = 1 + \frac{2}{y - 1},$$

which is real. As y ranges over the reals other than 1, the fraction $2/(y - 1)$ takes every non-zero real value, so w takes every real value except 1. (Conversely, if $w = u$ is real with $u \neq 1$, then $z = \mathrm{i}(u + 1)/(u - 1)$ is imaginary.) The locus is the real axis with the point $w = 1$ removed. Here $z = 0$ gives $w = -1$; if you read "imaginary" as excluding $z = 0$, then $w = -1$ is removed as well.

Sketch



The teal circle is the locus for (i), the amber circle (minus the point 1) is the locus for (ii), and the purple axis (minus the point 1) is the locus for (iii). The open dot marks $w = 1$, which is never attained.

Checks

- The circle in (i) meets the real axis at $z = \pm\sqrt{3}$. These should map to points on both the teal and amber circles, and indeed $w(\sqrt{3}) = (\sqrt{3}+\mathrm{i})^2/4 = \frac{1}{2} + \frac{\sqrt{3}}{2}\mathrm{i} \approx 0.5 + 0.866\mathrm{i}$, which lies on $|w| = 1$ and on $|w - 1| = 1$.
- The circle in (i) meets the imaginary axis at $3\mathrm{i}$ and $-\mathrm{i}$, which map to 2 and 0: exactly where the teal circle crosses the real axis.
- A numerical check of 10^4 random points on each locus confirmed $|w - 1| = 1$, $|w| = 1$ and $\operatorname{Im} w = 0$, and showed that the inverse map recovers z.

Common slips

- Forgetting that $w = 1$ is missing in (ii) and (iii). It corresponds to "$z = \infty$".
- Expanding into x and y in (i) and getting lost in the algebra. The parametrisation makes $z - \mathrm{i}$ a single exponential, and the rest is one line.

Beyond STEP

The map $z \mapsto (z+\mathrm{i})/(z-\mathrm{i})$ is a Möbius transformation. Such maps always send circles and lines to circles and lines, which is exactly what you found in all three parts. They are best understood on the Riemann sphere, where adding a "point at infinity" fills the gap at $w = 1$. Its relative, the Cayley transform $z \mapsto (z-\mathrm{i})/(z+\mathrm{i})$, maps the upper half-plane onto the unit disc. Needham's *Visual Complex Analysis* has a beautiful geometric treatment of all of this.

WORKED SOLUTION

STEP III 1988, Question 4

Pure • Arc Length Optimisation

STEP III 1988, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/88-s3-q4

The key idea

Write the total length of the road as a function $L(b)$ and differentiate. The two pieces in the vertical plane must run along the surface, so their length is an arc length of the parabola; the semicircle has length πb. Differentiating an arc-length integral with respect to its lower limit needs no integration at all, and the sign of $L'(b)$ settles everything.

(The printed formula for the cross-section has $x =$ on the left; it should of course read $z =$.)

Solution

Setting up the length

Let $h(b) = (a^2 - b^2)/(2a)$, the height of the hill at distance b from the axis. The road must lie on the ground, so a path in the (x, z) plane from $(-R, 0)$ to $(-b, h(b))$ has no choice: it runs along the flat plane from $x = -R$ to $x = -a$, and then up the parabola from $x = -a$ to $x = -b$. Segment (iii) is its mirror image.

On the hill $z = (a^2 - x^2)/(2a)$, so $\dfrac{\mathrm{d}z}{\mathrm{d}x} = -\dfrac{x}{a}$, and the arc length from $x = b$ to $x = a$ is

$$\int_b^a \sqrt{1 + \frac{x^2}{a^2}}\,\mathrm{d}x.$$

The semicircle (ii) is horizontal, at height $h(b)$, with radius b (it is the circle of latitude of the hill through $(\pm b, h(b))$), so it has length πb. Hence

$$L(b) = 2(R - a) + 2\int_b^a \sqrt{1 + \frac{x^2}{a^2}}\,\mathrm{d}x + \pi b, \qquad 0 \le b \le a.$$

This also covers $b = 0$ (no semicircle; the road goes straight over the summit) and $b = a$ (the road runs round the foot of the hill).

Minimising

By the fundamental theorem of calculus, differentiating with respect to the lower limit gives minus the integrand there:

$$L'(b) = \pi - 2\sqrt{1 + \frac{b^2}{a^2}}.$$

Ask yourself Where would the stationary point be, and is it allowed?

$L'(b) = 0$ would need $1 + b^2/a^2 = \pi^2/4$, that is $b = \frac{a}{2}\sqrt{\pi^2 - 4} \approx 1.21a$. That is larger than a, so it lies outside the allowed range $0 \le b \le a$. In fact, for $0 \le b \le a$ we have

$$2\sqrt{1 + \frac{b^2}{a^2}} \le 2\sqrt{2} \approx 2.83 < \pi,$$

so $L'(b) > 0$ throughout the interval. L is strictly increasing, and its minimum is at the left end-point, $b = 0$.

Advice: take $b = 0$. The road should go straight up one side of the hill and down the other, with no semicircular detour.

How much it saves

For interest, the antiderivative is

$$\int \sqrt{1 + \frac{x^2}{a^2}}\,\mathrm{d}x = \frac{x}{2}\sqrt{1 + \frac{x^2}{a^2}} + \frac{a}{2}\sinh^{-1}\frac{x}{a} + C,$$

(substitute $x = a\sinh t$), so

$$L(0) = 2(R - a) + a\big(\sqrt{2} + \ln(1 + \sqrt{2})\big) \approx 2(R - a) + 2.296a,$$

while going round the foot gives $L(a) = 2(R - a) + \pi a \approx 2(R - a) + 3.142a$. Going over the top saves about $0.85a$.

Checks

- The answer does not depend on R, as it shouldn't: the flat stretches have total length $2(R - a)$ whatever b is.
- A rough sanity check: the hill is only $a/2$ high and $2a$ wide, so going over it cannot add much to the straight distance $2a$; the semicircle round the foot is πa. Indeed $2.296a$ lies between $2a$ and πa.
- Numerically, with $a = 1$ and $R = 3$, evaluating $L(b)$ by quadrature at 201 points of $[0, 1]$ gives the smallest value at $b = 0$; and $L'(a) = \pi - 2\sqrt{2} \approx 0.313$ is still positive.

Common slips

- Setting $L'(b) = 0$ and quoting $b = \frac{a}{2}\sqrt{\pi^2 - 4}$ without noticing that it is bigger than a.
- Using straight-line (chord) lengths for segments (i) and (iii): those would tunnel through the hill.
- Forgetting that a minimum on a closed interval can occur at an end-point.

Beyond STEP

Looking for the shortest path on a surface is the problem of *geodesics*, met in a first course on the calculus of variations or differential geometry. On a surface of revolution, Clairaut's relation describes the geodesics neatly; the king's family of roads is a small, one-parameter slice of that problem. The two analysis facts used here, differentiating an integral with respect to a limit and the fact that a continuous function on a closed interval attains its minimum, are proved carefully in Abbott, *Understanding Analysis*.

WORKED SOLUTION

STEP III 1988, Question 5

Pure • Recurrence Relations and Loans

STEP III 1988, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/88-s3-q5

The key idea

Write down what happens to the debt in one day. Each day the debt is multiplied by $1+k$ (interest), and then a fixed amount is either added (digging cost) or taken away (earnings). A recurrence of the form $S_n = (1+k)S_{n-1} + \text{constant}$ unrolls into a geometric series. The final part then comes down to one comparison: is the daily income bigger than the interest on the debt at the moment the tunnel opens?

Solution

The debt while the tunnel is being dug

An interest rate of $100k\%$ means the daily interest on a debt of £S is £kS. Let S_n be the debt (in pounds) at the end of day n, with $S_0 = 0$. At the end of day $n \le T$ the firm borrows c for the digging and kS_{n-1} for the interest, so

$$S_n = S_{n-1} + kS_{n-1} + c = (1+k)S_{n-1} + c.$$

Applying this repeatedly, starting from $S_0 = 0$,

$$\begin{aligned} S_n &= c + (1+k)S_{n-1} \\ &= c + c(1+k) + (1+k)^2 S_{n-2} \\ &= \cdots = c[1 + (1+k) + (1+k)^2 + \cdots + (1+k)^{n-1}]. \end{aligned}$$

This is a geometric series with n terms and common ratio $1+k \neq 1$ (since $k > 0$), so

$$S_n = \frac{c[(1+k)^n - 1]}{k}, \qquad n \le T,$$

as required. (To be careful, check it by induction: it gives $S_0 = 0$, and if it holds for S_{n-1} then $(1+k)S_{n-1} + c = \frac{c}{k}[(1+k)^n - (1+k) + k]$, which is the formula for S_n.)

The debt after the tunnel opens

From day $T+1$ on there is no more digging. Each day the debt gains interest kS and the firm pays in d, so, for as long as the debt is still positive,

$$S_{T+j} = (1+k)S_{T+j-1} - d.$$

This is the same recurrence as before with c replaced by $-d$, but now it starts from S_T instead of from 0. Unrolling it m times,

$$\begin{aligned} S_{T+m} &= (1+k)^m S_T - d[1 + (1+k) + \cdots + (1+k)^{m-1}] \\ &= (1+k)^m S_T - \frac{d[(1+k)^m - 1]}{k}. \end{aligned}$$

Substituting S_T from the first part,

$$S_{T+m} = \frac{c[(1+k)^T - 1](1+k)^m - d[(1+k)^m - 1]}{k}.$$

We need $S_{T+m} > 0$ because the recurrence only applies while the debt is still being paid. On the day the debt is cleared, the firm pays only what is owed, not the full d.

When the debt is paid off

Collect the terms in $(1+k)^m$:

$$S_{T+m} = \frac{d}{k} - \frac{(1+k)^m}{k}\Big(d - c[(1+k)^T - 1]\Big).$$

Ask yourself What does the condition $d/c > (1+k)^T - 1$ actually mean in money terms?

Multiply it by c: it says $d > c[(1+k)^T - 1] = kS_T$. In other words, the first day's income is more than the interest on the debt when the tunnel opens. So the debt goes down on day $T+1$, and the interest then goes down too, so it keeps going down faster and faster. The algebra makes this precise.

Suppose $d > c[(1+k)^T - 1]$, and write $\delta = d - c[(1+k)^T - 1] > 0$. Then

$$S_{T+m} = \frac{d - \delta(1+k)^m}{k}.$$

Since $1+k > 1$, $(1+k)^m \to \infty$ as $m \to \infty$, so $\delta(1+k)^m$ eventually exceeds d. The formula is valid only while the debt is positive, so it can't stay valid for ever: there is a first day on which the remaining debt is at most d plus that day's interest, and on that day the firm pays it off. In fact the debt is cleared on day $T+m$ for the least m with

$$(1+k)^m \geq \frac{d}{\delta}, \qquad \text{that is,} \qquad m \geq \frac{\ln(d/\delta)}{\ln(1+k)}.$$

So, if $d/c > (1+k)^T - 1$, the firm eventually pays off the debt, as required.

The condition is also necessary. If $d \le kS_T$, then $\delta \le 0$, and since $S_T = (d-\delta)/k$,

$$S_{T+m} - S_T = \frac{-\delta[(1+k)^m - 1]}{k} \ge 0$$

for every m. The debt never falls below S_T, so it is never paid off.

Checks

- Put $n = 1$: $S_1 = c$, which is right since nothing was owed on day 1, so no interest is due.
- Put $m = 0$ in the formula for S_{T+m}: it gives S_T, as it should.
- If $d = kS_T$ exactly, then $\delta = 0$ and $S_{T+m} = d/k = S_T$ for every m: the income just covers the interest and the debt never moves. This is the boundary case.
- A day-by-day simulation with $c = 100$, $k = 0.01$, $T = 30$ gives $S_T \approx 3478.49$, and the formula agrees. The threshold is $d > 34.78$. With $d = 50$ the debt is cleared after $m = 120$ days, matching $\ln(50/15.22)\,/\,\ln 1.01 \approx 119.6$.

Common slips

- Charging interest on day 1, or counting $n+1$ terms in the series. With $S_0 = 0$ the first day costs exactly c.
- Forgetting that after day T the series starts from S_T, not from 0: the term $(1+k)^m S_T$ is easy to drop.
- Saying the debt is paid off because "d is bigger than c". What matters is d against the interest kS_T at opening time.

Beyond STEP

The recurrence $S_n = aS_{n-1} + b$ is the simplest *linear first-order difference equation*. Its solution is a fixed point $b/(1-a)$ plus a multiple of a^n, which is exactly the shape of our formula: d/k plus a multiple of $(1+k)^m$. Whether the multiple is positive or negative decides between growing debt and repayment, an idea that reappears in the stability of fixed points and in linear ODEs. Boyce and DiPrima, *Elementary Differential Equations*, treats difference equations alongside differential ones, and Siklos, *Advanced Problems in Mathematics*, has more STEP practice on recurrences.

WORKED SOLUTION

STEP III 1988, Question 6

Pure • Repeated Differentiation

STEP III 1988, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/88-s3-q6

The key idea

Differentiating a product 1988 times directly is hopeless, but a product of sines and cosines can be rewritten as a *sum*. Each term $\sin kx$ then differentiates in a four-step cycle, picking up a factor k each time. For the zero, the huge factor 3^{1988} in front of $\sin 3x$ means the derivative can only vanish where $\sin 3x$ is almost zero, so we look just past the first zero of $\sin 3x$ and linearise.

Solution

The 1988th derivative

By the product-to-sum formula $2\sin A\cos B = \sin(A+B)+\sin(A-B)$,

$$\mathrm{f}(x) = \sin 2x\cos x = \tfrac{1}{2}\left(\sin 3x + \sin x\right).$$

Differentiating $\sin kx$ four times gives

$$\sin kx \;\to\; k\cos kx \;\to\; -k^2\sin kx \;\to\; -k^3\cos kx \;\to\; k^4\sin kx,$$

so every four differentiations simply multiply $\sin kx$ by k^4. (Equivalently, the nth derivative of $\sin kx$ is $k^n\sin\left(kx+\tfrac{1}{2}n\pi\right)$.) Since $1988 = 4\times 497$,

$$\mathrm{f}^{(1988)}(x) = \tfrac{1}{2}\left(3^{1988}\sin 3x + \sin x\right).$$

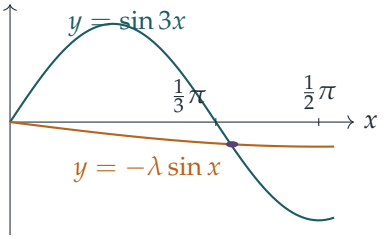
The smallest positive zero

We need the smallest $x>0$ with

$$3^{1988}\sin 3x + \sin x = 0. \qquad (*)$$

No zero in $(0,\tfrac{1}{3}\pi]$. For $0<x<\tfrac{1}{3}\pi$ we have $0<3x<\pi$, so both $\sin 3x$ and $\sin x$ are positive and the left-hand side of $(*)$ is positive. At $x=\tfrac{1}{3}\pi$ it equals $0+\tfrac{\sqrt{3}}{2}>0$.

A zero just after $\frac{1}{3}\pi$. At $x = \frac{1}{2}\pi$ the left-hand side is $3^{1988}\sin\frac{3}{2}\pi + 1 = 1 - 3^{1988} < 0$. It is continuous, so by the intermediate value theorem there is a zero in $(\frac{1}{3}\pi, \frac{1}{2}\pi)$, and the smallest positive zero is the first of these. Write it as $x = \frac{1}{3}\pi + \epsilon$ with $0 < \epsilon < \frac{1}{6}\pi$.



The sketch shows $(*)$ as $\sin 3x = -\lambda \sin x$ with $\lambda = 3^{-1988}$ (drawn far larger than it really is). The curves first meet just after $\sin 3x$ crosses zero at $\frac{1}{3}\pi$; for the true, tiny λ the crossing is pushed almost onto $\frac{1}{3}\pi$.

Since $\sin(\pi + 3\epsilon) = -\sin 3\epsilon$, equation $(*)$ becomes

$$3^{1988}\sin 3\epsilon = \sin\left(\tfrac{1}{3}\pi + \epsilon\right). \qquad (**)$$

Ask yourself How do we know ϵ is small enough to use $\sin 3\epsilon \approx 3\epsilon$?

The right-hand side of $(**)$ is at most 1, so $\sin 3\epsilon \le 3^{-1988}$. As $0 < 3\epsilon < \frac{1}{2}\pi$, and $\sin t \ge \frac{2t}{\pi}$ on $[0, \frac{1}{2}\pi]$ (the sine curve lies above its chord there), we get $3\epsilon \le \frac{\pi}{2}3^{-1988}$. So ϵ is astronomically small, which justifies the approximations below.

With ϵ this small, $\sin 3\epsilon = 3\epsilon\,(1 + O(\epsilon^2))$ and $\sin(\frac{1}{3}\pi + \epsilon) = \frac{\sqrt{3}}{2} + O(\epsilon)$. Substituting into $(**)$,

$$3^{1989}\,\epsilon \approx \frac{\sqrt{3}}{2}, \qquad \text{so} \qquad \epsilon \approx \frac{3^{-1989}\sqrt{3}}{2} = \frac{1}{3}\cdot\frac{3^{-1988}\sqrt{3}}{2},$$

with relative error of order 3^{-1988}.

A word about the printed value. The paper asks for $\epsilon \approx \frac{1}{2}3^{-1988}\sqrt{3}$, which is three times the value found here. The factor 3 comes from $\sin 3\epsilon \approx 3\epsilon$, and the numerical checks below confirm 3^{-1989}, not 3^{-1988}. So the question appears to contain a slip: the method it intends is exactly the one above, and only the power of 3 is off by one. If you reached 3^{-1989}, trust your working.

Checks

- The same argument works for any multiple n of 4: the first positive zero of $3^n\sin 3x + \sin x$ is at $\frac{1}{3}\pi + \epsilon_n$ with $\epsilon_n \approx \frac{1}{2}3^{-(n+1)}\sqrt{3}$. For $n = 4$, solving numerically gives $\epsilon_4 = 3.5713 \times 10^{-3}$, against the prediction $\frac{1}{2}\sqrt{3}/3^5 = 3.5640 \times 10^{-3}$ (ratio 1.0021). For $n = 8$ the ratio is 1.000025, and for $n = 40$ and beyond it is 1 to machine precision. The ratio to $\frac{1}{2}\sqrt{3}/3^n$ tends to $\frac{1}{3}$.

- The derivative formula can be checked on the first few cases: $f^{(4)}(x) = \frac{1}{2}(81\sin 3x + \sin x)$, which you can confirm by differentiating $\frac{1}{2}(\sin 3x + \sin x)$ four times.

Common slips

- Getting the sign or phase wrong: only after a multiple of four differentiations does $\sin kx$ return to $+k^n \sin kx$.
- Forgetting to rule out zeros in $(0, \frac{1}{3}\pi)$; the sign argument there is what makes $\frac{1}{3}\pi + \epsilon$ the *smallest* positive zero.
- Writing $\sin(\pi + 3\epsilon) \approx \epsilon$ instead of -3ϵ, losing either the sign or the factor 3.

Beyond STEP

The second part is a simple instance of *perturbation theory*: a large parameter forces the solution to sit very close to a known root, and linearising about that root gives the leading correction, with the size of the error controlled by the next term in the Taylor series. Making "approximately equal" precise with $O(\cdot)$ notation and error bounds is a central theme of a first analysis course; see Abbott, *Understanding Analysis*.

WORKED SOLUTION

STEP III 1988, Question 7

Pure • Differential Equations and Orthogonality

STEP III 1988, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/88-s3-q7

This question has a full guided treatment in *Beyond STEP*, Section 8.6, *The Quantum Oscillator*: https://beyondstep.keithmcnulty.org/ch08/quantum-oscillator.html

The key idea

The first part is pure substitution: differentiate the trial functions twice and compare coefficients of powers of x, which pins down λ. The second part is the classic "Wronskian trick": the combination $y_m y_n' - y_n y_m'$ has a derivative in which the awkward $\omega^2 x^2$ terms cancel, leaving only a multiple of $y_m y_n$. Integrating that derivative over the whole line is then easy, because the conditions at $\pm\infty$ kill the boundary terms.

Solution

Throughout, primes denote $\mathrm{d}/\mathrm{d}x$. It helps to write the equation as

$$y_n'' = \left(\omega^2 x^2 - (2n+1)\omega\right) y_n. \qquad (*)$$

The case $n = 0$

With $y_0 = \mathrm{e}^{-\lambda x^2}$,

$$y_0' = -2\lambda x\, \mathrm{e}^{-\lambda x^2}, \qquad y_0'' = (4\lambda^2 x^2 - 2\lambda)\, \mathrm{e}^{-\lambda x^2}.$$

Substituting into $(*)$ with $n = 0$ and cancelling the non-zero factor $\mathrm{e}^{-\lambda x^2}$, we need

$$4\lambda^2 x^2 - 2\lambda = \omega^2 x^2 - \omega \quad \text{for all } x.$$

Comparing constant terms gives $\lambda = \frac{1}{2}\omega$, and then the x^2 terms agree too, since $4\lambda^2 = \omega^2$. (The other root $\lambda = -\frac{1}{2}\omega$ of $4\lambda^2 = \omega^2$ fails the constant term.)

The case $n = 1$

With $y_1 = x\,\mathrm{e}^{-\lambda x^2}$,

$$y_1' = (1 - 2\lambda x^2)\,\mathrm{e}^{-\lambda x^2},$$
$$y_1'' = \big(-4\lambda x - 2\lambda x(1 - 2\lambda x^2)\big)\mathrm{e}^{-\lambda x^2} = (4\lambda^2 x^3 - 6\lambda x)\,\mathrm{e}^{-\lambda x^2}.$$

Equation $(*)$ with $n = 1$ requires $y_1'' = (\omega^2 x^2 - 3\omega)\,x\,\mathrm{e}^{-\lambda x^2}$, that is

$$4\lambda^2 x^3 - 6\lambda x = \omega^2 x^3 - 3\omega x \quad \text{for all } x,$$

which again holds exactly when $\lambda = \frac{1}{2}\omega$. So the *same* constant works for both functions:

$$\lambda = \tfrac{1}{2}\omega, \qquad y_0 = \mathrm{e}^{-\omega x^2/2}, \qquad y_1 = x\,\mathrm{e}^{-\omega x^2/2}.$$

The conditions at infinity

Since $\lambda = \frac{1}{2}\omega > 0$, every function of the form (polynomial)$\times \mathrm{e}^{-\lambda x^2}$ tends to 0 as $x \to \pm\infty$, because $\mathrm{e}^{-\lambda x^2}$ decays faster than any power of x grows. The functions y_0, y_0', y_1 and y_1' are all of this form (the polynomials being 1, $-2\lambda x$, x and $1 - 2\lambda x^2$), so all four tend to 0 at both ends, as required.

The identity

By the product rule, the terms $y_m' y_n'$ cancel:

$$\frac{\mathrm{d}}{\mathrm{d}x}\big(y_m y_n' - y_n y_m'\big) = y_m' y_n' + y_m y_n'' - y_n' y_m' - y_n y_m'' = y_m y_n'' - y_n y_m''.$$

Now use $(*)$ for both y_n'' and y_m'':

$$\begin{aligned} y_m y_n'' - y_n y_m'' &= \big(\omega^2 x^2 - (2n+1)\omega\big) y_m y_n - \big(\omega^2 x^2 - (2m+1)\omega\big) y_m y_n \\ &= \big((2m+1) - (2n+1)\big)\omega\, y_m y_n \\ &= 2(m-n)\,\omega\, y_m y_n, \end{aligned}$$

which is the required result.

The orthogonality

> **Ask yourself** How do we know the infinite integral even exists, before we try to show it is zero?

We work on a finite interval first and let the ends go to infinity separately; existence then comes for free. Write $W = y_m y_n' - y_n y_m'$. For any $S, R > 0$, integrating the identity from $-S$ to R gives

$$2(m-n)\,\omega \int_{-S}^{R} y_m y_n \,\mathrm{d}x = W(R) - W(-S).$$

As $R \to \infty$, each of y_m, y_n, y_m', y_n' tends to 0 by the given conditions, so $W(R) \to 0$. Likewise $W(-S) \to 0$ as $S \to \infty$. Hence the right-hand side tends to 0, so the integral $\int_{-\infty}^{\infty} y_m y_n \,\mathrm{d}x$ converges and

$$2(m-n)\,\omega \int_{-\infty}^{\infty} y_m y_n \,\mathrm{d}x = 0.$$

If $m \neq n$ then $2(m-n)\omega \neq 0$, because $\omega > 0$. Dividing by it gives

$$\int_{-\infty}^{\infty} y_m(x)\, y_n(x) \,\mathrm{d}x = 0,$$

as required. Notice that this needs only the equation and the end conditions: we never needed formulas for y_m or y_n.

Checks

- For $m = 0$, $n = 1$ the result is obvious directly: $y_0 y_1 = x\,\mathrm{e}^{-\omega x^2}$ is odd, so its integral over the whole line is 0.
- A less trivial test: $y_2 = (2\omega x^2 - 1)\mathrm{e}^{-\omega x^2/2}$ satisfies the equation with $n = 2$ (try it). Using $\int_{-\infty}^{\infty} \mathrm{e}^{-\omega x^2}\,\mathrm{d}x = \sqrt{\pi/\omega}$ and $\int_{-\infty}^{\infty} x^2 \mathrm{e}^{-\omega x^2}\,\mathrm{d}x = \frac{1}{2\omega}\sqrt{\pi/\omega}$, we get $\int y_0 y_2 \,\mathrm{d}x = 2\omega \cdot \frac{1}{2\omega}\sqrt{\pi/\omega} - \sqrt{\pi/\omega} = 0$.
- Numerically, with $\lambda = \frac{1}{2}\omega$ the equation's residual at random points was below 10^{-6} for $n = 0, 1, 2$; the derivative identity held to 10^{-4} by finite differences; and Simpson's rule on $[-30, 30]$ gave all of $\int y_0 y_1$, $\int y_0 y_2$, $\int y_1 y_2$ smaller than 10^{-15} for $\omega = 0.5, 1, 2.5$.

Common slips

- Matching only the x^2 coefficients and getting $\lambda = \pm\frac{1}{2}\omega$: the constant term rules out the negative sign (and so does the decay at infinity).
- Forgetting to say why the boundary terms vanish: this is exactly where both conditions, on y_n *and* on y_n', are used.
- Dividing by $m - n$ without noting that it is non-zero and that $\omega > 0$.

Beyond STEP

This is the quantum harmonic oscillator: the y_n are its energy states, $(2n+1)\omega$ are (up to scaling) its energy levels, and the functions are Hermite polynomials times a Gaussian. The orthogonality argument is a special case of Sturm–Liouville theory, which shows that eigenfunctions of a self-adjoint differential operator with different eigenvalues are always orthogonal, just as eigenvectors of a symmetric matrix are. Boyce and DiPrima, *Elementary Differential Equations*, covers boundary-value problems and Sturm–Liouville theory.

WORKED SOLUTION

STEP III 1988, Question 8

Pure • Parabola Tangents and Normals

STEP III 1988, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/88-s3-q8

The key idea

Write every line in terms of the parameter t, find where pairs of lines meet, and put those vertices into the determinant. The determinants look messy, but they are antisymmetric in t_1, t_2, t_3, so each must be a multiple of $(t_1 - t_2)(t_2 - t_3)(t_3 - t_1)$. Simple row and column operations bring that factor out and leave the comparison between the two areas.

(The printed question asks for A_3; this is a misprint for A_2, the area of the normal triangle.)

Solution

Tangent and normal

Differentiating $y^2 = 4ax$ with respect to t along $x = at^2$, $y = 2at$ gives $\dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{2a}{2at} = \dfrac{1}{t}$. The tangent at $(at^2, 2at)$ is $y - 2at = \frac{1}{t}(x - at^2)$, that is

$$ty = x + at^2.$$

This form also holds when $t = 0$ (the tangent at the vertex is $x = 0$). The normal has gradient $-t$:

$$y + tx = 2at + at^3.$$

We assume $a \neq 0$ throughout.

The tangent triangle

Subtracting the tangent equations at t_i and t_j gives $(t_i - t_j)y = a(t_i^2 - t_j^2)$. Since $t_i \neq t_j$ we may divide, so $y = a(t_i + t_j)$ and then $x = t_i y - at_i^2 = at_i t_j$. So

$$\text{tangents at } t_i, t_j \text{ meet at } \big(at_i t_j,\ a(t_i + t_j)\big).$$

The tangents have distinct gradients $1/t_i$ (or one is vertical), so no two are parallel and the triangle exists. Write $s = t_1 + t_2 + t_3$. The vertex opposite P_1 is $(at_2t_3,\ a(s - t_1))$, and similarly for the others. Taking out the factor a from the first two columns, then adding $-s$ times column 3 to column 2 and negating column 2,

$$\det \begin{pmatrix} at_2t_3 & a(s-t_1) & 1 \\ at_1t_3 & a(s-t_2) & 1 \\ at_1t_2 & a(s-t_3) & 1 \end{pmatrix} = -a^2 \det \begin{pmatrix} t_2t_3 & t_1 & 1 \\ t_1t_3 & t_2 & 1 \\ t_1t_2 & t_3 & 1 \end{pmatrix} = a^2 D,$$

where we swapped the first two columns and set

$$D = \det \begin{pmatrix} t_1 & t_2t_3 & 1 \\ t_2 & t_1t_3 & 1 \\ t_3 & t_1t_2 & 1 \end{pmatrix}.$$

To evaluate D, replace row 1 by row 1 − row 2 and row 2 by row 2 − row 3. Row 1 becomes $(t_1 - t_2,\ t_3(t_2 - t_1),\ 0) = (t_1 - t_2)(1, -t_3, 0)$, and row 2 becomes $(t_2 - t_3)(1, -t_1, 0)$. Expanding along the last column,

$$D = (t_1 - t_2)(t_2 - t_3) \det \begin{pmatrix} 1 & -t_3 \\ 1 & -t_1 \end{pmatrix} = (t_1 - t_2)(t_2 - t_3)(t_3 - t_1).$$

Hence

$$A_1 = \tfrac{1}{2}a^2 \left|(t_1 - t_2)(t_2 - t_3)(t_3 - t_1)\right|,$$

which is non-zero because the t_i are distinct.

The normal triangle

Subtracting the normal equations at t_i and t_j gives $(t_i - t_j)x = 2a(t_i - t_j) + a(t_i^3 - t_j^3)$, so dividing by $t_i - t_j \neq 0$,

$$x = 2a + a(t_i^2 + t_it_j + t_j^2), \qquad y = 2at_i + at_i^3 - t_ix = -at_it_j(t_i + t_j).$$

The normals have distinct gradients $-t_i$, so again every pair meets. For the vertex opposite P_1, use $t_2 + t_3 = s - t_1$ and write $e_1 = t_2t_3$ (and $e_2 = t_1t_3$, $e_3 = t_1t_2$). Then

$$t_2^2 + t_2t_3 + t_3^2 = (t_2 + t_3)^2 - t_2t_3 = (s - t_1)^2 - e_1, \qquad -t_2t_3(t_2 + t_3) = -se_1 + p,$$

where $p = t_1t_2t_3$. So the vertex is $a\big(2 + (s - t_1)^2 - e_1,\ -se_1 + p\big)$, and similarly for the others. Now take out a^2 and do column operations, which don't change the determinant: subtract 2 times column 3 from column 1, subtract p times column 3 from column 2, then take out the factor $-s$ from column 2:

$$\begin{aligned} \frac{1}{a^2}\det(\text{normal vertices}) &= \det\begin{pmatrix}(s-t_i)^2 - e_i & -se_i & 1\end{pmatrix}_{i=1,2,3} \\ &= -s\det\begin{pmatrix}(s-t_i)^2 - e_i & e_i & 1\end{pmatrix}_{i=1,2,3}, \end{aligned}$$

where each display shows the typical row i. Next add column 2 to column 1, which removes the $-e_i$, and expand $(s - t_i)^2 = s^2 - 2st_i + t_i^2$. The s^2 is a multiple of column 3 and can be removed, and the determinant is linear in column 1:

$$\det\begin{pmatrix}t_i^2 - 2st_i & e_i & 1\end{pmatrix} = \det\begin{pmatrix}t_i^2 & e_i & 1\end{pmatrix} - 2s\det\begin{pmatrix}t_i & e_i & 1\end{pmatrix} = E - 2sD,$$

say. The second determinant is exactly D.

Ask yourself How do we evaluate E, the determinant with rows $(t_i^2,\ e_i,\ 1)$?

Use the same row operations as for D. Row 1 − row 2 is $(t_1^2 - t_2^2,\ t_3(t_2 - t_1),\ 0) = (t_1 - t_2)(t_1 + t_2,\ -t_3,\ 0)$, and row 2 − row 3 is $(t_2 - t_3)(t_2 + t_3,\ -t_1,\ 0)$. So

$$\begin{aligned} E &= (t_1 - t_2)(t_2 - t_3)\left[-t_1(t_1 + t_2) + t_3(t_2 + t_3)\right] \\ &= (t_1 - t_2)(t_2 - t_3)\left[(t_3^2 - t_1^2) + t_2(t_3 - t_1)\right] \\ &= (t_1 - t_2)(t_2 - t_3)(t_3 - t_1)(t_1 + t_2 + t_3) = sD. \end{aligned}$$

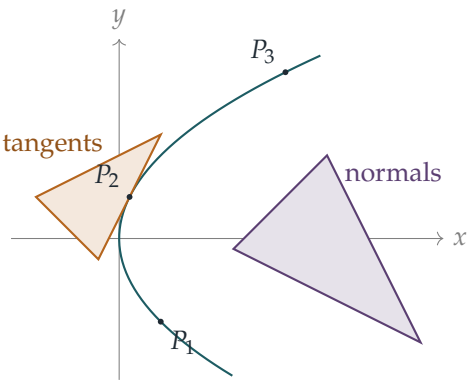
Putting everything together,

$$\det(\text{normal vertices}) = a^2 \cdot (-s)\,(sD - 2sD) = a^2 s^2 D.$$

Taking half the absolute value and comparing with $A_1 = \frac{1}{2}a^2|D|$,

$$A_2 = (t_1 + t_2 + t_3)^2 A_1,$$

which is the required result.



The figure takes $a = 1$ and $t_1, t_2, t_3 = -1, \frac{1}{2}, 2$, so $s = \frac{3}{2}$. The tangent triangle (amber) has area $\frac{27}{8}$, and the normal triangle (plum) has area $\frac{243}{32} = \left(\frac{3}{2}\right)^2 \cdot \frac{27}{8}$.

When are the normals concurrent?

No two normals are parallel, so they meet in pairs at three points Q_{12}, Q_{13}, Q_{23}. If the normals are concurrent, these points coincide and the triangle has zero area. Conversely, suppose the area is zero but the points are not all equal, say $Q_{12} \neq Q_{13}$. Both points lie on the normal at P_1, so the line through them is that normal. Since the points are collinear, Q_{23} also lies on the normal at P_1. Then Q_{23} lies on all three normals, so $Q_{23} = Q_{12}$ (the only common point of the normals at P_1 and P_2) and likewise $Q_{23} = Q_{13}$, which is a contradiction. So

$$\text{normals concurrent} \iff A_2 = 0 \iff (t_1 + t_2 + t_3)^2 A_1 = 0 \iff t_1 + t_2 + t_3 = 0,$$

using $A_1 \neq 0$ because the t_i are distinct. The required condition is $t_1 + t_2 + t_3 = 0$.

Checks

- With $t_1, t_2, t_3 = 0, 1, 2$ and $a = 1$, the tangent vertices are $(2, 3)$, $(0, 2)$, $(0, 1)$, giving area 1, and $\frac{1}{2}|D| = \frac{1}{2}|(-1)(-1)(2)| = 1$.
- The normals at $t = 1, 2, -3$ (sum zero, $a = 1$) all pass through $(9, -6)$. For example, the normal at $t = 1$ is $y + x = 3$.
- A computer computed the vertices exactly with fractions for about 2,800 random choices of a and distinct t_i, and confirmed $A_2 = (t_1 + t_2 + t_3)^2 A_1$ every time.

Common slips

- Forgetting to justify that the three lines really form a triangle. Pairwise non-parallel is what we need, and here it follows from the distinct gradients.
- Concluding concurrency from $A_2 = 0$ without a reason. Zero area only says the three meeting points are collinear, and the short argument above upgrades this to "equal".
- Expanding the 3×3 determinants in full. This is possible, but sign errors are almost certain. Factor out $(t_i - t_j)$ with row operations first.

Beyond STEP

The factor $(t_1 - t_2)(t_2 - t_3)(t_3 - t_1)$ is a Vandermonde determinant, and the cofactor $t_1 + t_2 + t_3$ in $E = sD$ is a symmetric polynomial. Any antisymmetric polynomial is the Vandermonde product times a symmetric one, a fact that reappears in the theory of determinants and symmetric functions. The triangle of normals is related to the *evolute* of the parabola, the curve traced out by its centres of curvature, which is the envelope of the normals. Strang's *Introduction to Linear Algebra* covers determinants and their row operations well.

WORKED SOLUTION

STEP III 1988, Question 9

Pure • Group Element Orders

STEP III 1988, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/88-s3-q9

The key idea

Part (i) is the division algorithm in disguise: write N as a multiple of n plus a remainder, and the minimality of n forces the remainder to vanish. Part (ii) works because conjugation by g respects products, so in a product of conjugates the inner $g^{-1}g$ pairs cancel. In part (iii) we apply conjugation by g again and again; since $g^5 = e$, five rounds take h back to itself, which pins down the order of h.

Solution

Part (i)

By the division algorithm we can write $N = qn + r$ with q an integer and $0 \leq r < n$. (This works for any integer N, including zero or negative ones.) Then, since $g^n = e$,

$$e = g^N = (g^n)^q g^r = e^q g^r = g^r.$$

If r were positive, it would be a positive integer smaller than n with $g^r = e$, which contradicts the definition of $n = o(g)$ as the *smallest* such integer. Hence $r = 0$, so $N = qn$ and n divides N.

Part (ii)

$m = 0$: both sides equal e.

$m \geq 1$: we use induction on m. The case $m = 1$ is immediate. If $gh^m g^{-1} = (ghg^{-1})^m$, then

$$\begin{aligned}(ghg^{-1})^{m+1} &= (ghg^{-1})^m (ghg^{-1}) = (gh^m g^{-1})(ghg^{-1}) \\ &= gh^m(g^{-1}g)hg^{-1} = gh^{m+1}g^{-1},\end{aligned}$$

which completes the induction. (Written out, $(ghg^{-1})(ghg^{-1})\cdots(ghg^{-1})$ telescopes: every $g^{-1}g$ in the middle cancels.)

$m \leq -1$: first, ghg^{-1} has inverse $gh^{-1}g^{-1}$, because

$$(ghg^{-1})(gh^{-1}g^{-1}) = gh(g^{-1}g)h^{-1}g^{-1} = ghh^{-1}g^{-1} = e.$$

Now write $m = -k$ with $k \geq 1$. Applying the positive case to the element h^{-1} in place of h,

$$gh^m g^{-1} = g(h^{-1})^k g^{-1} = (gh^{-1}g^{-1})^k = \left((ghg^{-1})^{-1}\right)^k = (ghg^{-1})^{-k} = (ghg^{-1})^m.$$

So the identity holds for every integer m.

Part (iii)

Using $ghg^{-1} = h^2$ and then part (ii) with $m = 2$,

$$g^2hg^{-2} = g\,(ghg^{-1})\,g^{-1} = gh^2g^{-1} = (ghg^{-1})^2 = (h^2)^2 = h^4,$$

as required.

The same step works at every stage. We claim that $g^khg^{-k} = h^{2^k}$ for all $k \geq 0$. This is true for $k = 0$, and if it holds for k, then by part (ii) with $m = 2^k$,

$$g^{k+1}hg^{-(k+1)} = g\,(g^khg^{-k})\,g^{-1} = gh^{2^k}g^{-1} = (ghg^{-1})^{2^k} = (h^2)^{2^k} = h^{2^{k+1}}.$$

Taking $k = 5$ and using $g^5 = e$ (so $g^{-5} = e$ too) gives

$$h = g^5hg^{-5} = h^{32}, \qquad \text{so} \qquad h^{31} = e$$

on multiplying by h^{-1}.

Ask yourself We know $h^{31} = e$. Why does that make $o(h)$ exactly 31, and not just at most 31?

By part (i), $o(h)$ divides 31. Since 31 is prime, $o(h)$ is 1 or 31. But $o(h) = 1$ would mean $h = e$, which we are told is not the case. Therefore

$$o(h) = 31.$$

Such a group really exists, so the conditions are consistent. Take the permutations of $\{0, 1, \ldots, 30\}$ given by $h(x) = x + 1$ and $g(x) = 2x$, working modulo 31. Then $g^5(x) = 32x = x$, and $ghg^{-1}(x) = 2\left(\frac{x}{2} + 1\right) = x + 2 = h^2(x)$, and h does have order 31.

Checks

- The prime 31 makes sense: conjugating by g acts on the powers of h as "multiply the exponent by 2", and five doublings must return to the start, so $2^5 \equiv 1$ modulo $o(h)$. In other words, $o(h)$ divides $2^5 - 1 = 31$.
- A computer check with the permutations $g(x) = 2x$ and $h(x) = x + 1$ modulo 31 confirmed $g^5 = e$, $ghg^{-1} = h^2$, $g^khg^{-k} = h^{2^k}$ for $k = 0, \ldots, 5$, and $o(h) = 31$. It also tested (i) and (ii), for $-12 \leq m \leq 12$, on hundreds of random permutations.

Common slips

- In (ii), proving only the case $m \geq 1$. The question says *any integer*, so $m = 0$ and negative m need their own (short) argument.
- Concluding $o(h) = 31$ straight from $h^{31} = e$. That only gives $o(h) \mid 31$; you need the primality of 31 and the condition $h \neq e$ as well.

Beyond STEP

The map $x \mapsto gxg^{-1}$ is an *automorphism* of G (a structure-preserving bijection), and part (ii) is one consequence of that. Part (iii) is a small instance of a *semidirect product*: the group generated by g and h has order $5 \times 31 = 155$, and its existence depends on 2 having order 5 modulo 31. This construction is central to classifying groups of small order. Allenby's *Rings, Fields and Groups* is a good place to read about orders, conjugation and Lagrange's theorem.

WORKED SOLUTION

STEP III 1988, Question 10

Pure • Pursuit Curves

STEP III 1988, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/88-s3-q10

This question has a full guided treatment in *Beyond STEP*, Section 8.5, *Greyhounds Chasing Greyhounds*: https://beyondstep.keithmcnulty.org/ch08/pursuit.html

The key idea

Don't try to track four moving dogs separately. The set-up is unchanged by a quarter-turn about the centre, so the dogs always stay at the corners of a square centred at O: it shrinks and rotates, but it stays a square. That fixes the angle between A's velocity and the line AO, and a fixed angle between velocity and radius is exactly what produces an equiangular spiral $r = \lambda \mathrm{e}^{-k\theta}$.

Solution

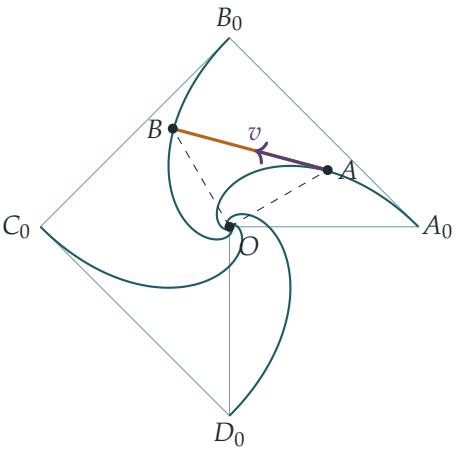
The dogs stay on a square

Take the origin O at the centre of the square and label the dogs anticlockwise, so that B is a quarter-turn anticlockwise from A, C from B, and so on.

Rotate the whole picture through $\frac{\pi}{2}$ about O. The starting positions of A, B, C, D go to the starting positions of B, C, D, A, and the rule "run at speed v straight at the next dog" is the same rule for every dog. So the rotated motion is also a motion obeying the same rules from the same starting configuration. The motion from a given start is unique (the velocities are smooth functions of the positions while the dogs are apart), so the rotated motion is the actual motion. In other words, at every instant B's position is A's position rotated through $\frac{\pi}{2}$ about O. The same goes for C and D. So $ABCD$ is always a square centred at O.

The angle between the velocity and the radius

Let A be at (r, θ) in polar coordinates. Then B is at $(r, \theta + \frac{\pi}{2})$, so $OA = OB = r$ and $\angle AOB = \frac{\pi}{2}$. Triangle OAB is right-angled and isosceles, so $\angle OAB = \frac{\pi}{4}$.



The figure shows the four paths spiralling in from the starting corners A_0, B_0, C_0, D_0, with the dogs A and B at one instant. A's velocity (plum) points along AB.

A's velocity has size v and points along AB, which makes an angle $\frac{\pi}{4}$ with the inward radius AO. Splitting it into the radial component (outwards positive) and the transverse component (in the direction of increasing θ):

$$\dot{r} = -v\cos\frac{\pi}{4}, \qquad r\dot{\theta} = v\sin\frac{\pi}{4}.$$

The transverse component is positive because B lies anticlockwise of A, so A is pulled round anticlockwise.

Ask yourself Why is it legitimate to divide these two equations?

Because $r\dot{\theta} = v/\sqrt{2} > 0$ while the dogs are apart, θ is strictly increasing in time. So we can use θ in place of t as the parameter along the path, and by the chain rule

$$\frac{\mathrm{d}r}{\mathrm{d}\theta} = \frac{\dot{r}}{\dot{\theta}} = \frac{-v/\sqrt{2}}{v/(\sqrt{2}\,r)} = -r.$$

Solving for the path

Since $r > 0$ we can separate variables: $\displaystyle\int \frac{\mathrm{d}r}{r} = -\int \mathrm{d}\theta$, so $\ln r = -\theta + c$, that is

$$r = \lambda \mathrm{e}^{-\theta}$$

with $\lambda = \mathrm{e}^{c}$ a constant. This is the required result. If we choose the initial line to be OA_0, so that $\theta = 0$ at the start, then λ is A's initial distance from O, namely $a/\sqrt{2}$ for a square of side a. With any other initial line λ just changes by a constant factor. (If we had labelled the dogs clockwise, measuring θ clockwise gives the same equation.)

Generalisation to n dogs

For n dogs at the vertices of a regular n-gon (labelled anticlockwise, each chasing the next) the same argument works with a rotation through $\frac{2\pi}{n}$. The dogs always form a regular n-gon centred at O. So if A is at (r,θ), the next dog B is at $(r,\theta+\frac{2\pi}{n})$.

Triangle OAB is isosceles with apex angle $\frac{2\pi}{n}$, so each base angle is

$$\angle OAB = \frac{1}{2}\left(\pi - \frac{2\pi}{n}\right) = \frac{\pi}{2} - \frac{\pi}{n}.$$

Resolving A's velocity as before,

$$\dot r = -v\cos\left(\frac{\pi}{2}-\frac{\pi}{n}\right) = -v\sin\frac{\pi}{n}, \qquad r\dot\theta = v\sin\left(\frac{\pi}{2}-\frac{\pi}{n}\right) = v\cos\frac{\pi}{n}.$$

Since $n \geq 3$, $\cos\frac{\pi}{n} > 0$, so again θ increases steadily and

$$\frac{\mathrm{d}r}{\mathrm{d}\theta} = -r\tan\frac{\pi}{n} \qquad \Longrightarrow \qquad r = \lambda\,\mathrm{e}^{-\theta\tan(\pi/n)}.$$

For $n = 4$, $\tan\frac{\pi}{4} = 1$ and we recover $r = \lambda\mathrm{e}^{-\theta}$. Each dog follows an equiangular (logarithmic) spiral, crossing every radius at the fixed angle $\frac{\pi}{2} - \frac{\pi}{n}$.

A nice by-product: $\dot r = -v\sin\frac{\pi}{n}$ is constant, so the dogs meet at O after time $r_0/(v\sin\frac{\pi}{n})$, where r_0 is the initial distance from O. In that time each dog runs the finite distance $r_0/\sin\frac{\pi}{n}$, even though $\theta\to\infty$ and it circles O infinitely often. (For the square this is a, the side length.)

Checks

- For $n = 4$ the result is the special case $\tan\frac{\pi}{4} = 1$ of the general formula.
- As $n\to\infty$, $\tan\frac{\pi}{n}\to 0$ and the path tends to a circle: each dog is chasing a neighbour almost alongside it on the circle.
- We simulated the chase numerically (with RK4 for $n = 3,4,5,6,8$ dogs, starting at radius 1). The computed r agreed with $\mathrm{e}^{-\theta\tan(\pi/n)}$ to a relative error of 10^{-11}. For $n = 4$ the time and path length to reach $r = 0.02$ came out as $1.386 = 0.98\sqrt{2}$, as predicted by $(1-0.02)/\sin\frac{\pi}{4}$.

Common slips

- Writing down $\dot r$ and $r\dot\theta$ without first justifying that the dogs stay on a regular polygon. The symmetry argument is what makes the angle at A known.
- Using the interior angle of the polygon ($\frac{\pi}{2}$ for a square) instead of the angle between AB and the radius AO ($\frac{\pi}{4}$).
- Forgetting that the transverse velocity component is $r\dot\theta$, not $\dot\theta$.

Beyond STEP

This is a classic *pursuit problem*. The symmetry step is an instance of a general principle in differential equations: if the equations and the initial data are both invariant under a symmetry, then uniqueness of solutions forces the solution to be invariant too. Curves meeting every radius at a fixed angle are the equiangular spirals. In complex form they are $z = \mathrm{e}^{(-k+\mathrm{i})\theta}$, so the whole motion is a continuous "spiral similarity" of the plane. Needham's *Visual Complex Analysis* develops this geometric view, and Boyce and DiPrima's *Elementary Differential Equations* covers existence and uniqueness.

WORKED SOLUTION

STEP III 1988, Question 11

Mechanics • Ladder Equilibrium with Friction

STEP III 1988, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/88-s3-q11

The key idea

First sort out the geometry: the ground, the ramp and the ladder make a triangle whose angles you already know, and that tells you which way the smooth ramp pushes on the top of the ladder. After that it is the standard ladder recipe: resolve twice, take moments about the foot, and compare the friction needed with μ times the normal reaction. Because the labourer's position enters only through the moment equation, one general formula answers both parts.

Solution

Setting up

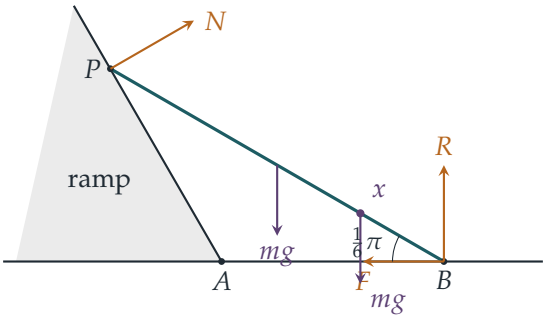
Model the ladder as a uniform rod and the labourer as a particle. Let A be the foot of the ramp, B the foot of the ladder and P its top, on the ramp.

Geometry. In triangle ABP the angle at B is $\frac{1}{6}\pi$ (the ladder's inclination). The ramp makes $\frac{1}{6}\pi$ with the vertical, so it makes $\frac{1}{3}\pi$ with the horizontal on the far side of A. The angle at A inside the triangle is therefore $\pi - \frac{1}{3}\pi = \frac{2}{3}\pi$, which leaves $\frac{1}{6}\pi$ at P. So the triangle is isosceles with $AP = AB$, and the sine rule gives

$$AB = \frac{l \sin \frac{1}{6}\pi}{\sin \frac{2}{3}\pi} = \frac{l}{\sqrt{3}},$$

which matches the given distance. The data are consistent, and P really is on the ramp.

Direction of the ramp's reaction. The ramp is smooth, so its reaction N on the ladder is perpendicular to the ramp. The ramp is at $\frac{1}{3}\pi$ to the horizontal, so its normal is at $\frac{1}{6}\pi$ to the horizontal, pointing up and away from the ramp, towards B. So N has horizontal component $\frac{\sqrt{3}}{2}N$ pushing the ladder away from the ramp, and vertical component $\frac{1}{2}N$ upwards.



Equations. Let the labourer be a distance x up the ladder from B. At B the ground gives normal reaction R upwards and friction F horizontally towards the ramp (the only horizontal force otherwise is the component of N, which pushes the foot away from the ramp, so friction must oppose that).

Resolving horizontally and vertically:

$$F = \tfrac{\sqrt{3}}{2}N, \qquad\qquad R + \tfrac{1}{2}N = 2mg.$$

Taking moments about B, so that R and F drop out. The horizontal distance from B to a point a distance s up the ladder is $s\cos\frac{1}{6}\pi = \frac{\sqrt{3}}{2}s$. The line of action of N is at $\frac{1}{6}\pi$ to the horizontal, while BP is at $\frac{1}{6}\pi$ on the other side of the horizontal, so N makes an angle $\frac{1}{3}\pi$ with the ladder. Its moment about B is $Nl\sin\frac{1}{3}\pi = \frac{\sqrt{3}}{2}Nl$. Hence

$$\tfrac{\sqrt{3}}{2}Nl = mg\cdot\tfrac{\sqrt{3}}{2}\cdot\tfrac{l}{2} + mg\cdot\tfrac{\sqrt{3}}{2}x \quad\Longrightarrow\quad N = mg\left(\frac{1}{2}+\frac{x}{l}\right).$$

Ask yourself Why does the moment of N come out so simply?

The perpendicular distance from B to the line of action of N is $l\sin\frac{1}{3}\pi$, because N meets the ladder at $\frac{1}{3}\pi$. Or split N into components: the horizontal part $\frac{\sqrt{3}}{2}N$ acts at height $\frac{1}{2}l$ and the vertical part $\frac{1}{2}N$ acts at horizontal distance $\frac{\sqrt{3}}{2}l$. Both give a moment of $\frac{\sqrt{3}}{4}Nl$, and these add to $\frac{\sqrt{3}}{2}Nl$.

Then $R = 2mg - \frac{1}{2}N = mg\left(\frac{7}{4} - \frac{x}{2l}\right) > 0$ for $0 \le x \le l$, so the foot stays on the ground. The ladder does not slip as long as $F \le \mu R$, which means

$$\mu \;\ge\; \frac{F}{R} = \frac{\sqrt{3}\,N}{4mg - N} = \frac{\sqrt{3}\,(l+2x)}{7l-2x}.$$

This ratio increases with x: the numerator increases and the positive denominator decreases. So the friction needed grows steadily as the labourer climbs.

Part (i)

The ladder slips as soon as the labourer reaches the midpoint. So limiting friction is reached exactly when $x = \frac{1}{2}l$, and since F/R increases with x, not before. Then $N = mg$,

$F = \frac{\sqrt{3}}{2}mg$ and $R = \frac{3}{2}mg$, so

$$\mu = \frac{F}{R} = \frac{\sqrt{3}/2}{3/2} = \frac{1}{\sqrt{3}}.$$

Part (ii)

The labourer reaches the top if $F \leq \mu R$ for every $x \in [0, l]$. Since F/R increases with x, the worst case is $x = l$. There $N = \frac{3}{2}mg$, $F = \frac{3\sqrt{3}}{4}mg$ and $R = \frac{5}{4}mg$, so the minimum coefficient is

$$\mu = \frac{3\sqrt{3}/4}{5/4} = \frac{3\sqrt{3}}{5} \approx 1.04.$$

Checks

- With nobody on the ladder ($x = 0$) the formula gives $\mu \geq \sqrt{3}/7 \approx 0.247$. That is smaller than both answers, as it should be.
- Substituting $l = 1$, $m = 1$, $g = 9.8$ into the three equilibrium equations numerically, with the forces written as vectors, reproduces $N/mg = 1$ and $\frac{3}{2}$ and the ratios $F/R = 0.57735\ldots$ and $1.03923\ldots$. It also confirms that F/R increases at 101 positions up the ladder.

Common slips

- Taking the ramp's reaction to be horizontal, as for a vertical wall. Here it is tilted $\frac{1}{6}\pi$ above the horizontal, and its upward component reduces R.
- Forgetting the labourer's weight when resolving vertically, so that $R + \frac{1}{2}N = mg$ instead of $2mg$.

Beyond STEP

Part (ii) is a small optimisation over a configuration: you need the worst case of F/R over all positions of the load. This is how engineers find the critical load case for a structure. A useful way to picture the condition $F \leq \mu R$ is the *cone of friction*: the total ground reaction must lie within an angle $\arctan \mu$ of the normal, and the ladder slips at the moment the required reaction leaves that cone. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats rigid-body statics and friction carefully.

WORKED SOLUTION

STEP III 1988, Question 12

Mechanics • Oblique Collisions

STEP III 1988, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/88-s3-q12

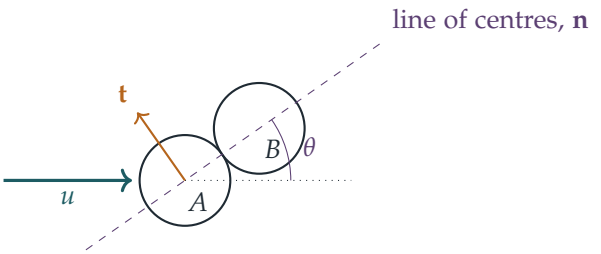
The key idea

For smooth spheres the impulse acts only along the line of centres at the moment of impact. So we split every velocity into a component along that line and one perpendicular to it: the perpendicular components are untouched, and along the line we have an ordinary one-dimensional collision. Everything then depends on a single unknown, the angle between the line of centres and the first ball's original velocity.

Solution

Setting up

Model the balls as smooth spheres of equal mass m, sliding on a smooth table, so the only horizontal force during impact is the contact impulse along the line of centres. Let the first ball A move with speed u, and let the line of centres make angle θ with A's velocity at impact, where $0 \le \theta < \frac{1}{2}\pi$ (if $\theta = \frac{1}{2}\pi$ the balls only graze and nothing happens). Take unit vectors $\mathbf{n}$ along the line of centres (from A to B) and $\mathbf{t}$ perpendicular to it.



Before impact A has components $u\cos\theta$ along $\mathbf{n}$ and $u\sin\theta$ along $\mathbf{t}$; B is at rest.

Along $\mathbf{t}$: no impulse acts in this direction on either ball, so A keeps $u\sin\theta$ and B keeps 0.

Along $\mathbf{n}$*:* let the components afterwards be v_A and v_B. Conservation of momentum (the masses cancel) and Newton's law of restitution give

$$v_A + v_B = u\cos\theta, \qquad v_B - v_A = e\,u\cos\theta,$$

so

$$v_A = \tfrac{1}{2}(1-e)\,u\cos\theta, \qquad v_B = \tfrac{1}{2}(1+e)\,u\cos\theta.$$

So afterwards

$$\mathbf{v}_A = \tfrac{1}{2}(1-e)u\cos\theta\,\mathbf{n} + u\sin\theta\,\mathbf{t}, \qquad \mathbf{v}_B = \tfrac{1}{2}(1+e)u\cos\theta\,\mathbf{n}.$$

The angle between the velocities

B moves along the line of centres, and the angle between the two velocities is acute exactly when their scalar product is positive. Here

$$\mathbf{v}_A \cdot \mathbf{v}_B = \tfrac{1}{4}(1-e)(1+e)\,u^2\cos^2\theta > 0,$$

because $e < 1$, $u \neq 0$ and $\cos\theta > 0$ for $0 \le \theta < \tfrac{1}{2}\pi$. Hence the angle between the velocities is less than $\tfrac{1}{2}\pi$, as required. (For a head-on impact, $\theta = 0$, both balls move along the same line and the angle is 0.)

Note where $e < 1$ was used: if $e = 1$ then $v_A = 0$ and the balls separate at exactly a right angle, the familiar snooker result.

The maximum deflection

Let ϕ be the angle through which A is deflected, i.e. the angle between $\mathbf{v}_A$ and the original direction $\cos\theta\,\mathbf{n} + \sin\theta\,\mathbf{t}$. Write $k = \tfrac{1}{2}(1-e)$, so $0 < k \le \tfrac{1}{2}$ and $\mathbf{v}_A = u(k\cos\theta\,\mathbf{n} + \sin\theta\,\mathbf{t})$. Using the scalar product and the two-dimensional cross product of the direction vectors $(\cos\theta, \sin\theta)$ and $(k\cos\theta, \sin\theta)$,

$$|\mathbf{v}_A|\cos\phi = u\,(k\cos^2\theta + \sin^2\theta),$$
$$|\mathbf{v}_A|\sin\phi = u\,(\cos\theta\sin\theta - k\cos\theta\sin\theta) = u\,(1-k)\sin\theta\cos\theta.$$

The first is positive, so ϕ is acute, and dividing (then dividing top and bottom by $\cos^2\theta$) with $t = \tan\theta \ge 0$,

$$\tan\phi = \frac{(1-k)t}{k+t^2}.$$

Ask yourself How do we maximise this without a messy derivative?

Since ϕ is acute, maximising ϕ is the same as maximising $\tan\phi$. For $t > 0$,

$$\tan\phi = \frac{1-k}{k/t + t} \quad \text{and} \quad \frac{k}{t} + t \ge 2\sqrt{k}$$

by the AM–GM inequality, with equality exactly when $t = \sqrt{k}$. (At $t = 0$, $\phi = 0$.) So the maximum is attained when $\tan\theta = \sqrt{k}$, and

$$\tan\phi_{\max} = \frac{1-k}{2\sqrt{k}}.$$

To convert to a sine, note $(1-k)^2 + (2\sqrt{k})^2 = 1 - 2k + k^2 + 4k = (1+k)^2$, so in the right-angled triangle with legs $1-k$ and $2\sqrt{k}$ the hypotenuse is $1+k$, and

$$\sin\phi_{\max} = \frac{1-k}{1+k} = \frac{\frac{1}{2}(1+e)}{\frac{1}{2}(3-e)} = \frac{1+e}{3-e},$$

using $1-k = \frac{1}{2}(1+e)$ and $1+k = \frac{1}{2}(3-e)$. Since $\phi_{\max}$ is acute,

$$\phi_{\max} = \sin^{-1}\left(\frac{1+e}{3-e}\right),$$

which is the required result. It occurs when the line of centres makes angle $\tan^{-1}\sqrt{\frac{1}{2}(1-e)}$ with the original motion.

Checks

- $e \to 1$: the formula gives $\sin^{-1} 1 = \frac{1}{2}\pi$, consistent with the perfectly elastic case: there A loses all its velocity along the line of centres, so it always moves at right angles to that line, and a nearly head-on blow sends it off at nearly a right angle to its original path.
- $e = 0$: $\sin\phi_{\max} = \frac{1}{3}$, so $\phi_{\max} \approx 0.3398$ rad (19.5°).
- $e = \frac{1}{2}$: $\sin\phi_{\max} = \frac{3}{5}$, $\phi_{\max} \approx 0.6435$ rad, at $\theta = \tan^{-1}\frac{1}{2} \approx 0.4636$. A numerical scan of the deflection over 200 000 values of θ reproduced both numbers, and 200 000 random collisions never gave an angle between the velocities of $\frac{1}{2}\pi$ or more.

Common slips

- Applying restitution to the full speeds rather than to the components along the line of centres.
- Forgetting that the perpendicular component of A's velocity is unchanged; this is what makes A deflect at all.
- Maximising ϕ itself by differentiating an inverse tangent; working with $\tan\phi$ (legitimate because ϕ is acute) is far cleaner.

Beyond STEP

The decomposition here is the impulse form of Newton's second law: impulses change momentum only in the direction they act. In the elastic case the right-angle result is a special case of a neat frame-of-reference argument: in the centre-of-mass frame an elastic collision keeps each ball's speed and only turns its direction, and the transformation back to the lab frame gives every possible outcome at once. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats collisions in the centre-of-mass frame this way.

WORKED SOLUTION

STEP III 1988, Question 13

Mechanics • Projectile with Air Resistance

STEP III 1988, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/88-s3-q13

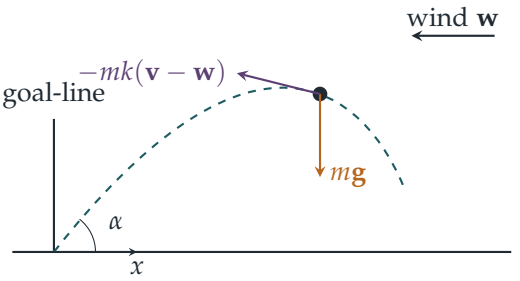
The key idea

Air resistance proportional to velocity gives a *linear* equation of motion, so each component can be solved with an integrating factor. The neat way to see the last part is to ask what the terminal velocity is. The velocity relaxes exponentially from its starting value towards the terminal velocity, so if those two vectors are parallel, the velocity never leaves one line through the origin. The ball then goes out and comes back along the same straight path.

Solution

The equation of motion

We model the ball as a particle of mass m. Two forces act on it: its weight $m\mathbf{g}$, and the air resistance. The velocity of the ball relative to the air is $\mathbf{v} - \mathbf{w}$, and resistance opposes this, so the resistance is $-mk(\mathbf{v} - \mathbf{w})$.



Newton's second law gives

$$m\frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t} = m\mathbf{g} - mk(\mathbf{v} - \mathbf{w}).$$

Dividing by m and rearranging,

$$\frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t} + k\mathbf{v} = \mathbf{g} + k\mathbf{w},$$

as required.

The position at time t

Take the origin at the point of the kick, with the x-axis horizontal in the direction of the kick and the y-axis vertically upwards. Write $V = |\mathbf{v}_0|$, $W = |\mathbf{w}|$ and $g = |\mathbf{g}|$. The ball is kicked *into* the wind, so

$$\mathbf{w} = (-W,\, 0), \qquad \mathbf{g} = (0,\, -g), \qquad \mathbf{v}_0 = (V\cos\alpha,\, V\sin\alpha).$$

With $\mathbf{v} = (u, v)$ the two components of the equation of motion are

$$\frac{\mathrm{d}u}{\mathrm{d}t} + ku = -kW, \qquad \frac{\mathrm{d}v}{\mathrm{d}t} + kv = -g.$$

Multiplying the first by the integrating factor e^{kt} gives $\frac{\mathrm{d}}{\mathrm{d}t}\left(u\,\mathrm{e}^{kt}\right) = -kW\mathrm{e}^{kt}$, so $u\,\mathrm{e}^{kt} = -W\mathrm{e}^{kt} + C$. Since $u = V\cos\alpha$ at $t = 0$, $C = V\cos\alpha + W$, and

$$u = -W + (V\cos\alpha + W)\,\mathrm{e}^{-kt}.$$

In exactly the same way,

$$v = -\frac{g}{k} + \left(V\sin\alpha + \frac{g}{k}\right)\mathrm{e}^{-kt}.$$

Integrating again from 0 to t, with $x = y = 0$ at $t = 0$ and $\int_0^t \mathrm{e}^{-ks}\,\mathrm{d}s = \frac{1}{k}\left(1 - \mathrm{e}^{-kt}\right)$,

$$x = -Wt + \frac{V\cos\alpha + W}{k}\left(1 - \mathrm{e}^{-kt}\right),$$
$$y = -\frac{g}{k}\,t + \frac{1}{k}\left(V\sin\alpha + \frac{g}{k}\right)\left(1 - \mathrm{e}^{-kt}\right).$$

In vector form, with $\mathbf{u}_\infty = \mathbf{w} + \mathbf{g}/k$ (the terminal velocity, the constant solution of the equation of motion), this is

$$\mathbf{r} = \mathbf{u}_\infty t + \frac{1}{k}\left(\mathbf{v}_0 - \mathbf{u}_\infty\right)\left(1 - \mathrm{e}^{-kt}\right).$$

The own goal

Ask yourself What does $\tan\alpha = g/(kW)$ say about the terminal velocity?

The terminal velocity is $\mathbf{u}_\infty = (-W,\, -g/k) = -W\left(1,\, g/(kW)\right)$. The condition says $g/(kW) = \tan\alpha$, so $\mathbf{u}_\infty = -W(1, \tan\alpha)$, which is parallel to $\mathbf{v}_0$ (pointing the opposite way). The vector formula shows that $\mathbf{r}$ is then always a combination of two vectors along the direction $(\cos\alpha, \sin\alpha)$, so the ball moves along the straight line $y = x\tan\alpha$.

To see this directly from the components, use $g/k = W\tan\alpha$:

$$V\sin\alpha + \frac{g}{k} = V\sin\alpha + W\tan\alpha = \tan\alpha\,(V\cos\alpha + W),$$

so

$$y = \tan\alpha\left[-Wt + \frac{V\cos\alpha + W}{k}\left(1 - \mathrm{e}^{-kt}\right)\right] = x\tan\alpha \quad \text{for all } t \geq 0.$$

So the ball is at ground level, $y = 0$, exactly when $x = 0$ (as $\tan\alpha \neq 0$). It remains to check that this happens at some $t > 0$. Let $f(t) = x(t)$. Then $f(0) = 0$ and $f'(0) = V\cos\alpha > 0$, so $x > 0$ just after the kick. Also

$$f''(t) = -k(V\cos\alpha + W)\,\mathrm{e}^{-kt} < 0,$$

so f is strictly concave, and $f(t) \to -\infty$ as $t \to \infty$ because of the $-Wt$ term. Hence f has exactly one positive zero T, and $f'(T) < 0$ there (a concave function that has risen from 0 must be decreasing when it falls back to 0).

At time T the ball is back at the origin, the point on the goal-line where it was kicked, moving with negative horizontal velocity, that is, towards the goal. It has come back down the line it went out on, and it crosses the goal-line at ground level, well under the crossbar. With the goalkeeper out of the way, it is an own goal.

Checks

- As $k \to 0$, $\frac{1}{k}(1 - \mathrm{e}^{-kt}) \to t$ and $\frac{1}{k}\left(t - \frac{1}{k}(1 - \mathrm{e}^{-kt})\right) \to \frac{1}{2}t^2$, so $x \to Vt\cos\alpha$ and $y \to Vt\sin\alpha - \frac{1}{2}gt^2$: ordinary projectile motion.
- As $t \to \infty$ the velocity tends to $(-W, -g/k)$, the constant solution of the equation of motion: the ball eventually drifts with the wind while falling at its terminal speed.
- A Runge–Kutta integration of the equation of motion agreed with the formulae for x and y to about 10^{-13} over many random choices of V, α, W, k. With $\tan\alpha = g/(kW)$ the computed path stayed on $y = x\tan\alpha$ and landed at $x = 0$ with $u < 0$; with, say, $V = 20$, $\alpha = 0.6$, $W = 5$, $k = 0.5$ it landed 17.1 m downfield instead.

Common slips

- Taking the resistance as $-mk\mathbf{v}$ (relative to the ground) instead of $-mk(\mathbf{v} - \mathbf{w})$.
- Getting the sign of $\mathbf{w}$ wrong: the ball is kicked *into* the wind, so $\mathbf{w}$ points back towards the goal.
- Stopping at $y = x\tan\alpha$ without checking that the ball really does come back to $x = 0$, and is moving towards the goal when it does.

Beyond STEP

The equation $\dot{\mathbf{v}} + k\mathbf{v} = \mathbf{F}$ with constant $\mathbf{F}$ is the simplest example of a linear system relaxing to equilibrium: the deviation $\mathbf{v} - \mathbf{F}/k$ decays like e^{-kt}. Linear drag is realistic only for slow, small objects; a football really feels drag roughly proportional to the square of its speed, which makes the equations nonlinear and couples the horizontal and vertical motions. Kleppner and Kolenkow, *An Introduction to Mechanics*, discusses both kinds of drag.

WORKED SOLUTION

STEP III 1988, Question 14

Mechanics • Motion on a Smooth Wire

STEP III 1988, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

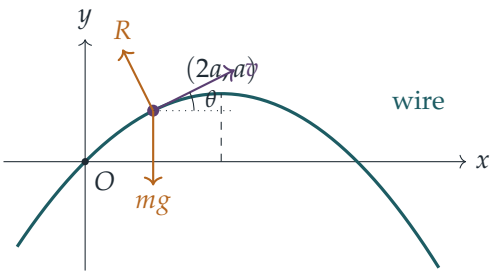
step.maths.org/questions/88-s3-q14

The key idea

A bead that feels no force from the wire is simply a projectile, moving under gravity alone. So part (i) asks: can a free projectile, launched along the wire, follow the wire's parabola exactly? Match the two parabolas. For part (ii), the wire is smooth, so energy is conserved. The work is in writing the speed and the height in terms of the slope angle θ, using $\tan\theta = \mathrm{d}y/\mathrm{d}x$.

Solution

We model the bead as a particle of mass m on a smooth fixed wire. The only forces are its weight mg, acting downwards, and the normal reaction R of the wire, which is perpendicular to the wire.



Two facts about the wire are used throughout. At a general point,

$$\frac{\mathrm{d}y}{\mathrm{d}x} = 1 - \frac{x}{2a},$$

so at the origin the slope is 1 and the bead starts at 45° above the x-axis. Since the wire is a parabola opening downwards, its highest point is at $x = 2a$, $y = a$.

Part (i)

Suppose $R = 0$ throughout the motion. Then gravity is the only force, so the bead moves exactly as a projectile with initial speed V at $45°$. Its horizontal velocity is constant at $V/\sqrt{2}$, so $x = Vt/\sqrt{2}$. Also

$$y = \frac{V}{\sqrt{2}}\,t - \tfrac{1}{2}gt^2.$$

Eliminating $t = \sqrt{2}\,x/V$ gives the trajectory

$$y = x - \frac{gx^2}{V^2}.$$

This is the path of the bead only if it is the wire, $y = x - x^2/(4a)$, for every x the bead reaches. That needs

$$\frac{g}{V^2} = \frac{1}{4a}, \qquad \text{that is,} \qquad V = 2\sqrt{ag}.$$

Conversely, take $V = 2\sqrt{ag}$. A free particle projected from O at $45°$ with this speed follows the curve $y = x - x^2/(4a)$, which is the wire itself. So it moves along the wire with $R = 0$ at every instant, and this motion satisfies the bead's equation of motion with zero reaction. Because $x = Vt/\sqrt{2}$ increases steadily, the bead passes over the top at $(2a, a)$ and continues down the far side for all time. Throughout, it exerts no pressure on the wire, as required, and the suitable speed is $V = 2\sqrt{ag}$.

? **Ask yourself** How can a bead on a wire move with no reaction at all? Doesn't the wire have to hold it on?

The wire only has to push when gravity alone would carry the bead off the curve. For any other speed it would. A larger V gives a flatter projectile parabola, so the wire must pull the bead down. A smaller V gives a more sharply curved one, so the wire must hold it up. At exactly $V = 2\sqrt{ag}$, gravity already supplies the acceleration the curve requires, and the wire has nothing to do.

Part (ii)

Now V is general. The velocity is along the tangent to the wire, so

$$\tan\theta = \frac{\mathrm{d}y}{\mathrm{d}x} = 1 - \frac{x}{2a}, \qquad \text{so} \qquad x = 2a(1 - \tan\theta).$$

(If the bead slides back, its velocity points the opposite way along the tangent and its angle changes by π. Then $\tan\theta$ and $\cos^6\theta$ are unchanged, so nothing below is affected.)

Speed in terms of θ. Differentiating $x = 2a(1 - \tan\theta)$ with respect to t gives

$$\dot{x} = -2a\sec^2\theta\,\dot{\theta}.$$

Since $\dot{y} = \tan\theta\,\dot{x}$, the speed v satisfies

$$v^2 = \dot{x}^2 + \dot{y}^2 = \dot{x}^2(1 + \tan^2\theta) = \dot{x}^2\sec^2\theta = \frac{4a^2\dot{\theta}^2}{\cos^6\theta}.$$

Height in terms of θ. Writing $T = \tan\theta$, so that $x = 2a(1-T)$,

$$\begin{aligned} y = x - \frac{x^2}{4a} &= 2a(1-T) - a(1-T)^2 \\ &= a(1-T)\big(2-(1-T)\big) \\ &= a(1-T)(1+T) = a(1-\tan^2\theta). \end{aligned}$$

Energy. The wire is smooth and R is perpendicular to the velocity, so R does no work and mechanical energy is conserved. The bead starts at O with speed V, so

$$\tfrac{1}{2}mv^2 + mgy = \tfrac{1}{2}mV^2, \qquad \text{that is,} \qquad v^2 + 2gy = V^2.$$

Substituting the expressions for v^2 and y gives

$$\frac{4a^2\dot\theta^2}{\cos^6\theta} + 2ga(1-\tan^2\theta) = V^2,$$

which is the required result.

Checks

- At $t = 0$ we have $\theta = 45^\circ$, so $\tan^2\theta = 1$ and the second term vanishes. The first term is then $v^2 = V^2$, as it should be.
- With $V^2 = 4ag$ from part (i), $\dot x = V/\sqrt{2}$ is constant, so $\dot\theta = -\dot x\cos^2\theta/(2a)$. The first term becomes $2ag\sec^2\theta = 2ag(1+\tan^2\theta)$. Adding $2ga(1-\tan^2\theta)$ gives $4ag = V^2$. The two parts agree.
- Numerically, we stepped the bead's equation of motion along the wire with a Runge–Kutta method. With $a = 1$, $g = 9.8$ and $V = 2\sqrt{9.8}$, the computed normal reaction stayed below 10^{-15} (per unit mass) all the way to $x \approx 13$. With V increased by 30%, it was about 4.8 at the start. For random values of V, including some where the bead slides back, the identity in (ii) held to a relative error of about 10^{-13}.

Common slips

- Showing only that $R = 0$ at the instant of projection. The question wants a motion with no reaction *throughout*, which is why we match the whole trajectory to the wire.
- Forgetting the chain rule factor $\sec^2\theta$ when differentiating $\tan\theta$. This leaves $\cos^2\theta$ instead of $\cos^6\theta$ in the first term.

Beyond STEP

Part (ii) is a first integral of the motion: energy conservation reduces a second-order equation to a first-order one in a single coordinate, here θ. In a first-year course this becomes the Lagrangian approach to constrained motion, where you pick one

coordinate that describes the constraint and never need to calculate the reaction at all. Kleppner and Kolenkow, *An Introduction to Mechanics*, is a good place to see constrained motion and energy methods treated carefully.

WORKED SOLUTION

STEP III 1988, Question 15

Probability and Statistics • Recurrence for Expectations

STEP III 1988, Question 15

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/88-s3-q15

The key idea

Condition on the *first* book of the day. Once it is on the shelf, the rest of the day looks exactly like a fresh day with a shorter shelf, so the expected count for a shelf of length n is tied to the counts for lengths $n-1$ and $n-2$. That gives a linear difference equation, and we solve it just like a second-order recurrence: an auxiliary quadratic for the homogeneous part, plus a particular solution for the constant 1 on the right.

Solution

Throughout, let N_n be the number of books put on the shelf on a day when the shelf has length n, so that $M(n) = \mathrm{E}(N_n)$. Books are thick or thin independently of one another, each thick with probability p. We are told there are always more books than the shelf can hold, so the day always ends with some book failing to fit.

Part (i)

With $n = 0$ there is no room at all. The first book, thick or thin, does not fit, so it and every later book go on the trolley. Hence $N_0 = 0$ always, and $M(0) = 0$.

Part (ii)

With $n = 1$, look at the first book.

- If it is thin (probability q) it fits and fills the shelf. The next book needs at least one unit, so it does not fit, and from then on everything goes on the trolley. So $N_1 = 1$.
- If it is thick (probability p) it needs two units and does not fit, so $N_1 = 0$.

Therefore $M(1) = q \cdot 1 + p \cdot 0 = q$.

Part (iii)

Let $n \geq 2$ and again condition on the first book. Whatever it is, it fits, since it needs at most $2 \leq n$ units.

- If it is thin (probability q), it is shelved and $n-1$ units remain.
- If it is thick (probability p), it is shelved and $n-2$ units remain.

> **Ask yourself** Why is the expected number of *further* books shelved exactly $M(n-1)$ in the first case?

After the first book, the later books arrive in order, are thick or thin independently of the first book with the same probabilities, and are placed by the same rule: shelve until one does not fit, then stop. The only thing the first book has changed is the free space, which is now $n-1$ units. So the remainder of the day is a copy of a whole day with a shelf of length $n-1$, and the expected number of further books shelved is $M(n-1)$. Likewise it is $M(n-2)$ after a thick first book. (This is also true when $n-2=0$, by part (i).)

Counting the first book as well, the law of total expectation gives

$$\begin{aligned} M(n) &= q\big(1+M(n-1)\big) + p\big(1+M(n-2)\big) \\ &= (p+q) + qM(n-1) + pM(n-2), \end{aligned}$$

and since $p+q=1$,

$$M(n) - qM(n-1) - pM(n-2) = 1 \qquad (n \geq 2),$$

as required.

The closed form

First the homogeneous equation $M(n) - qM(n-1) - pM(n-2) = 0$. Trying $M(n) = \lambda^n$ gives the auxiliary equation

$$\lambda^2 - q\lambda - p = 0.$$

Since $q = 1-p$, $\lambda = 1$ is a root ($1-q-p=0$), and the product of the roots is $-p$, so

$$\lambda^2 - q\lambda - p = (\lambda - 1)(\lambda + p).$$

So $A(-p)^n$ and the constant B satisfy the homogeneous equation for any A and B.

Because 1 is a root, a constant cannot produce the right-hand side 1, so we try Cn. Substituting,

$$\begin{aligned} Cn - qC(n-1) - pC(n-2) &= C\big[(1-q-p)n + q + 2p\big] \\ &= C(q+2p) = C(1+p). \end{aligned}$$

This equals 1 when $C = \dfrac{1}{1+p}$, which is fine since $1+p > 0$. By linearity, then, $M(n) = A(-p)^n + B + Cn$ satisfies the equation in (iii) for every $n \geq 2$, whatever A and B are.

Now fit the starting values. Part (i) gives

$$M(0) = A + B = 0, \qquad \text{so } B = -A,$$

and part (ii) gives

$$M(1) = -Ap + B + C = -A(1+p) + \frac{1}{1+p} = 1 - p.$$

Hence

$$A(1+p) = \frac{1}{1+p} - (1-p) = \frac{1-(1-p^2)}{1+p} = \frac{p^2}{1+p},$$

so

$$A = \frac{p^2}{(1+p)^2}, \qquad B = -\frac{p^2}{(1+p)^2}, \qquad C = \frac{1}{1+p}.$$

That is,

$$M(n) = \frac{n}{1+p} - \frac{p^2}{(1+p)^2}\bigl(1 - (-p)^n\bigr).$$

Since (i), (ii) and (iii) fix $M(0)$, $M(1)$ and then each later value in turn, this is not just a possible solution but the only one.

Checks

- $p = 0$ (all thin): $M(n) = n$, as it must be.
- $p = 1$ (all thick): $M(n) = \frac{n}{2} - \frac{1}{4}\bigl(1 - (-1)^n\bigr)$, which is $\frac{n}{2}$ for even n and $\frac{n-1}{2}$ for odd n, i.e. $\lfloor n/2 \rfloor$. Correct.
- Direct from the recurrence with $p = q = \frac{1}{2}$: $M(2) = 1 + \frac{1}{4} = \frac{5}{4}$, and the formula gives $\frac{2}{3} - \frac{1}{9} \cdot \frac{3}{4} = \frac{5}{4}$.
- For large n each book uses on average $1 + p$ units, so we expect about $n/(1+p)$ books; the formula agrees, with a bounded correction from the wasted space at the end of the shelf.

We checked the closed form against exact enumeration of the shelving process for $n \leq 12$ and several values of p, and by simulation; for example, with $p = \frac{1}{2}$ and $n = 5$ both give $M(5) = \frac{103}{32} \approx 3.219$.

Common slips

- Thinking a thin book can still be shelved after a thick one fails to fit. The question says all later books go on the trolley, which is exactly why $M(1) = q$ and not something larger.

- Trying a constant as the particular solution: because $\lambda = 1$ solves the auxiliary equation, a constant is swallowed by the homogeneous part, and you need Cn.

Beyond STEP

Conditioning on the first step and noticing that what remains is a smaller copy of the same problem is called *first-step analysis*. It is the standard way to find expected hitting times and absorption probabilities in Markov chains. Our answer also illustrates the *renewal theorem*: if items of random size are packed into a length n, the number that fit grows like n divided by the mean size. Grimmett and Welsh, *Probability: An Introduction*, covers both difference equations for random walks and an introduction to Markov chains.

WORKED SOLUTION

STEP III 1988, Question 16

Probability and Statistics • Hypergeometric Distribution

STEP III 1988, Question 16

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/88-s3-q16

The key idea

Drawing without replacement makes the draws dependent, so working through them one at a time is messy. Two ideas get round this. First, count unordered selections to get the distribution of the number of reds. Second, use symmetry: if we imagine the whole urn being emptied in a random order, every ball is equally likely to be in any given position. Writing X as a sum of indicators X_i then gives the mean and variance with no need for the distribution at all. We only need the covariances between pairs.

Solution

The probability of exactly k reds

Think of the balls as distinguishable. The n balls drawn form a set of size n, and every such set is equally likely, so there are $\binom{M}{n}$ equally likely outcomes. A set with exactly k red balls is made by choosing k of the m reds and $n-k$ of the $M-m$ greens. Hence

$$\mathrm{P}(\text{exactly } k \text{ red}) = \frac{\binom{m}{k}\binom{M-m}{n-k}}{\binom{M}{n}},$$

with the usual convention that $\binom{a}{b} = 0$ if $b < 0$ or $b > a$. (This covers the impossible cases, such as $n-k > M-m$.) This is the *hypergeometric* distribution.

Part (i)

Imagine continuing to draw until the urn is empty. The order in which the M (distinguishable) balls come out is a uniformly random arrangement: all $M!$ orders are equally likely. Stopping after n draws doesn't change the first n positions, so we may work with the full arrangement.

Fix a position i. The number of arrangements with a particular ball in position i is

$(M-1)!$, the same for every ball. So the ball in position i is equally likely to be any of the M balls, and

$$\mathrm{P}(X_i = 1) = \frac{m\,(M-1)!}{M!} = \frac{m}{M},$$

as required.

> **Ask yourself** Surely the second ball's colour depends on what the first ball was?

It does, given the first ball. But we want the *unconditional* probability. For $i = 2$ the law of total probability gives

$$\mathrm{P}(X_2 = 1) = \frac{m}{M}\cdot\frac{m-1}{M-1} + \frac{M-m}{M}\cdot\frac{m}{M-1} = \frac{m(M-1)}{M(M-1)} = \frac{m}{M}.$$

The symmetry argument does the same calculation for every i at once.

Part (ii)

Fix positions $i \neq j$ (so $M \geq 2$). Count the arrangements with red balls in both positions. There are m choices for the ball in position i, then $m-1$ for position j, and $(M-2)!$ ways to arrange the rest. Hence

$$\mathrm{P}(X_i = 1 \text{ and } X_j = 1) = \frac{m(m-1)(M-2)!}{M!} = \frac{m(m-1)}{M(M-1)},$$

as required. In other words, the ordered pair of balls in positions (i, j) is equally likely to be any of the $M(M-1)$ ordered pairs of distinct balls.

Mean and variance of X

X counts the red balls among the first n drawn. By linearity of expectation, which needs no independence, and since $\mathrm{E}(X_i) = \mathrm{P}(X_i = 1)$,

$$\mathrm{E}(X) = \sum_{i=1}^{n} \mathrm{E}(X_i) = \frac{nm}{M}.$$

For the variance, use $\mathrm{Var}(X) = \mathrm{E}(X^2) - [\mathrm{E}(X)]^2$. Expanding the square of the sum, and using $X_i^2 = X_i$ because X_i is 0 or 1,

$$X^2 = \sum_{i=1}^{n} X_i^2 + \sum_{i\neq j} X_i X_j = X + \sum_{i\neq j} X_i X_j,$$

where the second sum runs over the $n(n-1)$ ordered pairs (i, j) with $i \neq j$. Now $X_i X_j = 1$ exactly when both balls are red, so by part (ii) $\mathrm{E}(X_i X_j) = \dfrac{m(m-1)}{M(M-1)}$.

Therefore, for $M \geq 2$,

$$\begin{aligned}\mathrm{Var}(X) &= \frac{nm}{M} + \frac{n(n-1)m(m-1)}{M(M-1)} - \frac{n^2m^2}{M^2} \\ &= \frac{nm}{M^2(M-1)}\Big[M(M-1) + (n-1)(m-1)M - nm(M-1)\Big].\end{aligned}$$

The bracket expands to

$$\begin{aligned}&M^2 - M + nmM - nM - mM + M - nmM + nm \\ &\quad = M^2 - nM - mM + nm = (M-m)(M-n).\end{aligned}$$

Hence

$$\mathrm{Var}(X) = \frac{nm(M-m)(M-n)}{M^2(M-1)}.$$

(If $M = 1$ then $n \leq 1$, there are no pairs, and X is constant, so $\mathrm{Var}(X) = 0$.)

It is illuminating to write $p = m/M$. Then

$$\mathrm{E}(X) = np, \qquad \mathrm{Var}(X) = np(1-p) \cdot \frac{M-n}{M-1}.$$

This is the binomial variance multiplied by a correction factor, which is less than 1 when $n \geq 2$.

Checks

- $n = M$: we take every ball, so $X = m$ for certain. The formula gives mean m and variance 0, because of the factor $M - n$.
- $n = 1$: X is a single Bernoulli variable with $p = m/M$, and the variance reduces to $p(1-p)$.
- $M = 6$, $m = 2$, $n = 3$: the formula gives $\mathrm{Var}(X) = \frac{3 \cdot 2 \cdot 4 \cdot 3}{36 \cdot 5} = \frac{2}{5}$. Enumerating all $6!$ orders exactly agrees, as does every case with $M \leq 6$.

Common slips

- Treating the X_i as independent and getting the binomial variance $np(1-p)$. They are negatively correlated: a red ball drawn early makes later reds less likely.
- Forgetting that the double sum $\sum_{i \neq j}$ has $n(n-1)$ terms, not $\binom{n}{2}$, when the pairs are ordered.

Beyond STEP

Writing a count as a sum of indicator variables is one of the most powerful tricks in probability. It gives means and variances even when the distribution itself is hopeless to

write down. The factor $\frac{M-n}{M-1}$ is the *finite population correction* used in survey sampling. It shows why a poll of 1000 people is almost as informative about a country as it would be about an infinite population. Blitzstein and Hwang, *Introduction to Probability*, uses indicators throughout and treats the hypergeometric distribution in detail.

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